Variants of genuine G-spectra for profinite groups

with an outlook towards endomorphism K-theory

von

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Introduction

Equivariant (stable) homotopy theory is an umbrella term for the study of topological spaces (or spectra) with group action. Here one finds a spectrum of notions between *naive*, or "up to homotopy", and *genuine* equivariance. The genuine theory is more intricate and has underlain many recent developments on trace invariants such as THH and TC, with important applications to algebraic K-theory, geometric topology and arithmetic geometry.

The current manuscript studies equivariant stable homotopy theory for profinite groups. Our motivation is provided by the study of endomorphism and automorphism K-theory. Although THH and KEnd of a ring are known to carry an S^1-action, the same isn't true when they are taken with coefficients in a bimodule. Instead, one obtains an *(epi)cyclic* structure on the relevant spectra, with an underlying genuine $\widehat{\mathbb{Z}}$-spectra. Another important source of profinite group actions is Galois theory, such as the $K(n)$-local pro-$\mathbb{G}_n$-Galois extension $\mathbb{S}_{K(n)} \to E_n$ [DH04]) from chromatic homotopy theory.

Equivariant (stable) homotopy theory of various flavors

One way of approaching genuine G-spectra Sp_G for a finite group G is to start with the symmetric monoidal category $\mathrm{An}_{G,*}$ of pointed G-spaces under smash product, and invert all representation spheres. [CMNN24, Theorem A.2]. Classically this is achieved using the model category of orthogonal G-spectra, which is often still the canonical approach for compact Lie groups, although it has been shown in [AMR17] that one can recover Sp_G from geometric fixed points and gluing datum.

For a profinite group, all of its representation spheres are inflated from one of its finite discrete quotients.

Proposition 0.1 *Any continuous group homomorphism $f : G \to H$ from a profinite G to a Lie group H factors through a discrete quotient. Consequently, every representation sphere of G is a representation sphere of one of its discrete quotients.*

Proof. Every Lie group has a neighborhood basis at e_H which contains no non-trivial subgroups. Choose such a neighborhood U of e_H. As G is profinite, it has a neighborhood basis at e_G consisting of open normal subgroups. Hence $f^{-1}(U)$ contains an open normal subgroup of G which must be mapped to e_H. □

This suggests that the category Sp_G should be "assembled" from Sp_Q, where Q runs over the discrete quotients of G.

In the case of finite groups, the role of orthogonal spectra is in recent years more frequently being replaced by spectral Mackey functors, or spectral presheaves with transfers, in the wake of work by Guillou–May, partly summarised in [CMNN24, Appendix A].

The theory of Mackey functors confirms our intuition from representation spheres.

Proposition 0.2 (cf. Proposition 2.1.1) *For a profinite group G, the category of (spectral) G-Mackey functors is the Pr^L-colimit of the G/N-Mackey functors along inflation, namely the $\widehat{\mathrm{Cat}}$-limit along categorical fixed points.*

As a variant of Sp_G, we define a category $\mathrm{Sp}_G^{\mathrm{geo}}$ using geometric fixed points, which we call *geometric G-spectra*. For $G = \widehat{\mathbb{Z}}$, this encodes a notion of completeness with respect to the *Verschiebungen* on $\widehat{\mathbb{Z}}$-spectra, and is instrumental to the theory of topological Cartier modules [McC23; AN21].

In Chapter 2, we prove the following relation between geometric and genuine G-spectra. See there for the definitions of concepts.

Theorem 0.3 (Geometric convergence, 2.3.4, 2.4.11) *Let G be a topologically finitely generated pro-p-group. The category $\mathrm{Sp}_G^{\mathrm{geo},-}$ of uniformly bounded below geometric G-spectra is equivalent to the category $L^{\mathrm{geo}}\mathrm{Sp}_G^-$ of uniformly bounded below geometrically local G-spectra.*

For any topologically finitely generated profinite group G, the category $\mathrm{Sp}_G^{\mathrm{geo},-}$ is equivalent to the category $\mathrm{Sp}_G^{\mathrm{qfgen},-}$ of uniformly bounded below quasifinitely genuine G-spectra as in [KMN23].

Equivariant K-theory in topology and automorphisms

It shouldn't be surprising that many interesting applications of genuine equivariant spectra have to do the study of actual equivariant geometric objects.

In geometric topology, one naturally encounters spaces carrying a group action by a discrete group G, often the fundamental group of a space. The *algebraic K-theory* groups $\mathrm{K}_\bullet(\mathbb{Z}[G])$, or their spherical variants $\mathrm{K}_\bullet(\mathbb{S}[\Omega G])$ known as Waldhausen A-theory accommodate invariants of geometric origin such as Wall's finiteness obstruction and Whitehead torsion. As G varies, one obtains restriction and transfer maps between K-theory spectra, resulting in a Mackey functor known as equivariant algebraic K-theory.

The modern construction of the equivariant K-theory Mackey functor works equally well to recover a special case of endomorphism and automorphism K-theory which are twisted by an endofunctor. We will explain this construction in Chapter 3.

Structure of the manuscript

In chapter 1, we will concern ourselves with the theory of spectral Mackey functors, with genuine G-spectra as a guiding thread. Most content of this chapter is present in the literature in some form, though a detailed treatment which pertains to the Mackey approach is harder to come by.

We provide a version of the classical Segal–tom Dieck splitting in section 1.3 for Mackey functors on an orbital category, including the interaction of the splitting under with transfers. This is a minor extension of [McC23, 3.2.11] and was used during the proof of a previous version of theorem 0.3. We also use it to supply a simple argument identifying geometric fixed point $\Phi^N : \mathrm{Sp}_G \twoheadrightarrow \mathrm{Sp}_{G/N}$ as a smashing localisation.

In chapter 2, we explore the divergence between the geometric and categorical approach. Here we prove theorem 0.3, and study a relevant finiteness property of profinite groups. We also review the work of Krause–McCandless–Nikolaus [KMN23], and show how their construction circumvents this finiteness property.

Chapter 3 is a survey on endomorphism K-theory and its correct habitat of *epicyclic Mackey functors*, a notion related to $\mathrm{Sp}_{\widehat{\mathbb{Z}}}$, and highlights the conceptual roots of the usage of spans in the study

of localising invariants and K-theory, as is evident from its usage in [CMNN24; MackI; MackII]. As a side product, we also obtain equivariant K-theory from the same machinery.

Appendix A is concerned with notions in higher algebra, which are important but also distract from the main topic.

Conventions and notations

- We view ordinary $(1,1)$-categories, such as Set, and $(2,1)$-categories, such as $\mathrm{Span}(G)$, as $(\infty,1)$-categories, which we simply refer to as *category*, or 1-*category*. We make heavy use of the theory of presentable, stable and symmetric monoidal categories as laid down in respective chapters of [HTT; HA].
- For a category $\mathcal{C}$, the notation $\mathrm{Map}_{\mathcal{C}}(-,-)$ denotes mapping anima or commutative monoid when $\mathcal{C}$ is semiadditive, whereas $\mathrm{map}_{\mathcal{C}}(-,-)$ stands for mapping spectra when $\mathcal{C}$ is stable.
- Duals in a symmetric monoidal category are denoted $(-)^{\vee}$.

  In a presentable symmetric monoidal category $\mathcal{C}$, we write $\underline{\mathrm{map}}_{\mathcal{C}}(-,-)$ for the internal mapping object in $\mathcal{C}$. For a presentable category $\mathcal{D}$ tensored over $\mathcal{C}$, we write $\mathcal{C}\text{-}\underline{\mathrm{map}}_{\mathcal{D}}(-,-)$ for the $\mathcal{C}$-mapping object in $\mathcal{D}$. Exceptions are when $\mathcal{C} = \mathrm{Sp}$, Cat or $\widehat{\mathrm{Cat}}$, where we write $\mathrm{map}_{\mathcal{D}}(-,-)$ and $\mathrm{Fun}_{\mathcal{D}}(-,-)$ for conciseness.
- The phrase 2-*category* means $(\infty,2)$-category, or complete Segal 1-categories as in [GH15, 6.1.5, 5.4.3].
- The word *space* refers to objects of the category An, also known as anima, weak homotopy types, 0-categories or ∞-groupoids. We mean by *groupoid* a 1-groupoid, or $(1,0)$-category. The underlying space, also known as the core of a category $\mathcal{C}$, is denoted $\mathcal{C}^{\simeq}$.
- We always prepend 2-categories with 2. For instance, the notation Cat_2 denotes the 1-category of 2-categories, while 2Cat stands for the 2-category of 1-categories.

Symbol	Objects	Morphisms
An	spaces	maps
CMon	$\mathbb{E}_\infty$-spaces/commutative monoids	—
Sp	spectra	—
$\mathrm{Cat}/\widehat{\mathrm{Cat}}$	small/large categories	functors
$\mathrm{Pr}^L/\mathrm{Pr}^R$	presentable categories	left/right adjoint functors
$\mathrm{Pr}^L_{\mathrm{st},\omega}$	compactly generated stable categories	strongly continuous functors
$\mathrm{Cat}^{\mathrm{perf}}$	small idempotent complete stable categories	exact functors
$\mathrm{coCart}(\mathcal{C})$	(large) coCartesian fibrations over $\mathcal{C}$	functors over $\mathcal{C}$
$\mathrm{RFib}(\mathcal{C})$	right fibrations over $\mathcal{C}$	—

Table 1.: Commonly used categories

Recall that unstraightening identifies $\mathrm{Fun}(\mathcal{C},\widehat{\mathrm{Cat}})$ with the subcategory of $\mathrm{coCart}(\mathcal{C})$ consisting of functors that preserve coCartesian morphisms.

- In the chapters 1 and 2, all groups are profinite and subgroups are closed unless otherwise stated. In chapter 2, we also consider discrete groups. The unit and the trivial subgroup of a group G are both denoted e_G or simply e.

  A group G acts on the left by default, and we view right actions as G^{op}-actions. Composition are read from right to left, so $f \circ g$ means "f after g". Automorphisms of the G-set G/H are given by the Weyl group $W_G(H) := \mathrm{Aut}_{G\text{-Set}}(G/H)$, which is $(N_G(H)/H)^{\mathrm{op}}$ in our convention.

- Actions of groups on sets, topological spaces etc. are always continuous. A topological space with G-action is called a topological G-space.

- For a group G, we write $\mathbf{B}G$ for its classifying space, the unique connected pointed space with $\Omega\mathbf{B}G \simeq G$. For a monoid M, we write $\mathbf{B}M$ for its classifying category, the unique connected pointed category with $\mathrm{End}_{\mathbf{B}M}(*) \simeq M$.

- For an object X of a category $\mathcal{C}$ with G-action, given by a functor $\mathbf{B}G \to \mathcal{C}$, its colimit, the homotopy G-orbits of X, is denoted by $X_{\mathrm{h}G}$. When $\mathcal{C} = \mathrm{An}$, this is the space over $\mathbf{B}G$ corresponding to $X : \mathbf{B}G \to \mathrm{An}$ under unstraightening.

  For a category $\mathcal{C}$ with an action of a monoid M, given as a functor $\mathbf{B}M \to \widehat{\mathrm{Cat}}$, we write $M\backslash\backslash\mathcal{C} \to \mathbf{B}M$ for its coCartesian unstraightening.

1. Equivariant spectra as Mackey functors

This chapter recounts the basic theory of genuine equivariant spectra for profinite groups. Although the Mackey functor approach we take extends to any *orbital category*, our stress will fall upon the special case, with a minor bias towards finite groups.

We aim at developing the theory completely from the categorical-algebraic perspective, and provide proofs of several classical results in equivariant stable homotopy theory.

1.1. G-Mackey functors

Let G be a group and A an abelian group with G-action. The concept of G-Mackey functors is an abstraction of the structure on the fixed points of A by subgroups of G.

Example 1.1.1 (Mackey functors in ordinary algebra) Consider an abelian group A acted upon by a discrete group G. For any G-set X, the "X-points" of A is the abelian group

$$\underline{A}(X) = \mathrm{Map}_{\mathbb{Z}[G]\text{-Mod}}(\mathbb{Z}[X], A) = \mathrm{Map}_{G\text{-Set}}(X, A).$$

When $X = G/H$ is G-orbit, we have $\underline{A}(G/H) \simeq A^H$ by evaluation at $eH \in G/H$.

The following structure is carried by these fixed-point groups:

1. For a map $X \xrightarrow{f} Y$ of G-sets, there is a *restriction* morphism

   $$f^* : \underline{A}(Y) \to \underline{A}(X)$$

   given by composition:

   $$(\alpha : Y \to A) \mapsto \big(f^*\alpha = \alpha \circ f : X \to Y \to A\big).$$

   This gives a functor $\underline{A}^* : G\text{-Set}^{\mathrm{op}} \to \mathrm{Ab}$.

   In the case of a map $f : G/H \to G/K, H \mapsto gK$, one can write $f^* : A^K \to A^H$ as $x \mapsto gx$.

2. Writing $X \xleftarrow{\sim} \coprod_{i\in I} G/H_i$ as a sum or orbits, $\underline{A}^*$ takes this sum to a product:

   $$\underline{A}(X) \xrightarrow{\sim} \prod_{i\in I} \underline{A}(G/H).$$

3. For a map $X \xrightarrow{f} Y$ of G-sets with finite fibers, there is a *transfer* morphism

   $$f_! : \underline{A}(X) \to \underline{A}(Y)$$

   given by summing over fibers:

   $$(\alpha : X \to A) \mapsto \left(f_!(\alpha) : Y \to A, y \mapsto \sum_{x\in f^{-1}(y)} \alpha(x)\right).$$

   This gives a functor $\underline{A}_! : G\text{-Set}^{\text{ét}} \to \mathrm{Ab}$, where $G\text{-Set}^{\text{ét}}$ is the category of finite fiber maps between G-sets. The functor $\underline{A}^!$ agrees with $\underline{A}^*$ on objects and bijections (up to taking inverse).

4. (Mackey formula) For a pullback of G-sets, where f has finite fibers,

$$\begin{array}{ccc} X & \xrightarrow{f'} & Y \\ \downarrow{\scriptstyle g'} & \ulcorner & \downarrow{\scriptstyle g} \\ Z & \xrightarrow{f} & W \end{array}$$

one has

$$g^* \circ f_! = f'_! \circ g'^* : \underline{A}(Z) \to \underline{A}(Y).$$

The datum of the two functors $\underline{A} := (\underline{A}^*, \underline{A}!)$ is called the *fixed point Mackey functor* of A.

The structure existing upon such a pair of functors can be encoded using the construction of *span categories*. We start with a special case of Barwick's "adequate triples" [MackI, 5.4].

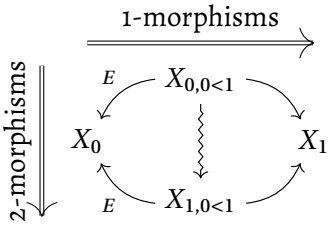
Definition 1.1.2 *We let* $\mathrm{Cat}^{\mathrm{pbPair}}$ *denote the category of* ***pullback pairs****, whose objects are pairs* $(\mathcal{C}, E)$ *of a small category* $\mathcal{C}$ *and a wide subcategory* $\mathcal{C}^{\simeq} \subseteq E \subseteq \mathcal{C}$*, such that the pullback of any morphism of* E *along morphisms of* $\mathcal{C}$ *exists and belongs to* E*, and whose morphisms are functors which preserve the wide subcategory* E *and pullbacks along* E*.*

We also let $\mathrm{Cat}^{\mathrm{pb}}$ *be the full subcategory of* $\mathrm{Cat}^{\mathrm{pbPair}}$ *on those pairs with* $E = \mathcal{C}$*.*

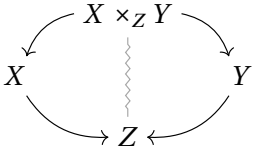
The span construction is a functor $\mathrm{Span} : \mathrm{Cat}^{\mathrm{pbPair}} \to \mathrm{Cat}$, taking $(\mathcal{C}, E)$ to the **span 1-category** $\mathrm{Span}_E(\mathcal{C})$ with $\mathrm{Span}_E(\mathcal{C})^{\simeq} \simeq \mathcal{C}^{\simeq}$, whose morphism spaces are the space of spans with **backward**[1] morphisms coming from E as follows, and whose composition is given by forming pullbacks:

$$\left(\begin{array}{ccc} & X_{1<2} & \\ {\scriptstyle E\ni h}\swarrow & & \searrow{\scriptstyle k} \\ X_1 & & X_2 \end{array} \right) \circ \left(\begin{array}{ccc} & X_{0<1} & \\ {\scriptstyle E\ni f}\swarrow & & \searrow{\scriptstyle g} \\ X_0 & & X_1 \end{array} \right) = \left(\begin{array}{ccccc} & & X_{0<1<2} & & \\ & {\scriptstyle h'}\swarrow & \wedge & \searrow{\scriptstyle g'} & \\ & X_{0<1} & & X_{1<2} & \\ {\scriptstyle f}\swarrow & & \searrow{\scriptstyle g}\ \ {\scriptstyle h}\swarrow & & \searrow{\scriptstyle k} \\ X_0 & & X_1 & & X_2 \end{array} \right)$$

Such spans naturally form the morphism categories of a **span 2-category** $2\mathrm{Span}_E(\mathcal{C})$, whose underlying 1-category is $\mathrm{Span}_E(\mathcal{C})$. The 2-morphisms in $2\mathrm{Span}_E(\mathcal{C})$ are given by commuting diagrams between spans, which we depict using squiggly arrows to distinguish from the 1-morphisms:



, with composition denoted:



.

[1] We adopt a convention opposite to the literature, where it is common to restrict the forward morphisms, and to define Mackey functors as covariant functors on spans.

There is a forwards inclusions of $\mathcal{C}$ and a backwards inclusion of E into $\mathrm{Span}_E(\mathcal{C})$, denoted

$$(-)_\dagger : \mathcal{C} \to \mathrm{Span}_E(\mathcal{C}), \quad (X \xrightarrow{f} Y) \mapsto \left(\begin{array}{ccc} & X & \\ \swarrow & & \searrow^{f} \\ X & & Y \end{array} \right),$$

$$(-)^\dagger : E^{\mathrm{op}} \to \mathrm{Span}_E(\mathcal{C}), \quad (X \xleftarrow{f} Y) \mapsto \left(\begin{array}{ccc} & Y & \\ \swarrow_{f} & & \searrow \\ X & & Y \end{array} \right).$$

Remark 1.1.3. More general versions of the span categories are studied, including restricting both forward and backward morphisms to a smaller class ([MackI]), up until with spans between spans as 2-morphisms (as in [CLL25b]).

We sketch their construction below. The reader who wishes for more details may consult [MackI, Sec. 2–3].

Construction 1.1.4 (Span 2- and 1-categories) *We construct* $2\mathrm{Span}_E(\mathcal{C})$ *as a complete Segal category. Let* $\mathrm{TwAr}[n]$ *be the twisted arrow poset of the poset* $[n] = \{0 < 1 < \ldots < n\}$. *An object in* $\mathrm{TwAr}[n]$ *is an arrow* $(i \leqslant j)$ *where* $0 \leqslant i \leqslant j \leqslant n$, *and there is an arrow* $(i \leqslant j) \to (k \leqslant l)$ *precisely when*[2] $i \leqslant k$ *and* $l \leqslant j$.

The n*-th component of* $2\mathrm{Span}_E(\mathcal{C})$ *is the full subcategory*

$$2\mathrm{Span}_E(\mathcal{C})_n \subseteq \mathrm{Fun}(\mathrm{TwAr}[n], \mathcal{C})$$

of those functors which are right Kan extended from the sub-poset on $\{(i \leqslant j), j - i = 0, 1\} \subseteq [n]$, *sending each twisted arrow* $(i \leqslant j+1) \to (i \leqslant j)$ *into* E. *Intuitively, this category consists of* n *composable spans and coherently chosen pullbacks.*

Assuming the existence of pullbacks along E, *the resulting simplicial category is Segal. It being complete Segal amounts to the statement that the subcategory of* $\mathrm{Fun}(\mathrm{TwAr}[1], \mathcal{C})$ *consisting of invertible spans is equivalent to* $\mathcal{C}$. *The datum of an inverse to a span implies that both of its legs are split epimorphisms, with sections also being a split epimorphism, which implies invertibility. Since* $\mathrm{TwAr}[1]$ *is contractible, the category they form is indeed equivalent to* $\mathcal{C}$.

Remark 1.1.5. If $E = \mathcal{C}$, then $\mathrm{Span}(\mathcal{C})$ is equivalent to its own opposite, by exchanging the "forward" and "backward" directions of spans. The same can be said for the span 2-category, if we only reverse the direction of 1-morphisms.

If $E = \mathcal{C}^{\simeq}$ consists of only isomorphisms, then the complete Segal category $2\mathrm{Span}_{\mathrm{iso.}}(\mathcal{C})$ is constant and thus equivalent to the 1-category $\mathcal{C}$ one started with.

Example 1.1.6 The category of spans of finite sets with injections as the backward morphisms is the category of partially defined maps between finite sets. This category occurs as the commutative operad, and is equivalent to the category of pointed finite sets.

The following example hints at the use of span categories at encoding semiadditive structure, and in fact gives the "correct" answer equivariantly.

[2] We warn the reader that the twisted arrow categories of [HA, §5.2.1] and [MackI, §2.] are opposite to each other. Here we follow the former convention, so that $(0 < n)$ is the initial vertex ("top") of a span.

Example 1.1.7 The Lawvere theory ([Cra09]) of $\mathbb{E}_\infty$-monoids is the $\mathrm{Span}(\mathrm{Set}^\omega)$, where Set^ω denotes the category of *finite* sets. The mapping spaces are 1-groupoids of matrices of finite sets which can be presented with matrices of natural numbers as objects, and products of symmetric groups as automorphisms.

Definition 1.1.8 *The span (2-)category of a profinite group G is the span (2-)category on the category G-Set^ω of finite (continuous) G-sets:*

$$\mathrm{Span}(G) := \mathrm{Span}(G\text{-}\mathrm{Set}^\omega), \qquad 2\mathrm{Span}(G) := 2\mathrm{Span}(G\text{-}\mathrm{Set}^\omega).$$

Note that $\mathrm{Span}(G)$ is in fact a $(2,1)$-category, meaning its mapping spaces of spans are in fact $(1,0)$-categories (a.k.a. 1-groupoid). Similarly $2\mathrm{Span}(G)$ is a $(2,2)$-category.

Proposition 1.1.9 ([MackI, Prop. 4.3]) *The span category $\mathrm{Span}(G)$ is semiadditive. Biproducts in $\mathrm{Span}(G)$ are given by the disjoint union of G-sets.*

Proof. The category $\mathcal{C} := G\text{-}\mathrm{Set}^\omega$ is *disjunctive* in the sense that the slice category $\mathcal{C}_{/x \amalg y}$ is equivalent to $\mathcal{C}_{/x} \times \mathcal{C}_{/y}$ through pulling back along the inclusions $x \to x \amalg y \leftarrow y$. Since any equivalence can be made adjoint, the inverse is seen to be given by the coproduct functor.

In the span category, the mapping spaces are equivalent to the core of a variant of these slice categories:

$$\mathrm{Map}_{\mathrm{Span}(G)}(T, X \amalg Y) \simeq \mathcal{C}^{\simeq}_{\{T, X\amalg Y\}} \simeq \mathcal{C}^{\simeq}_{\{T,X\}} \times \mathcal{C}^{\simeq}_{\{T,Y\}} \simeq \mathrm{Map}_{\mathrm{Span}(G)}(T,X) \times \mathrm{Map}_{\mathrm{Span}(G)}(T,Y).$$

Moreover, the equivalence is given by pullbacks, which is exactly composition in $\mathrm{Span}(G)$. Hence the backward spans

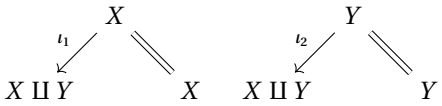


exhibit $X \amalg Y$ as a product of X and Y in $\mathrm{Span}(G)$. Self-symmetry then implies that the mirrored spans exhibit it as the coproduct.

Similarly we see that the zero object is $\emptyset$ and the natural map from the coproduct to the product is the identity of $X \amalg Y$, so that $\mathrm{Span}(G)$ is semiadditive. □

Similarly $\mathrm{Span}_E(\mathcal{C})$ is semiadditive when $\mathcal{C}$ is disjunctive and the relevant morphisms belong to E, see [MackI, 5.11].

Definition 1.1.10 (Mackey functors) *Let G be a finite group and $\mathcal{V}$ be a category with finite products. The category of $\mathcal{V}$-**valued** G-**Mackey functors** consists of product-preserving functors from $\mathrm{Span}(G)^{op}$ to $\mathcal{V}$.*

$$\mathrm{Mack}(G;\mathcal{V}) := \mathrm{Fun}^\times(\mathrm{Span}(G)^{op}, \mathcal{V}).$$

For a subgroup $H \leqslant G$ and $M \in \mathrm{Mack}(G;\mathcal{V})$, we call

$$M^H := M(G/H)$$

*the **(categorical)** H-**fixed points** of M.*

A map $X \xrightarrow{f} Y$ in G-Set^ω induces then the two functorialities.

$$f^* : M(X) \longleftarrow M(Y) \quad \text{“restriction”}, \qquad f_! : M(X) \longrightarrow M(Y) \quad \text{“transfer”}$$

Remark 1.1.11 ([Ram, Theorem 2.15]). As the span categories are semiadditive, Mackey functors lift uniquely to $\mathrm{CMon}(\mathcal{V})$, and the category of Mackey functors is automatically semiadditive.

Observe that $\mathrm{Mack}(G;\mathrm{An})$ is the non-abelian derived category $\mathcal{P}_\Sigma(\mathrm{Span}(G))$.

Proposition 1.1.12 *For presentable $\mathcal{V}$, the category of Mackey functors are given by the Lurie tensor product*

$$\mathrm{Mack}(G;\mathcal{V}) \simeq \mathcal{P}_\Sigma(\mathrm{Span}(G)) \otimes \mathcal{V}.$$

Thus if $\mathcal{V}$ is stable or compactly generated, then so is $\mathrm{Mack}(G;\mathcal{V})$.

Let $y_{G/H}$ denote the Yoneda image of G/H in $\mathcal{P}_\Sigma(\mathrm{Span}(G))$. If $\mathcal{V}$ is presentably symmetric monoidal, then $\mathrm{Mack}(G;\mathcal{V})$ *is tensored over $\mathcal{V}$, and we have*

$$M(G/H) \overset{\text{Yoneda}}{=\!=} \mathcal{V}\text{-}\underline{\mathrm{map}}_{\mathrm{Mack}(G;\mathcal{V})}(y_{G/H} \otimes \mathbb{1}_{\mathcal{V}}, M).$$

Henceforth, when $\mathcal{V}$ is presentably symmetric monoidal, we will often write $G/H \in \mathrm{Mack}(G;\mathcal{V})$ in place of the more proper notation $\mathbb{1}_{\mathcal{V}}[G/H]$ for $y_{G/H} \otimes \mathbb{1}_{\mathcal{V}}$.

Proof. From [HA, 4.8.1.17] and [HTT, 5.5.8.15(3)], we obtain the following equivalences.

$$\mathcal{P}_\Sigma(\mathrm{Span}(G)) \otimes \mathcal{V} \simeq \mathrm{RFun}(\mathcal{P}_\Sigma(\mathrm{Span}(G))^{\mathrm{op}}, \mathcal{V}) \simeq \mathrm{Fun}^{\times}(\mathrm{Span}(G)^{\mathrm{op}}, \mathcal{V}).$$

Compact generation of $\mathcal{P}_\Sigma(\mathrm{Span}(G))$ is [HTT, 5.5.8.10(6)]. By [HA, 4.8.1.4], applied to the class of finite colimits and idempotents, a tensor product of two compactly generated categories is compactly generated.

As for second paragraph, consider the two-variable adjunction

$$\otimes : \mathcal{P}_\Sigma(\mathrm{Span}(G)) \times \mathcal{V} \to \mathrm{Mack}(G;\mathcal{V}), \qquad \hom_{\mathcal{V}} : \mathcal{P}_\Sigma(\mathrm{Span}(G))^{\mathrm{op}} \times \mathrm{Mack}(G;\mathcal{V}) \to \mathcal{V}.$$

Seeing that $\hom_{\mathcal{V}}$ is equivalent to $\mathrm{ev} : \mathcal{P}_\Sigma(\mathrm{Span}(G))^{\mathrm{op}} \times \mathrm{RFun}(\mathcal{P}_\Sigma(\mathrm{Span}(G))^{\mathrm{op}}, \mathcal{V}) \to \mathcal{V}$, there holds $\hom_{\mathcal{V}}(y_{G/H}, M) \simeq M(G/H)$. Thus when $\mathcal{V}$ is presentably symmetric monoidal, we have

$$\mathcal{V}\text{-}\underline{\mathrm{map}}_{\mathrm{Mack}(G;\mathcal{V})}(y_{G/H} \otimes \mathbb{1}_{\mathcal{V}}, M) \simeq \underline{\mathrm{map}}_{\mathcal{V}}(\mathbb{1}_{\mathcal{V}}, \hom_{\mathcal{V}}(y_{G/H}, M)) \simeq M(G/H). \qquad \square$$

Much of the structure on the categories of Mackey functors will have to do with functors between them. As it turns out, computing (co)limits of Mackey functors is quite simple. This is an important fact, as it implies that many functors have both adjoints, and are (bi)monadic as soon as they are conservative.

Proposition 1.1.13 ([MackI, Prop. 6.7]) *Let $\mathcal{V}$ be a presentable semiadditive category, then* $\mathrm{Mack}(G;\mathcal{V})$ *is presentable and (co)limits of Mackey functors are computed objectwise.*

In particular, for any product preserving functor $f : \mathrm{Span}(H) \to \mathrm{Span}(G)$, the induced functor

$$f^* : \mathrm{Mack}(H;\mathcal{V}) \leftarrow \mathrm{Mack}(G;\mathcal{V})$$

is both a left and right adjoint. $\square$

In light of the following theorems, we will also refer to G-Mackey functors valued in spectra as **genuine G-spectra**, and those valued in spaces as **genuine G-commutative monoids**.

Theorem 1.1.14 (Guillou–May, see [CMNN24, Theorem A.1], [BBB24, Prop. A.8]) *For a profinite group G, the category* $\mathrm{Mack}(G;\mathrm{Sp})$ *underlies the model category* Sp_G *of orthogonal G-spectra.* $\square$

Theorem 1.1.15 ([CLL25a, Cor. 9.16]) *For finite groups G, the category* $\mathrm{Mack}(G;\mathrm{An})$ *underlies the model category* CMon_G *of Shimakawa's G-Γ-spaces.* $\square$

Most classically among such results, the category of **genuine G-spaces** also admits a description as a functor category.

Theorem 1.1.16 (Elmendorf, [LNP25, Example 3.40]) *For finite G, the category $\mathrm{PSh}(\mathrm{Orb}_G)$ of presheaves on the category of finite transitive G-sets underlies the model category An_G of topological G-spaces.* □

In the above equivalences (implicit in the statement), a (bifibrant) topological G-space, or orthogonal G-spectrum X, is taken to the functor whose value at the G-set G/H is the set-theoretical, or *genuine* H-fixed points of X. In the case of G-spectra, the values of the corresponding Mackey functor is also known as the *categorical* or Lewis–May fixed points.

For convenience, we collect the above notations appearing above, and anachronistically make them definitions. We also extrapolate the latter two in notation onto profinite groups.

Notation 1.1.17 *For a profinite group G, we let An_G, CMon_G and Sp_G denote the equivalent functor categories as in the previous theorems. The value of such functor X at the orbit G/H is called the **categorical H-fixed points** of X, and denoted X^H.*

*We write $\mathrm{Orb}_G \subseteq G\text{-Set}$ for the category of transitive (continuous) G-sets, also called the **G-orbits**.*

Remark 1.1.18. If G weren't profinite but infinite discrete, pullbacks need not exist in $G\text{-Set}^\omega$. In any cases, only pullbacks along maps of *finite fiber* are guaranteed to remain a finite union of orbits. This produces a span category which we call $\mathrm{Span}_{\text{ét}}(G)$.

Alternatively, one can consider the G-sets with finite subgroups as isotropy, which produces $\mathrm{Span}_{\text{proper}}(G)$, related to "proper equivariant algebra".

Note that $G\text{-Set}^\omega \simeq (\mathrm{Orb}_G)^{\amalg}$ is the category obtained from Orb_G by freely adjoining finite coproducts[3], so that $\mathcal{P}_\Sigma(G\text{-Set}^\omega) \simeq \mathrm{PSh}(\mathrm{Orb}_G)$. In fact, we will see that this is a symmetric monoidal equivalence. But we must first introduce the monoidal structures.

1.2. Monoidal structures and suspension spectra

In this section, we discuss the monoidal structure on span categories, and equip An_G and Sp_G with a symmetric monoidal structure, in analogy to the ones on An and Sp. We'll show that the symmetric monoidal structure on An_G is Cartesian and describe the symmetric monoidal left adjoint functor, commonly denoted $\Sigma^\infty_+ : \mathrm{An}_G \to \mathrm{Sp}_G$, which by [CMNN24, Theorem A.1], coincides with the suspension functor in the model of orthogonal spectra.

Construction 1.2.1 *Equip the category $G\text{-Set}^\omega$ with the Cartesian symmetric monoidal structure and write*

$$\iota := (-)_\dagger : G\text{-Set}^\omega \to 2\mathrm{Span}(G)$$

for the forward inclusion. We equip $2\mathrm{Span}(G)$ with a symmetric monoidal structure that makes ι a symmetric monoidal functor.

Note that the Cartesian monoidal structure consists of pullback-preserving functors, i.e.

$$(a \times x) \times_{b\times y} (c \times z) \simeq (a \times_b c) \times (x \times_y z) \quad \textit{etc.}$$

[3] This is the category $\mathcal{C}^{\amalg}$ under $\mathcal{C}$, such that functors $\mathcal{C} \to \mathcal{D}$, where $\mathcal{D}$ admits coproducts, uniquely determine a coproduct preserving functor $\mathcal{C}^{\amalg} \to \mathcal{D}$. By the construction of [HTT, 5.3.6.2], $\mathcal{C}^{\amalg}$ is equivalent to the full subcategory of $\mathrm{PSh}(\mathcal{C})$ spanned by the finite coproducts of representables.

Hence G-Set^{ω} is a commutative algebra in $\mathrm{Cat}^{\mathrm{pb}}$, so that $2\mathrm{Span}(G)$ is a commutative algebra in Cat_2 by Lemma 1.2.2 below.

Equivalently ι is induced by the symmetric monoidal functor $(G\text{-}\mathrm{Set}^{\omega}, \mathrm{iso.}) \to (G\text{-}\mathrm{Set}^{\omega}, \mathrm{all})$ in $\mathrm{Cat}^{\mathrm{pbPair}}$.

Lemma 1.2.2 *The span 2-category construction $2\mathrm{Span} : \mathrm{Cat}^{\mathrm{pbPair}} \to \mathrm{Cat}_2$ preserves filtered colimits, coproducts and products.*

Proof. Filtered colimits and coproducts in $\mathrm{Cat}^{\mathrm{pbPair}}$ are formed underlying. The functor $2\mathrm{Span} : \mathrm{Cat}^{\mathrm{pbPair}} \to \mathrm{Cat}_2$ is defined by mapping out of twisted arrow categories. Since each $\mathrm{TwAr}[n]$ is connected and finite, mapping out of them preserves filtered colimits, disjoint unions and products. □

The monoidal structures of both G-Set^{ω} and $\mathrm{Span}(G)$ preserve coproducts in each variable separately. We equip $\mathcal{P}_{\Sigma}(G\text{-}\mathrm{Set}^{\omega})$ and $\mathrm{Sp}_G = \mathcal{P}_{\Sigma}(\mathrm{Span}(G); \mathrm{Sp})$ with the presentable symmetric monoidal structure of (localised) Day convolution.

Lemma 1.2.3 *The symmetric monoidal category $\mathcal{P}_{\Sigma}(G\text{-}\mathrm{Set}^{\omega})$ and the presheaf ∞-topos $\mathrm{PSh}(\mathrm{Orb}_G)$ with its Cartesian monoidal structure are symmetric monoidally equivalent.*

Henceforth we use An_G for both interchangeable.

Proof. The embedding $y : G\text{-}\mathrm{Set}^{\omega} \simeq (\mathrm{Orb}_G)^{\amalg} \overset{y}{\hookrightarrow} \mathrm{PSh}(\mathrm{Orb}_G)$ is symmetric monoidal and coproduct-preserving by definition of $(-)^{\amalg}$. Thus the symmetric monoidal initiality of Day convolution on $\mathcal{P}_{\Sigma}(-)$ implies that the equivalence $\mathcal{P}_{\Sigma}(G\text{-}\mathrm{Set}^{\omega}) \simeq \mathrm{PSh}(\mathrm{Orb}_G)$ is symmetric monoidal. □

Recall [HA, 4.8.2.11/18] that An_* and Sp are idempotent algebras in Pr^L. The tensor products $\mathcal{C} \otimes \mathrm{An}_*$ and $\mathcal{C} \otimes \mathrm{Sp}$ are the categories $\mathcal{C}_*$ and $\mathrm{Sp}(\mathcal{C})$ of pointed objects And spectrum objects in $\mathcal{C}$ respectively. Let us reiterate that Sp_G is not the stabilisation of An_G.

Notation 1.2.4 *The unreduced and reduced suspension spectrum functors are denoted respectively by*

$$\mathbb{S}[-] : \mathrm{An} \to \mathrm{Sp} \quad \textit{and} \quad \mathbb{S}\langle - \rangle : \mathrm{An}_* \to \mathrm{Sp}.$$

For a group G, we define the category of pointed G-spaces and "Bredon" G-spectra by

$$\mathrm{An}_{G,*} := \mathrm{An}_G \otimes \mathrm{An}_* \quad \textit{and} \quad \mathrm{Sp}_G^{\mathrm{Bredon}} := \mathrm{An}_G \otimes \mathrm{Sp}.$$

*The corresponding **equivariant suspension spectrum** functors are defined as*

$$\mathbb{S}_G[-] := \left(\iota_{\sharp} \otimes \mathbb{S}[-]\right) : \mathrm{An}_G \to \mathrm{Sp}_G \quad \textit{and} \quad \mathbb{S}_G\langle - \rangle : \mathrm{An}_{G,*} \to \mathrm{Sp}_G,$$

*where $(-)_{\sharp}$ denotes the left Kan extension functoriality of $\mathcal{P}_{\Sigma}(-)$, which we call **Mackey induction**.*

There is an induction functor $\mathrm{Sp}_G^{\mathrm{Bredon}} \to \mathrm{Sp}_G$, which we denote by σ^{∞}.

We thus have the following commutative diagram of symmetric monoidal functors:

$$\begin{array}{ccccccc}
G\text{-}\mathrm{Set}^{\omega} & \xrightarrow{y} & \mathrm{An}_G & \xrightarrow{(-)_+} & \mathrm{An}_{G,*} & \longrightarrow & \mathrm{Sp}_G^{\mathrm{Bredon}} \\
\downarrow{\scriptstyle \iota} & & \downarrow{\scriptstyle \iota_{\sharp}} & & \downarrow{\scriptstyle \mathbb{S}_G\langle - \rangle} & & \downarrow{\scriptstyle \sigma^{\infty}} \\
\mathrm{Span}(G) & \xrightarrow{y} & \mathrm{CMon}_G & \xrightarrow{(-)^{\mathrm{grp}}} & \mathrm{Sp}_G & \overset{\mathrm{def.}}{=\!=} & \mathrm{CMon}_G \otimes \mathrm{Sp}
\end{array}$$

where y stands for the respective Yoneda embeddings.

As such, we obtain an An_G-module structure on Sp_G and may speak of $- \otimes G/H$ on Sp_G as an abbreviation for $- \otimes \mathbb{S}_G[G/H]$.

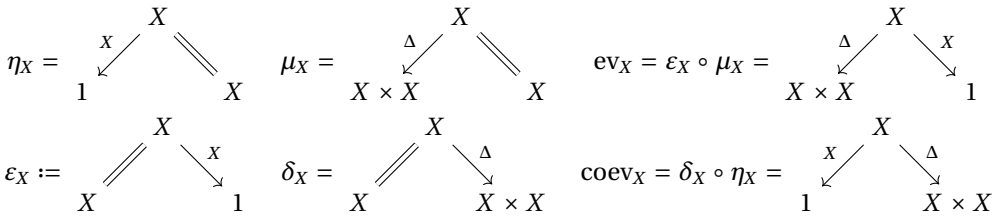
Proof. The sifted cocompletion $\mathcal{P}_\Sigma(-)$ from $\mathrm{Cat}^{\amalg}$ of small categories with coproducts to Pr^L is symmetric monoidal. This follows from a result similar to [HA, 4.8.1.14] with sifted colimits, using [HTT, 5.5.8.15] instead of [HTT, 5.5.1.9]. This provides the left-most square. The remaining parts of the diagram come from proposition 1.1.12. □

Lemma 1.2.5 *The genuine suspension spectra of orbits $\mathbb{S}_G[G/H] \in \mathrm{Sp}_G$ are self-dual. Thus Sp_G is compactly generated by dualisable objects.*

For any $X \in \mathrm{Sp}_G$, we have $X^H \simeq (X \otimes G/H)^G \simeq \left(X \otimes (G/H)^\vee\right)^G$. Moreover, for a map $G/H \xrightarrow{f} G/K$ of G-sets, the restriction and transfer along f is given by

$$f^* = \left(X \otimes f^\dagger\right)^G \quad : \quad \left(X \otimes (G/K)^\vee\right)^G \longrightarrow \left(X \otimes (G/H)^\vee\right)^G,$$
$$f_! = (X \otimes f_\dagger)^G \quad : \quad (X \otimes G/H)^G \longrightarrow (X \otimes G/K)^G.$$

Proof. By monoidality of the Yoneda embedding, it suffices to know the following property of spans in a category with products: for each G-set X, the spans

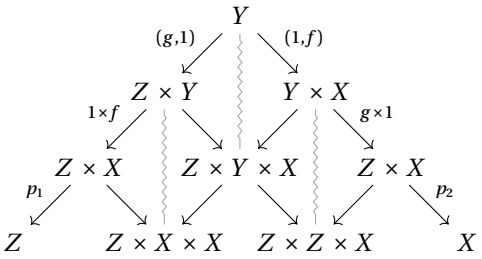


establish $\iota(X)$ as a Frobenius algebra in $\mathrm{Span}(G)$ and in particular self-dual. Thus self-duality of G/H gives:

$$\mathrm{map}_{\mathrm{Sp}_G}(G/G, X \otimes (G/H)^\vee) \simeq \mathrm{map}_{\mathrm{Sp}_G}(G/H \otimes G/G, X) \simeq \mathrm{map}_{\mathrm{Sp}_G}(G/H, X)$$

which proves the claim for f^*.

For $f_!$, we just need that the dual to a span $s = X \xleftarrow{f} Y \xrightarrow{g} Z$ is the mirrored span $Z \xleftarrow{g} Y \xrightarrow{f} X$:



$$\mathrm{id}_Z \times \mathrm{coev}_X \quad \mathrm{id}_Z \times s \times \mathrm{id}_X \quad \mathrm{ev}_Z \times \mathrm{id}_X$$

□

The monoidal unit of Sp_G is called the **genuine G-sphere** and denoted

$$\mathbb{S}_G := \mathbb{S}_G[G/G].$$

Proposition 1.2.6 (equivariant Barratt–Priddy–Quillen) *For any profinite group G and subgroup H of G, the categorical H-fixed points of $\mathbb{S}_G$ are given by*

$$\mathbb{S}_G^H \simeq \mathrm{K}(H\text{-Set}) := (H\text{-Set}^{\omega,\simeq}, \amalg)^{\mathrm{grp}} \simeq \bigoplus_{(K) \leqslant H} \mathbb{S}[\mathbf{B}W_H(K)],$$

where the last sum runs over H-conjugacy classes of open subgroups $K \leqslant H$.

The fixed points admit ring structures corresponding to disjoint unions and products of H-sets.

In particular, when G is finite "underlying" spectrum $\mathbb{S}_G^e$ of the genuine G-sphere is the sphere spectrum $\mathbb{S}$. We will see how to make sense of this for profinite G as well.

Proof. Since $\mathrm{Mack}(G;\mathrm{An}) \simeq \mathrm{Mack}(G;\mathrm{CMon})$ and $(-)^{\mathrm{grp}} : \mathrm{CMon} \to \mathrm{Sp}$ preserves products, we have that

$$\mathrm{map}_{\mathrm{Sp}_G}(\mathbb{S}_G[G/H], \mathbb{S}_G) \overset{\text{Yoneda}}{=\!=} \mathrm{Map}_{\mathrm{Span}(G)}(G/H, *)^{\mathrm{grp}} = (H\text{-}\mathrm{Set}^{\omega,\simeq}, \amalg)^{\mathrm{grp}},$$

where we identify the $\mathbb{E}_\infty$-structure on the mapping space with disjoint union using the description given in Proposition 1.1.9. We can furthermore identify a span $G/H \leftarrow X \to *$ with the H-set that is the fiber of X over the coset eH.

The commutative monoid structure on the mapping space comes from the coalgebra structure on G/H. Computing the pullbacks along the algebra structure maps of Lemma 1.2.5, we see that the commutative ring structure can be described as given by products of H-sets. □

1.3. The tom Dieck splitting

In this section, we prove the tom Dieck splitting for Mackey functors on orbital categories, first obtained by McCandless [McC23, 3.2.11]. In addition to loc. cit., we observe the compatibility of the splitting with transfers maps.

Recall that a category $\mathcal{C}$ is *orbital* if $\mathcal{C}^{\amalg}$ admits all pullbacks. We may thus speak of $\mathcal{C}$-Mackey functors as referring to

$$\mathrm{Mack}(\mathcal{C};\mathcal{V}) := \mathcal{P}_\Sigma(\mathrm{Span}(\mathcal{C}^{\amalg});\mathcal{V}).$$

Theorem 1.3.1 (General tom Dieck splitting, at the final object) *Let $\mathcal{C}$ be an orbital category with a terminal object 1.*

There is a commutative diagram of functors:

$$\begin{array}{ccc} \mathrm{PSh}(\mathcal{C}) & \xrightarrow{\iota} & \mathrm{Mack}(\mathcal{C};\mathrm{Sp}) \\ {\scriptstyle \amalg_{(T)}(\mathrm{ev}_T)_{\mathrm{hAut}(T)}}\downarrow & & \downarrow{\scriptstyle \mathrm{ev}_1} \\ \mathrm{An} & \xrightarrow{\mathbb{S}[-]} & \mathrm{Sp} \end{array}$$

where ev_1 denotes evaluation at the terminal object, the coproduct is taken over the isomorphism classes of objects in $\mathcal{C}$, and $\mathrm{Aut}(T)$ denotes the automorphism group of T in $\mathcal{C}$.

In other words, we have for any $X \in \mathrm{PSh}(\mathcal{C})$:

$$(\iota X)(1) \simeq \operatorname*{colim}_{T \in \mathcal{C}^{\simeq}} X(T) \simeq \bigoplus_{(T)} \mathbb{S}[X(T)]_{\mathrm{hAut}(T)}$$

We can moreover encode the transfer maps between the above splitting in a similar factor-by-factor fashion. The proof of the following will be deferred to a subsection 1.3.1, after a discussion of the special case of Sp_G.

Theorem 1.3.2 (General tom Dieck splitting with transfers, cf. [McC23, Lemma 3.2.11]) *Let $\mathcal{C}$ be an orbital category. Define* $\mathrm{evtr} : \mathcal{C} \times \mathrm{Mack}(\mathcal{C};\mathrm{Sp})$ *as evaluating a Mackey functor along the covariant functoriality (transfers):*

$$\mathrm{evtr} : \mathcal{C} \times \mathrm{Mack}(\mathcal{C};\mathrm{Sp}) \xrightarrow{((-)^\dagger)^{op} \times \mathrm{id}} \mathrm{Span}(\mathcal{C})^{op} \times \mathrm{Mack}(\mathcal{C};\mathrm{Sp}) \xrightarrow{\mathrm{ev}} \mathrm{Sp}.$$

There is a commutative diagram of functors:

$$\begin{array}{ccc} \mathcal{C} \times \mathrm{PSh}(\mathcal{C}) & \xrightarrow{\mathrm{id}_{\mathcal{C}} \times \iota} & \mathcal{C} \times \mathrm{Mack}(\mathcal{C}; \mathrm{Sp}) \\ \downarrow{\scriptstyle \mathrm{utD}} & & \downarrow{\scriptstyle \mathrm{evtr}} \\ \mathrm{An} & \xrightarrow{\mathbb{S}[-]} & \mathrm{Sp} \end{array}$$

where the functor utD *is the "unstable tom Dieck splitting", given on objects by*

$$\begin{array}{rccc} \mathrm{utD}: & \mathcal{C} \times \mathrm{PSh}(\mathcal{C}) & \longrightarrow & \mathrm{An} \\ & (S, X) & \longmapsto & \mathrm{colim}_{T \in (\mathcal{C}_{/S})^{\simeq}} X(T). \end{array}$$

and the functoriality in S is given by composition.

Remark 1.3.3. We record the current version to stress that the functoriality of the splitting is induced factor-by-factor. In an earlier version of the current manuscript, it was used during the proof of Theorem 0.3.

It is trickier to encode the restriction functoriality, which are induced by pullbacks which exist only in $\mathcal{C}^{\amalg}$, not necessarily in $\mathcal{C}$.

Recall our convention that $W_G(H)$ are the automorphism groups of the orbit G/H. We recover the classical splitting of Segal–tom Dieck, together with its compatibility with transfer maps:

Corollary 1.3.4 (Segal–tom Dieck splitting) *Let G be a profinite group and H an open subgroup. For $X \in \mathrm{An}_G$, one has*

$$\mathbb{S}_G[X]^H \simeq \bigoplus_{(K) \leqslant H} \mathbb{S}[X^K]_{\mathrm{h}W_H(K)},$$

where the sum is taken over the H-conjugacy classes of subgroups K of H.

Moreover, the transfer map

$$\mathbb{S}_G[X]^H \to \mathbb{S}[X]^G$$

is given by the sum over all conjugacy classes $(L) \leqslant G$ of maps

$$\bigoplus_{\substack{(K) \leqslant H \\ K \sim_G L}} \mathbb{S}[X^K]_{\mathrm{h}W_H(K)} \to \mathbb{S}[X^L]_{\mathrm{h}W_G(L)},$$

where the first sum is taken over all H-orbit types which are equivalent to G/L after inducing up to G. Moreover, this sum is finite.

Proof. This is the special case of 1.3.2 with $\mathcal{C} = \mathrm{Orb}_G$, in light of the equivalence $\mathrm{Orb}_H \simeq (\mathrm{Orb}_G)_{/(G/H)}$. Here an H-orbit H/K is identified with $G/K \to G/H$ in the slice category, with the inverse identification given by taking fiber over the coset eH.

The finiteness of the sum follows from the fact that L is of finite index in G, so that it has only finitely many conjugate subgroups, some of which are contained in H. □

Remark 1.3.5. The result for the sphere spectrum appeared in [MackI, Prop. B.8]. In [Fau08, Prop. 7.10], such a splitting is constructed using the model of orthogonal G-spectra.

A result similar to the general splitting appeared as [MackI, A.9], although it's also only stated for the sphere spectrum.

Remark 1.3.6. If we were only working with finite groups, it is wise to remove the ambient group G from the picture, and replace Orb_G with the category of finite connected 1-groupoids, also known as the *global orbit category*. Related methods has been studied under the name of *parametrised higher algebra/category theory*, in the work of Nardin, Shah, Cnossen–Lenz–Linskens among others.

1.3.1. Proof of the general tom Dieck splitting

We now prove Theorem 1.3.2, starting with the statement which were assumed in the proof of both [MackI, A.9] and [McC23, Lemma 3.2.11].

Recall that the $\mathcal{C}^{\amalg}$ is the full subcategory of $\mathrm{PSh}(\mathcal{C})$ spanned by disjoint unions of representables functors y_x, $x \in \mathcal{C}$. We can speak of the slice categories $\mathcal{C}_{/S}$ and $\mathcal{C}^{\amalg}_{/S}$ for any $S \in \mathrm{PSh}(\mathcal{C})$.

Lemma 1.3.7 *Let $\mathcal{C}$ be a small category and S an object of $\mathrm{PSh}(\mathcal{C})$, then the core $((\mathcal{C}^{\amalg})_{/S})^{\simeq}$, with the monoid structure given by coproducts, is equivalent to the free $\mathbb{E}_\infty$-monoid generated by $(\mathcal{C}_{/S})^{\simeq}$. This equivalence is compatible with postcompsition in the variable S.*

Proof. The inclusions $\mathcal{C}_{/S} \to (\mathcal{C}^{\amalg})_{/S}$ are functorial with respect to postcomposition in S. Since postcomposition preserves coproducts, there are comparison maps

$$\mathrm{Free}_{\mathrm{CMon}}\left((\mathcal{C}_{/S})^{\simeq}\right) \simeq \coprod_{n\geqslant 0} \left((\mathcal{C}_{/S})^{\simeq}\right)^{\times n}_{h\Sigma_n} \to \left(\mathcal{C}^{\amalg}_{/S}\right)^{\simeq},$$

which are functorially in S. We verify that they are isomorphisms.

As $\mathrm{PSh}(\mathcal{C})_{/S} \simeq \mathrm{PSh}(\mathcal{C}_{/S})$[4], and coproducts are preserved by this equivalence, we may replace $\mathcal{C}$ with $\mathcal{C}_{/S}$.

An object of $\mathcal{C}^{\amalg}$ is of the form $\coprod_{(x)} y_x^{\amalg n_x}$, where the sum is taken over isomorphism classes of objects in $\mathcal{C}$, and $(n_x)_{(x)} \in \mathbb{N}^{\oplus \pi_0(\mathcal{C}^{\simeq})}$. Since maps in $\mathrm{PSh}(\mathcal{C})$ from a representable functor y_x into a coproduct must land in a coproduct factor, we see by Yoneda lemma that an isomorphism in $\mathcal{C}^{\amalg}$ is given by a permutation of each of the n_x many copies of y_x, together with isomorphisms between the factors.

It follows that the fiber of the map of spaces

$$(\mathcal{C}^{\amalg})^{\simeq} \to (*^{\amalg})^{\simeq} \simeq \mathrm{Free}_{\mathrm{CMon}}(*) \simeq \coprod_{n\geqslant 0} \mathbf{B}\Sigma_n$$

over $\mathbf{B}\Sigma_n$ is given by $(\mathcal{C}^{\simeq})^{\times n}$. Thus the map $\mathrm{Free}_{\mathrm{CMon}}\left(\mathcal{C}^{\simeq}\right) \to (\mathcal{C}^{\amalg})^{\simeq}$ of spaces over $(*^{\amalg})^{\simeq}$ is a fiberwise equivalence of spaces, therefore an equivalence. □

Proof of theorem 1.3.2. Note that for each $S \in \mathcal{C}$, the functor

$$\mathrm{PSh}(\mathcal{C}) \to \mathrm{Sp}, \quad X \mapsto (\iota X)(S)$$

is colimit-preserving (proposition 1.1.13), so that it is determined by its restriction along the Yoneda embedding. We observe the following commutative diagram:

$$\begin{array}{ccccc}
\mathcal{C}\times\mathcal{C} & \xrightarrow{\mathrm{id}_{\mathcal{C}}\times y} & \mathcal{C}\times\mathrm{PSh}(\mathcal{C}) & & \\
\downarrow{\scriptstyle \mathrm{id}_{\mathcal{C}}\times\iota} & & \downarrow{\scriptstyle \iota_\sharp} & & \\
\mathcal{C}\times\mathrm{Span}(\mathcal{C}^{\amalg}) & \xrightarrow{\mathrm{id}_{\mathcal{C}}\times y} & \mathcal{C}\times\mathrm{Mack}(\mathcal{C};\mathrm{CMon}) & \xrightarrow{\mathrm{id}_{\mathcal{C}}\times(-)^{\mathrm{grp}}} & \mathcal{C}\times\mathrm{Mack}(\mathcal{C};\mathrm{Sp}) \\
\downarrow{\scriptstyle F} & & \downarrow{\scriptstyle \mathrm{evtr}} & & \downarrow{\scriptstyle \mathrm{evtr}} \\
\mathrm{An} & \xrightarrow{\mathrm{Free}_{\mathrm{CMon}}} & \mathrm{CMon} & \xrightarrow{(-)^{\mathrm{grp}}} & \mathrm{Sp}
\end{array}$$

The upper left square is given by Yoneda embedding, and the lower right square is by group completion of Mackey functors and evaluation along transfers. The upper and right edges of the lower left square thus composes to the functor

$$\mathrm{Map}_{\mathrm{Span}(\mathcal{C}^{\amalg})}((-_1)^{\dagger}, -_2) \simeq \left(\mathcal{C}^{\amalg}_{/\{-_1,-_2\}}\right)^{\simeq} \overset{\text{Lemma 1.3.7}}{=\!=} \mathrm{Free}_{\mathrm{CMon}}\left(\left(\mathcal{C}_{/\{-_1,-_2\}}\right)^{\simeq}\right),$$

[4] by unstraightening, $\mathrm{RFib}(\mathcal{C})_{/\mathrm{El}(S)} \simeq \mathrm{RFib}(\mathrm{El}(S)) \simeq \mathrm{RFib}(C_{/S})$.

where we write $\mathcal{C}_{/\{A,B\}}$ for $\mathcal{C}_{/A\times B}$ as in [MackI, 3.7]. This constructs the factorisation that is the lower left square, where

$$F(S,X) \simeq \left(\mathcal{C}_{/\{S,X\}}\right)^{\simeq}, \text{ with functoriality given by composition.}$$

The map $\mathrm{utD} : \mathcal{C} \times \mathrm{PSh}(\mathcal{C}) \to \mathrm{An}$ is thus left Kan extended from

$$(F \circ \iota)(-_1,-_2) = \mathrm{Map}_{\mathrm{Span}(\mathcal{C}^{\mathrm{ll}})}((-_1)^{\dagger},(-_2)_{\dagger}) \simeq \left(\mathcal{C}_{/\{-_1,-_2\}}\right)^{\simeq}$$

We claim the following, the proof of which we postpone.

$$\left(\mathcal{C}_{/\{S,X\}}\right)^{\simeq} \simeq \operatorname*{colim}_{T\in(\mathcal{C}_{/S})^{\simeq}} \mathrm{Map}_{\mathcal{C}}(T,X) \simeq \coprod_{(T)\leqslant S} \mathrm{Map}_{\mathcal{C}}(T,X)_{\mathrm{hAut}_S(T)}. \tag{!}$$

By Yoneda lemma, the unique colimit-preserving extension of $\mathrm{Map}_{\mathcal{C}}(T,-)$ is evaluation at T. Since homotopy orbits and coproducts in An are colimit-preserving, the Yoneda extension of $F \circ \iota$ is therefore given by

$$\mathrm{utD} : \mathcal{C} \times \mathrm{PSh}(\mathcal{C}) \longrightarrow \mathrm{An}, \qquad (S,X) \longmapsto \operatorname*{colim}_{T\in(\mathcal{C}_{/S})^{\simeq}} X(T).$$

This, composed with the colimit-preserving functor $\mathbb{S}[-] : \mathrm{An} \to \mathrm{Sp}$, provides the desired factorisation.

Proof of the claim (!)**:** Consider the following pullback square of anima.

$$\begin{array}{ccc} \left(\mathcal{C}_{/\{S,X\}}\right)^{\simeq} & \longrightarrow & \left(\mathcal{C}_{/\{X\}}\right)^{\simeq} \\ \downarrow & \ulcorner & \downarrow \\ \left(\mathcal{C}_{/\{S\}}\right)^{\simeq} & \longrightarrow & \mathcal{C}^{\simeq} \end{array}$$

The right hand map has fibers over $T \in \mathcal{C}$ given by $\mathrm{Map}_{\mathcal{C}}(T,-)$, hence so are the fibers of the left hand map. By the straightening equivalence, the upper left space is the colimit of the fibers over the lower left space, hence the first equivalence.

The second equivalence comes from first computing the colimit over each connected component (T) of $\mathcal{C}^{\simeq}_{/S}$ and taking their coproduct, the former being a homotopy quotient by the automorphisms of T in $\mathcal{C}_{/S}$. □

1.4. Borel and free G-spectra, homotopy fixed points

For the current section, the group G is taken to be finite, as notion of free and Borel G-spectra requires the presence of the free orbit G/e. This section is meant to highlight the special relation between Borel and naive G-spectra (Proposition 1.4.5), and also serves as a precursor to isotropy separation.

Recall our convention that the automorphism group of the transitive free G-set is G^{op}, so that $\mathbf{B}G^{\mathrm{op}}$ naturally embed into Orb_G as the full subcategory on G/e.

Definition 1.4.1 *Let $\mathcal{C}$ be a category. We call the functor category*

$$\mathcal{C}^{\mathbf{B}G} := \mathrm{Fun}(\mathbf{B}G, \mathcal{C})$$

*the category of $\mathcal{C}$-objects **with G-action**. When $\mathcal{C} = \mathrm{Sp}$, we speak of **naive** G-spectra.*

Every $\mathcal{C}$-valued presheaf on Orb_G and, if $\mathcal{C}$ admits products, every $\mathcal{C}$-valued Mackey functor induces a $\mathcal{C}$-object with G-action by restriction along

$$\mathbf{B}G \xrightarrow{\mathrm{free}^{op}} \mathrm{Orb}_G^{op} \subseteq G\text{-}\mathrm{Set}^{\omega,op} \xrightarrow{((-)_{\dagger})^{op}} \mathrm{Span}(G)^{op},$$

*which we call the **underlying** object with G-action.*

Proposition 1.4.2 *For presentable $\mathcal{C}$, restriction $\mathcal{C}^{\mathbf{B}G} \leftarrow \mathrm{PSh}(\mathrm{Orb}_G; \mathcal{C})$ is both a left and right Bousfield localisation.*

$$\mathcal{C}^{\mathbf{B}G} \underset{\text{free}_*}{\overset{\text{free}_\sharp}{\rightleftarrows}} \mathrm{PSh}(\mathrm{Orb}_G; \mathcal{C}) : \text{free}^*$$

For $\mathcal{C} = \mathrm{An}$, the right hand side is An_G, and the above localisations are induced by the idempotent coalgebra

$$\mathbf{E}G := \left(G/H \mapsto \begin{cases} \emptyset & H \neq e \\ \mathrm{pt} & H = e \end{cases}\right) \simeq (G/e)_{\mathrm{h}G^{op}} \in \mathrm{An}_G$$

in the sense that

$$\begin{aligned} \text{free}_\sharp \text{free}^*(-) &\simeq (\mathbf{E}G \times -), \\ \text{free}_* \text{free}^*(-) &\simeq \underline{\mathrm{map}}_{\mathrm{An}_G}(\mathbf{E}G, -). \end{aligned}$$

Proof. Fully faithfulness of either Kan extension follows from $\mathbf{B}G^{\mathrm{op}}$ being a full subcategory of Orb_G. In the case of An_G, note the inclusion $\mathbf{B}G^{\mathrm{op}} \subseteq \mathrm{Orb}_G$ is a right fibration, since every map in Orb_G into G/e is an isomorphism. As the fibers of the inclusion are contractible over the free orbit and empty otherwise, it is the unstraightening of the (-1)-truncated G-space $\mathbf{E}G \in \mathrm{An}_G$ given above. For the equivalence $\mathbf{E}G \simeq (G/e)$, see Lemma 1.5.13. Moreover $\mathrm{An}^{\mathbf{B}G} \simeq \mathrm{RFib}(\mathbf{B}G^{\mathrm{op}})$ identifies with the open subtopos $(\mathrm{An}_G)_{/\mathbf{E}G}$. □

Definition 1.4.3 *In the case of An_G, we call the essential image of $\text{free}_\sharp$ and free_*) the **free** and **Borel** G-spaces respectively.*

*The fixed points of a Borel G-space are called the **homotopy fixed points** of its underlying G-space (with respect to the various subgroups) and denoted $X^{\mathrm{h}H}$.*

It's easy to see that free G-spaces have empty fixed points at non-trivial subgroups.

Under the symmetric monoidal functor $\mathbb{S}_G[-]$, we obtain an idempotent coalgebra $\mathbb{S}_G[\mathbf{E}G]$ and a recollement. See Appendix A for a review of these concepts.

$$\mathrm{Sp}_G^{\text{free}} \underset{\underline{\mathrm{map}}(\mathbb{S}_G[\mathbf{E}G], -)}{\overset{-\otimes \mathbb{S}_G[\mathbf{E}G]}{\rightleftarrows}} \mathrm{Sp}_G \underset{\underline{\mathrm{map}}(\mathbb{S}_G\langle \widetilde{\mathbf{E}G}\rangle, -)}{\overset{-\otimes \mathbb{S}_G\langle \widetilde{\mathbf{E}G}\rangle}{\rightleftarrows}} \mathrm{Sp}_G^{\text{ac}}$$

Definition 1.4.4 (Free and Borel G-spectra) *The essential image of the upper and lower left functor are respectively called the **free** and **Borel** G-spectra. For $X \in \mathrm{Sp}_G$, we call $X_{\mathrm{Borel}} := \underline{\mathrm{map}}(\mathbf{E}G, X)$ the* Borel-isation *of X.*

Analogously to G-spaces, the category of free G-spectra is equivalent to spectra with G-action.

Proposition 1.4.5 *The functor $\sigma^\infty : \mathrm{Sp}_G^{\mathrm{Bredon}} \to \mathrm{Sp}_G$ restricts to an equivalence between $\mathbf{E}G$-comodules, which is furthermore equivalent to $\mathrm{Sp}^{\mathbf{B}G}$.*

Thus the categories of free and Borel G-spectra are both equivalent to naive G-spectra.

Proof. As σ^∞ is symmetric monoidal, it restricts to the categories of $\mathbf{E}G$-comodules, which for $\mathrm{Sp}_G^{\mathrm{Bredon}}$ is equivalent to $\mathrm{Sp}^{\mathbf{B}G}$ by copying the proof for spaces (1.4.2).

The free orbit G/e is a compact generator of $\mathbf{E}G$-comodules in both Sp_G and $\mathrm{Sp}_G^{\mathrm{Bredon}}$. Indeed, in either category, orbits are compact and form a set of generators. Since $\mathbf{E}G \simeq (G/e)_{\mathrm{h}G^{\mathrm{op}}}$, any comodule $X \otimes \mathbf{E}G$ belongs to the localising subcategory generated by all $G/H \otimes G/e$, each of which is a sum of copies of G/e.

By the Schwede–Shipley theorem (case $k = 1$ of [HA, 7.1.2.6]), it suffices to see that σ^∞ induces an equivalence between the endomorphism spectra of the Bredon and genuine suspension of G/e.

The endomorphism ($\mathbb{E}_1$-)space of G/e in An_G is the group G^{op}, and maps by applying $(-)_\dagger$ to the endomorphism object in CMon_G. On the other hand, every span from G/e to itself is a sum of spans whose middle term is a free orbit, and whose two legs are automorphisms. As such, the space $\mathrm{End}_{\mathrm{Span}(G)}(G/e)$ is the free $\mathbb{E}_\infty$-monoid generated by G^{op}, under the natural map $(-)_\dagger : \mathrm{Orb}_G \to \mathrm{Span}(G)$. Thus after tensoring with Sp, which on CMon_G acts as group completion, both endomorphism spectra become $\mathbb{S}[G^{\mathrm{op}}]$. □

Remark 1.4.6. The right adjoint $\omega^\infty : \mathrm{Sp}_G \to \mathrm{Sp}_G^{\mathrm{Bredon}}$, taking a genuine G-spectrum to its underlying Bredon G-spectrum, induces an equivalence between the respective subcategories of Borel objects.

$$\begin{aligned}\mathrm{Map}_{\mathrm{Bredon}}(-, \omega^\infty(X_{\mathrm{Borel}})) &\simeq \mathrm{Map}_{\mathrm{Sp}_G}(\sigma^\infty(-) \otimes \mathbf{E}G, X)\\ &\simeq \mathrm{Map}_{\mathrm{Sp}_G}(\sigma^\infty(- \otimes \mathbf{E}G), X)\\ &\simeq \mathrm{Map}_{\mathrm{Bredon}}(-, \omega^\infty(X)_{\mathrm{Borel}})\end{aligned}$$

It's easy to see that free Bredon G-spectra takes zero value on the non-free orbits. Meanwhile, the categorical fixed points of free G-spectra are, somewhat surprisingly, given by the homotopy **orbits**. In particular, the underlying Bredon spectrum of a genuine free G-spectrum is *not* Bredon-free.

Theorem 1.4.7 (Adams isomorphism) *For $X \in \mathrm{Sp}_G^{\mathrm{free}}$ and $Y \in \mathrm{Sp}_G^{\mathrm{Borel}}$, we have*

$$X^G \simeq (X^e)_{\mathrm{h}G^{\mathrm{op}}}, \quad Y^G \simeq (Y^e)^{\mathrm{h}G}.$$

We omit the proof here, and refer instead to the more general version, Theorem 1.5.5. In the literature it is customary to conflate G with G^{op}.

Once we learn about induction and restriction, we can deduce immediately that $(X \otimes \mathbf{E}G)^H \simeq (X^e)_{\mathrm{h}H^{\mathrm{op}}}$ for all subgroups $H \leqslant G$.

Remark 1.4.8. In case X is a suspension spectrum of $T \in \mathrm{An}_G$, the above is a special case of the Segal–tom Dieck splitting 1.3.4.

1.5. Isotropy separation

For finite groups we just saw that naive G-spectra are comodules over the idempotent coalgebra $\mathbf{E}G$ in both An_G and Sp_G. The idempotent coalgebras in An_G are central to equivariant homotopy theory: They are exactly the universal G-spaces $\mathbf{E}\mathcal{F}$ associated to families $\mathcal{F}$ of subgroups of G.

In this section, the profinite group G no longer needs to be finite.

Lemma 1.5.1 *The poset of idempotent coalgebras of An_G is equivalent to the poset of **families of subgroups** of G, by which we mean, subsets $\mathcal{F}$ of the set of* open *subgroups which are closed under subconjugation.*

Note that every open subgroup in a profinite group has only finitely many conjugates.

Proof. By Lemma 1.2.3, the monoidal structure of An_G is Cartesian. Thus every object is canonically a cocommutative coalgebra, and idempotence amounts to the diagonal $X \to X \times X$ being an equivalence, i.e. X being (-1)-truncated. This amounts to the functor $X : \mathrm{Orb}_G^{\mathrm{op}} \to \mathrm{An}$ taking values in the poset $\{\varnothing \leqslant *\}$ of truncated anima. It's elementary to verify that such functors correspond to downward closed subsets of (the (-1)-truncation of) Orb_G, which is equivalent to the lattice of such families of subgroups. □

Definition 1.5.2 *For a family of subgroups $\mathcal{F}$ of G, we let $\mathbf{E}\mathcal{F}$ denote the corresponding idempotent coalgebra in An_G, and $\mathrm{Orb}_{\mathcal{F}} \subseteq \mathrm{Orb}_G$ the full subcategory of orbits with stabiliser in $\mathcal{F}$, or equivalently the unstraightening of $\mathbf{E}\mathcal{F}$.*

By the obvious symmetric monoidal functors, we obtain idempotent coalgebras $\mathbf{E}\mathcal{F}_+$ and $\mathbb{S}_G[\mathbf{E}\mathcal{F}]$ in $\mathrm{An}_{G,*}$ and Sp_G respectively, along with a complementary idempotent *algebra* and on Sp_G a recollement, known as *isotropy separation* for the family $\mathcal{F}$.

Definition 1.5.3 (Isotropy separation) *We define $\widetilde{\mathbf{E}\mathcal{F}} \in \mathrm{An}_{G,*}$ as the idempotent algebra complementary to the idempotent coalgebra $\mathbf{E}\mathcal{F}_+$, i.e. $\widetilde{\mathbf{E}\mathcal{F}} := \mathrm{cofib}(\mathbf{E}\mathcal{F}_+ \xrightarrow{\varepsilon} *_+)$.*

For a family $\mathcal{F}$ of subgroups of G, the recollement on Sp_G induced by $\mathbb{S}_G[\mathbf{E}\mathcal{F}]$ is depicted as follows:

$$\mathrm{Sp}_G^{\mathcal{F}\text{-free}} \underset{\underline{\mathrm{map}}(\mathbf{E}\mathcal{F}_+,-)}{\overset{-\otimes\mathbf{E}\mathcal{F}_+}{\rightleftarrows}} \mathrm{Sp}_G \underset{\underline{\mathrm{map}}(\widetilde{\mathbf{E}\mathcal{F}},-)}{\overset{-\otimes\widetilde{\mathbf{E}\mathcal{F}}}{\rightleftarrows}} \mathrm{Sp}_G[\mathcal{F}^{-1}]$$

*We call the three full subcategories $\mathcal{F}$-**free**, $\mathcal{F}$-**Borel** and $\mathcal{F}$-**acyclic** G-spectra.*

The functors marked with "•" are those whose kernel is a localising $\otimes$-ideal, whence both $\mathrm{Sp}_G^{\mathcal{F}\text{-Borel}}$ and $\mathrm{Sp}_G[\mathcal{F}^{-1}]$ are symmetric monoidal localisations of Sp_G.

The following is an immediate corollary of tom Dieck splitting 1.3.2.

Proposition 1.5.4 *The G-fixed points of the above cofiber sequence in Sp_G associated to a family $\mathcal{F}$ are*

$$\begin{array}{ccccc}
\mathbb{S}_G[\mathbf{E}\mathcal{F}]^G & \longrightarrow & (\mathbb{S}_G)^G & \longrightarrow & \mathbb{S}_G\langle\widetilde{\mathbf{E}\mathcal{F}}\rangle^G \\
\wr & & \wr & & \wr \\
\bigoplus_{(H)\in\mathcal{F}} \mathbb{S}[\mathbf{B}W_G(H)] & \longrightarrow & \bigoplus_{(H)} \mathbb{S}[\mathbf{B}W_G(H)] & \longrightarrow & \bigoplus_{(H)\notin\mathcal{F}} \mathbb{S}[\mathbf{B}W_G(H)]
\end{array}$$

The maps are given by the inclusion of, as well as projection onto the corresponding tom Dieck summands. □

We obtain the "Adams isomorphisms" of $\mathcal{F}$-free and $\mathcal{F}$-Borel spectra:

Theorem 1.5.5 ($\mathcal{F}$-Adams isomorphism) *For $X \in \mathrm{Sp}_G^{\mathcal{F}\text{-free}}$ and $Y \in \mathrm{Sp}_G^{\mathcal{F}\text{-Borel}}$, there holds*

$$X^G \simeq \operatorname*{colim}_{G/H\to\mathbf{E}\mathcal{F}} (X \otimes G/H)^G \simeq \operatorname*{colim}_{G/H\to\mathbf{E}\mathcal{F}} X^H, \qquad Y^G \simeq \lim_{G/H\to\mathbf{E}\mathcal{F}} \left(Y \otimes (G/H)^\vee\right)^G \simeq \lim_{G/H\to\mathbf{E}\mathcal{F}} Y^H,$$

where the colimit is taken over $\mathrm{Orb}_{\mathcal{F}}$ along transfer, and the limit over $\mathrm{Orb}_{\mathcal{F}}^{op}$ along restriction.

Proof. By assumption, there are equivalences $X \simeq X \otimes \mathbf{E}\mathcal{F}$ and $Y \simeq \underline{\mathrm{map}}(\mathbf{E}\mathcal{F}, Y)$. Yoneda lemma provides an equivalence $\mathbf{E}\mathcal{F} \simeq \mathrm{colim}_{G/H\in\mathrm{Orb}_{\mathcal{F}}} G/H$. The claim thus follows from the description of restriction and transfers as in Lemma 1.2.5, and that $(-)^G$ commutes with limits (Proposition 1.1.13). □

Example 1.5.6 (Isotropy separation and Davis–Lück assembly) For each $X \in \mathrm{Sp}_G$, the upper part of the recollement provides a cofiber sequence

$$X \otimes \mathbb{S}_G[\mathbf{E}\mathcal{F}] \longrightarrow X \longrightarrow X \otimes \mathbb{S}_G\langle\widetilde{\mathbf{E}\mathcal{F}}\rangle.$$

The left map $X \otimes \mathbb{S}_G[\mathbf{E}\mathcal{F}] \longrightarrow X$ behaves on the categorical G-fixed points as:

$$\operatorname*{colim}_{G/H \to \mathbf{E}\mathcal{F}} X^H \to X^G.$$

In the context of K-theory of group rings, this map is known as the *Davis–Lück assembly map* for the family $\mathcal{F}$. Naturally, the categorical fixed points of $X \otimes \mathbb{S}_G\langle\widetilde{\mathbf{E}\mathcal{F}}\rangle$ are called the cofibers of assembly.

Analogously there is a *coassembly* from the lower part of the recollement.

Remark 1.5.7 (The trivial family). When G is finite, consider the family $\mathbf{Triv} = \{e \leqslant G\}$. We have that $\mathbf{ETriv} \simeq \mathbf{E}G$ and $\mathrm{Orb}_{\mathbf{Triv}} \simeq \mathbf{B}G^{\mathrm{op}}$, from which we obtain the Adams isomorphism 1.4.7.

Proposition 1.5.8 *For a finite group G and a Borel G-spectrum X, the assembly map for the family* **Triv** *is given on H-fixed points by by the semiadditive H-norm:*

$$X_{\mathrm{h}H} \xrightarrow{\mathrm{Nm}} X^{\mathrm{h}H}$$

Consequently, we have $\left(X \otimes \widetilde{\mathbf{E}G}\right)^H \simeq X^{\mathrm{t}H}$, the H-Tate construction.

Proof following [NS18, footnote during II.2.13]. By [NS18, I.4.1 (iii)], the transformation $(-)^{\mathrm{h}G} \to (-)^{\mathrm{t}G}$, as a map between functors $\mathrm{Sp}^{\mathbf{B}G} \to \mathrm{Sp}$, is characterised by its fiber being cocontinuous, together with $(-)^{\mathrm{t}G}$ vanishing on compact objects.

In the isotropy separation sequence

$$X \otimes \mathbf{E}G \longrightarrow X \longrightarrow X \otimes \widetilde{\mathbf{E}G},$$

the fiber is clearly cocontinuous in X, and thus also upon taking categorical H-fixed points. It suffices to check that $X \otimes \widetilde{\mathbf{E}G}$ vanishes on the compact generator of $\mathrm{Sp}_G^{\mathrm{Borel}}$, being $X = \underline{\mathrm{map}}(\mathbf{E}G, \mathbb{S}_G[G/e])$. It follows from dualisability of $\mathbb{S}_G[G/e]$ that

$$\underline{\mathrm{map}}(\mathbf{E}G, \mathbb{S}_G[G/e]) \otimes \widetilde{\mathbf{E}G} \simeq \underline{\mathrm{map}}(\mathbb{S}_G[\mathbf{E}G], \mathbb{S}_G) \otimes \underbrace{\mathbb{S}_G[G/e] \otimes \widetilde{\mathbf{E}G}}_{=0} = 0 \qquad \square$$

Proposition 1.5.9 *With the above notations, the category of $\mathcal{F}$-free G-spectra is the localising subcategory of Sp_G generated by (desuspensions of) the orbits $\{G/H\}_{H \in \mathcal{F}}$, and is a $\otimes$-ideal of Sp_G.*

Consequently, the category of $\mathcal{F}$-acyclic spectra consists exactly of those G-spectra whose categorical H-fixed point spectra are zero for all $H \in \mathcal{F}$.

Proof. Since the category of G-spectra with trivial H-fixed points for $H \in \mathcal{F}$ is the right orthogonal complement to the localising subcategory $\mathcal{C} \subseteq \mathrm{Sp}_G$ generated by $\mathcal{F}$-orbits, it suffices to show the first claim.

Each $\mathcal{F}$-orbit G/H is an $\mathbf{E}\mathcal{F}$-comodule in An_G, whence also in Sp_G by monoidality. Thus $\mathcal{C}$ is contained in $\mathrm{Sp}_G^{\mathcal{F}\text{-free}}$, as the latter is closed under colimits. For the opposite inclusion, we must show that $X \otimes \mathbf{E}\mathcal{F} \in \mathcal{C}$ for every $X \in \mathrm{Sp}_G$. Since $\mathbf{E}\mathcal{F}$ is a colimit of $\mathcal{F}$-orbits, it suffices to show that

$$\mathbb{S}_G[G/H] \otimes \mathbb{S}_G[G/K] \in \mathcal{C}$$

for $K \in \mathcal{F}$ and any subgroup $H \leqslant G$. This follows from the formula

$$G/H \times G/K \simeq \coprod_{HgK \in H\backslash G/K} G/(H \cap gKg^{-1}),$$

where each $H \cap gKg^{-1}$ belongs to $\mathcal{F}$. Thus $\mathcal{C} = \mathrm{Sp}_G^{\mathcal{F}\text{-free}}$, and is clearly a $\otimes$-ideal. $\square$

1.5.1. The $\mathcal{F}$-Adams isomorphism and the tom Dieck splitting

For $X \in \mathrm{An}_G$, consider the transfer map of categorical fixed points

$$\mathbb{S}_G[X]^H \simeq (\mathbb{S}_G[X] \otimes G/H)^G \xrightarrow{\mathrm{tr}} \mathbb{S}_G[X]^G.$$

By Segal–tom Dieck splitting 1.3.4, this is a map of type

$$\bigoplus_{(K)\leqslant H} \mathbb{S}[X^K]_{\mathrm{h}W_H(K)} \to \bigoplus_{(L)\leqslant G} \mathbb{S}[X^L]_{\mathrm{h}W_G(L)},$$

which sends the factor of each H-conjugacy class to its corresponding G-conjugacy class.

Proposition 1.5.10 *Let $X \in \mathrm{An}_G$ and $\mathcal{F}$ be a family of open subgroups of G. The Adams isomorphism, applied to $\mathbb{S}_G[X] \otimes \mathbf{E}\mathcal{F} \simeq \mathbb{S}_G[X \times \mathbf{E}\mathcal{F}]$,*

$$\operatorname*{colim}_{G/H \to \mathbf{E}\mathcal{F}} \bigoplus_{(K)\leqslant H} \mathbb{S}[X^K]_{\mathrm{h}W_H(K)} \xrightarrow{\simeq} \bigoplus_{(L)\in\mathcal{F}} \mathbb{S}[X^L]_{\mathrm{h}W_G(L)}$$

decomposes as the direct sum of the equivalences

$$\operatorname*{colim}_{G/H \to \mathbf{E}\mathcal{F}} \bigoplus_{(K)\leqslant H, K\sim_G L} \mathbb{S}[X^K]_{\mathrm{h}W_H(K)} \xrightarrow{\simeq} \mathbb{S}[X^L]_{\mathrm{h}W_G(L)},$$

where $K \sim_G L$ denotes that the subgroup K of H becomes conjugate to L when regarded as a subgroup of G.

Remark 1.5.11. For any finite index subgroup $L \leqslant G$, the number of subgroup of G conjugate to L is finite. In particular the direct sum on the left hand side is finite.

Proof. By the self-duality of $\mathbb{S}_G[G/H]$, the colimit involved in the Adams isomorphism lifts through evtr of 1.3.2, and for suspension spectra, through utD. As utD decomposes as a disjoint union over G-conjugacy classes, so does the Adams isomorphism. In particular, each direct sum factor is an isomorphism. □

1.5.2. Two families associated to a subgroup

To subgroups $H \leqslant G$, we can associate two families of subgroups.

Definition 1.5.12 ($\mathcal{F}_H$ and $\mathcal{P}_H$) *For a* open *subgroup $H \leqslant G$, define the family of subconjugates by*

$$\mathcal{F}_{H\leqslant G} := \{K \leqslant G : \exists g \in G, gKg^{-1} \subseteq H\}.$$

For a closed *subgroup $H \leqslant G$, define the family of non-superconjugates by*

$$\mathcal{P}_{H\leqslant G} := \{K \leqslant G : \nexists g \in G, gKg^{-1} \supseteq H\}.$$

When the group G is clear from context, we write $\mathcal{F}_H$ and $\mathcal{P}_H$ respectively.

In other words $\mathrm{Orb}_{\mathcal{F}_H}$ consists of those orbits which *admits a map to G/H*, while $\mathrm{Orb}_{\mathcal{P}_H}$ consists of those orbits which *doesn't admit a map from G/H*.

Lemma 1.5.13 *If the subgroup H is normal, then in the category of G-spaces, we have an equivalence*

$$\mathbf{E}\mathcal{F}_H \simeq (G/H)_{\mathrm{h}(G/H)^{op}},$$

where the Weyl group $W_G(H) \simeq (G/H)^{op}$ acts on the orbit.

Proof. For a normal subgroup $H \trianglelefteq G$ and any $K \leqslant G$, the K-fixed points of G/H consists of either all or none of the cosets. We can thus compute the above colimit objectwise:

$$((G/H)_{\mathrm{h}(G/H)^{\mathrm{op}}})^K \simeq ((G/H)^K)_{\mathrm{h}(G/H)^{\mathrm{op}}} \simeq \begin{cases} \emptyset_{\mathrm{h}(G/H)^{\mathrm{op}}} \simeq \emptyset & K \not\subseteq H \\ (G/H)_{\mathrm{h}(G/H)^{\mathrm{op}}} \simeq * & K \subseteq H \end{cases}.$$

Hence the projection from $(G/H)_{\mathrm{h}(G/H)^{\mathrm{op}}} \times \mathbf{E}\mathcal{F}_H$ to either factor is an equivalence. □

Remark 1.5.14. We can rephrase the above as follows:

$$\text{The full subcategory of } \mathrm{Orb}_{\mathcal{F}_H} \text{ spanned by } G/H \text{ is cofinal.}$$

By Quillen's Theorem A [HTT, 4.1.3.1], this reduces to the condition that for each $K \in \mathcal{F}_H$, the fixed point set $(G/H)^K$ is a free transitive $W_G(H)$-set. In particular, for $K = e$ the trivial group, this forces H to be normal.

The union of two families of subgroups is still a family, and every family is a union of $\mathcal{F}_H$'s. In the case of normal subgroups, we can describe the classifying space of a finite union. For instance, the binary case is:

Corollary 1.5.15 *Let $N, K \trianglelefteq G$ be normal subgroups, then*

$$\mathbf{E}(\mathcal{F}_N \cup \mathcal{F}_K) \simeq (G/N)_{\mathrm{h}(G/N)^{op}} \underset{(G/N\cap K)_{\mathrm{h}(G/(N\cap K))^{op}}}{\cup} (G/K)_{\mathrm{h}(G/K)^{op}}.$$

Proof. Combine the previous description with Lemma A.11. Note that $\mathbf{E}\mathcal{F}_N \times \mathbf{E}\mathcal{F}_K \simeq \mathbf{E}\mathcal{F}_{N\cap K}$ if either N or K is normal. □

Example 1.5.16 Consider the profinite abelian group $A = \mathbb{Z}/p^a q^b \mathbb{Z}$ for two distinct primes p, q and exponents $1 \leqslant a, b \leqslant \infty$, where $a = \infty$ is to be interpreted as a factor $\mathbb{Z}_p$. The family $\mathcal{P}_A$ of proper subgroups is the union of $\mathcal{F}_{pA}$ and $\mathcal{F}_{qA}$, thus

$$\mathbf{E}\mathcal{P}_A \simeq (\mathbb{Z}/p\mathbb{Z})_{\mathrm{h}\mathbb{Z}/p\mathbb{Z}} \underset{(\mathbb{Z}/pq\mathbb{Z})_{\mathrm{h}\mathbb{Z}/pq\mathbb{Z}}}{\cup} (\mathbb{Z}/q\mathbb{Z})_{\mathrm{h}\mathbb{Z}/q\mathbb{Z}}.$$

For the profinite integers $\widehat{\mathbb{Z}}$, one has

$$\mathbf{E}\mathcal{P}_{\widehat{\mathbb{Z}}} \simeq \operatorname*{colim}_{n\in(\mathbb{Z}_{\geqslant 1},|)^{\mathrm{op}}} (\mathbb{Z}/n\mathbb{Z})_{\mathrm{h}\mathbb{Z}/n\mathbb{Z}},$$

where the colimit is taken over the division poset of positive integers.

1.6. Induction and restriction

For this section, consider an open subgroup H of G. Between the categories H-Set and G-Set, we have a sequence of adjunctions

$$H\text{-Set} \underset{\mathrm{coind}_H^G}{\overset{\mathrm{ind}_H^G}{\underset{\longleftarrow}{\overset{\longrightarrow}{\bot}}{\bot}}} G\text{-Set} : \ \mathrm{res}_H^G$$

On under lying sets, the functors act as

$$\mathrm{ind}_H^G(S) \simeq G \times_H S, \mathrm{res}_H^G(S) \simeq S, \mathrm{coind}_H^G(S) \simeq S^{\times(G/H)}.$$

Since $H \leqslant G$ is open, and thus of finite index, the three functors above restrict to G-Set$^\omega$. Note that ind_H^G and res_H^G preserve coproducts and pullbacks, so that they induce a pair of adjoint functors between An_G and An_H through left Kan extension:

$$\mathrm{ind}_H^G \coloneqq (\mathrm{ind}_H^G)_\sharp : \mathrm{An}_H \rightleftarrows \mathrm{An}_G : (\mathrm{res}_H^G)_\sharp =: \mathrm{res}_H^G,$$

with further adjoints to the right.

We also have induced functors on the level of spans and genuine spectra.

Definition 1.6.1 *For an open subgroup H of a profinite group G, define **induction** and **restriction** between Sp_G and Sp_G by Mackey induction:*

$$\mathrm{ind}_H^G \coloneqq \mathrm{Span}(\mathrm{ind}_H^G)_\sharp : \mathrm{Sp}_H \longrightarrow \mathrm{Sp}_G,$$
$$\mathrm{res}_H^G \coloneqq \mathrm{Span}(\mathrm{res}_H^G)_\sharp : \mathrm{Sp}_G \longrightarrow \mathrm{Sp}_H.$$

Similarly we define induction and restriction between An_G and An_H.

As res_H^G between G- and H-sets is Cartesian monoidal, the functors induced between equivariant spaces or spectra along the Mackey induction functoriality, all of which are denoted "res_H^G" above, are symmetric monoidal.

The next step is to lift, from finite sets to the level of span categories, the unit-counit pair of the adjunction:

$$\mathrm{id}_{H\text{-Set}^\omega} \Rightarrow \mathrm{res}_H^G \circ \mathrm{ind}_H^G \quad \mathrm{ind}_H^G \circ \mathrm{res}_H^G \Rightarrow \mathrm{id}_{G\text{-Set}^\omega}.$$

Naturally, we want to study the following question.

When does $\mathcal{C} \overset{f}{\underset{g}{\Downarrow}} \mathcal{D}$ induce $\mathrm{Span}(\mathcal{C}) \overset{\mathrm{Span}(f)}{\underset{\mathrm{Span}(g)}{\Downarrow}} \mathrm{Span}(\mathcal{D})$?

Equivalently, the datum on the left can be encoded as one functor $H : \mathcal{C} \times I \to \mathcal{D}$, similarly for the right. We note that $\mathrm{Span}(\mathcal{C}) \times I$ is a span category with a restricted class of backward morphisms,

Lemma 1.6.2 *The projections $\mathcal{C} \leftarrow \mathcal{C} \times I \to I$ induces an equivalence $\mathrm{Span}_{E\times iso}(\mathcal{C} \times I) \simeq \mathrm{Span}_E(\mathcal{C}) \times I$.*

Moreover, a functor $H : \mathcal{C} \times I \to \mathcal{D}$, encoding a natural transformation $\tau : H_0 \Rightarrow H_1$, preserves pullbacks along $E \times \mathrm{iso}$ iff both H_0 and H_1 preserve pullbacks along E, and τ is Cartesian along E, in the sense that for every map $x \to y$ in E, the following square in $\mathcal{D}$ is Cartesian.

$$\begin{array}{ccc} H_0(x) & \xrightarrow{E} & H_0(y) \\ {\scriptstyle \tau_x}\downarrow & \ulcorner & \downarrow{\scriptstyle \tau_y} \\ H_1(x) & \xrightarrow{E} & H_1(y) \end{array}$$

Proof. By Lemma 1.2.2, taking span categories preserves products of pullback pairs. Note also that for any category $\mathcal{D}$, there holds $\mathrm{Span}_{\mathrm{iso}}(\mathcal{D}) \simeq \mathcal{D}$.

Any morphism in $\mathcal{C}\times I$ is either a map in $\mathcal{C}\times\{i\}$, where $i = 0, 1$, or such a map composed with $(\mathrm{id}_x, 0 < 1)$. Thus H preserves pullbacks along $E \times$ iso iff pullback squares involving one morphism of the three types above are preserved. The case of the first two types amounts to H_0 and H_1 preserving pullbacks along E, and the remaining case amounts to H preserving the following pullback.

$$\begin{array}{ccc} (x,0) & \xrightarrow{E} & (y,0) \\ {\scriptstyle (\mathrm{id}_x,0<1)}\downarrow & \ulcorner & \downarrow{\scriptstyle (\mathrm{id}_y,0<1)} \\ (x,1) & \xrightarrow{E} & (y,1) \end{array}$$

This is the case precisely when τ is Cartesian along E. □

Corollary 1.6.3 (Wirthmüller isomorphism) *There is a two-sided adjunction*

$$\mathrm{Sp}_G \underset{\mathrm{ind}_H^G \simeq \mathrm{coind}_H^G}{\overset{\mathrm{res}_H^G}{\underset{\curvearrowleft}{\overset{\curvearrowright}{\top\,\bot}}}} \mathrm{Sp}_H.$$

The functor $\mathrm{ind}_H^G : \mathrm{Sp}_H \to \mathrm{Sp}_G$ *admits the structure of a lax symmetric monoidal functor* as well as *the structure of an oplax symmetric monoidal functor. It satisfies the projection formula*

$$\mathrm{ind}_H^G(-_1 \otimes \mathrm{res}_H^G(-_2)) \simeq \mathrm{ind}_H^G(-_1) \otimes -_2.$$

Proof. The unit and counit transformations exhibiting the adjunction $\mathrm{ind}_H^G \dashv \mathrm{res}_H^G$ between H-Set and G-Set are Cartesian transformation. In other words, for any $X \to Y \in H$-Set and $Z \to W \in G$-Set, the squares

$$\begin{array}{ccc} X & \longrightarrow & Y \\ \downarrow & \ulcorner & \downarrow \\ G \times_H X & \longrightarrow & G \times_H Y \end{array} \qquad \text{and} \qquad \begin{array}{ccc} G \times_H Z & \longrightarrow & G \times_H W \\ \downarrow & \ulcorner & \downarrow \\ Z & \longrightarrow & W \end{array}$$

are pullbacks, an easy exercise which we omit. Using Lemma 1.6.2 and the 2-functoriality of $\mathcal{P}_\Sigma$, we obtain the adjunction $\mathrm{ind}_H^G \dashv \mathrm{res}_H^G$, as well as $\mathrm{res}_H^G \dashv \mathrm{ind}_H^G$ using the self-antiequivalence $\mathrm{Span}(-) \simeq \mathrm{Span}(-)^{\mathrm{op}}$.

The projection formula is Lemma A.17, a general feature of symmetric monoidal left adjoint functors between rigid-compactly generated categories.

□

Remark 1.6.4 (Closed subgroup). When $H \leqslant G$ is an *infinite index* (closed) subgroup, the restriction functor $\mathrm{res}_H^G : G\text{-Set}^\omega \to H\text{-Set}^\omega$ exists between finite sets, whereas induction and coinduction do not preserve finiteness. As such, one merely obtains a double adjunction from Proposition 1.1.13, in which res_H^G is monoidal.

$$\mathrm{Sp}_G \underset{\bot}{\overset{\mathrm{res}_H^G}{\overset{\curvearrowright}{\underset{\curvearrowleft}{\overset{\bot}{\leftarrow}}}}} \text{“}\mathrm{coind}_H^G\text{”} \underset{\bot}{\overset{}{\overset{\curvearrowright}{\underset{\curvearrowleft}{\overset{}{-}}}}} \mathrm{Sp}_H.$$

Thus we can make sense of the underlying spectrum of a genuine G-spectrum in terms of restriction to the trivial subgroup.

Induction and restriction of $\mathbf{E}\mathcal{F}$ We can now describe various categorical fixed points using the simple observation that every G-orbit G/H is obtained by inducing the trivial orbit H/H from H.

Proposition 1.6.5 *Let $K \leqslant H \leqslant G$ be open subgroups. For any G-space X, G-spectrum M and family $\mathcal{F}$ of subgroups of G, we have*

1. $X^K \simeq \left(\mathrm{res}^G_H X\right)^K$, $M^K \simeq \left(\mathrm{res}^G_H M\right)^K$.
2. $\mathrm{res}^G_H \mathbb{S}_G[X] \simeq \mathbb{S}_H\left[\mathrm{res}^G_H X\right]$,
3. $\mathrm{res}^G_H(\mathbf{E}\mathcal{F}) \simeq \mathbf{E}(\mathcal{F}|_H)$, *where $\mathcal{F}|_H$ is the family of subgroups of H which belong to $\mathcal{F}$.*

Proof. The claim 3. is a special case of 1., which follows from $\mathrm{ind}^G_H(H/K) \simeq G/K$. The claim 2. follows from the naturality of the inclusion $\mathcal{C} \xrightarrow{\iota} \mathrm{Span}(\mathcal{C})$ and of $\mathcal{P}_\Sigma$. □

Corollary 1.6.6 *If $N \trianglelefteq G$ is an open normal subgroup, and $H \leqslant G$ is an open subgroup, then*

$$\mathrm{res}^G_H\left(\mathbf{E}\mathcal{F}_{N\leqslant G}\right) \simeq \mathbf{E}\mathcal{F}_{(H\cap N)\leqslant H}, \quad \mathrm{res}^G_H\left(\mathbf{E}\mathcal{P}_{N\leqslant G}\right) \simeq \begin{cases} * & H \not\supseteq N \\ \mathbf{E}\mathcal{P}_{N\leqslant H} & H \supseteq N \end{cases}.$$

1.7. Geometric fixed points and inflation

The multitude of fixed points functors is a notorious feature of genuine equivariant spectra. Let's first recall how fixed points on G-sets work.

Let G be a profinite group and $N \trianglelefteq G$ a closed normal subgroup. There are the following functors at the level of finite sets.

$$(-)_N = \mathrm{ind}^{G/N}_G$$
$$G\text{-Set}^\omega \underset{\perp}{\overset{\perp}{\leftarrow\!\!\!\rightarrow}} G/N\text{-Set}^\omega : \ \mathrm{res}^{G/N}_G, \text{ also known as } \textit{inflation } \mathrm{Inf}^N$$
$$(-)^N = \mathrm{coind}^{G/N}_G$$

Among the three, only restriction and coinduction preserve pullbacks and coproducts. Thus they induce symmetric monoidal semiadditive functors between the corresponding span categories.

Lemma 1.7.1 *In the adjunction $\mathrm{res}^{G/N}_G \dashv \mathrm{coind}^{G/N}_G$, the unit $\mathrm{id}_{G/N\text{-Set}} \Rightarrow \mathrm{coind}^{G/N}_G \circ \mathrm{res}^{G/N}_G$ is an equivalence, and in particular a Cartesian transformation.*

The counit $\mathrm{res}^{G/N}_G \circ \mathrm{coind}^{G/N}_G \Rightarrow \mathrm{id}_{G\text{-Set}}$ is not a Cartesian transformation unless $N = e$.

Proof. The first assertion is obvious. If $N \neq e$, choose an orbit T with $T^N = \emptyset$. Then the counit is not Cartesian:

$$\begin{array}{ccccc} \mathrm{res}^{G/N}_G\mathrm{coind}^{G/N}_G(T) & = & \emptyset & \xrightarrow{\varepsilon} & T \\ \downarrow & & & & \downarrow \\ \mathrm{res}^{G/N}_G\mathrm{coind}^{G/N}_G(*) & = & * & \xrightarrow[\varepsilon]{\sim} & * \end{array}$$

□

Definition 1.7.2 *For a closed normal subgroup $N \trianglelefteq G$, the functors **inflation** Inf^N and **geometric N-fixed points** Φ^N are defined respectively as Mackey induction of $\mathrm{res}^{G/N}_G$ and $\mathrm{coind}^{G/N}_G$. Their right adjoints are called **categorical N-fixed points** $(-)^N$ and **geometric inflation** Ξ^N.*

$$\mathrm{Inf}^N := \left(\mathrm{Span}(\mathrm{res}^{G/N}_N)\right)_\sharp \quad : \quad \mathrm{Sp}_{G/N} \rightleftharpoons \mathrm{Sp}_G \quad : \quad \left(\mathrm{Span}(\mathrm{res}^{G/N}_N)\right)^* =: F^N$$
$$\Phi^N := \left(\mathrm{Span}(\mathrm{coind}^{G/N}_N)\right)_\sharp \quad : \quad \mathrm{Sp}_G \rightleftharpoons \mathrm{Sp}_{G/N} \quad : \quad \left(\mathrm{Span}(\mathrm{coind}^{G/N}_N)\right)^* =: \Xi^N$$

We define $(-)^N : \mathrm{An}_G \to \mathrm{An}_{G/N}$ and $\mathrm{Inf}^N : \mathrm{An}_{G/N} \to \mathrm{An}_G$ by the left Kan extension functoriality. All functors above given by left Kan extension or Mackey extension are canonically symmetric monoidal, thus F^N and Ξ^N are canonically lax symmetric monoidal. There is a natural equivalence

$$\Phi^N \circ \mathrm{Inf}^N \simeq \mathrm{id}_{\mathrm{Sp}_{G/N}} \simeq F^N \circ \Xi^N \quad \textit{adjoint to natural transformations} \quad F^N \Rightarrow \Phi^N \textit{ and } \mathrm{Inf}^N \Rightarrow \Xi^N.$$

As the right adjoint to Mackey induction is precomposition, we see that F^N forgets the categorical fixed points and Mackey structure involving subgroups $H \not\supseteq N$, whereas Ξ^N is the extension-by-zero functor.

The following result is immediately deduced from the corresponding compatibility of functors between G-sets and H-sets.

Corollary 1.7.3 *Geometric fixed points and inflation are compatible with suspension, while all four functors above are compatible with induction and restriction.*

$$\mathbb{S}_{G/N}[(-)^N] \simeq \Phi^N(\mathbb{S}_G[-]) \qquad \mathbb{S}_G[\mathrm{Inf}^N(-)] \simeq \mathrm{Inf}^N(\mathbb{S}_{G/N}[-])$$
$$\mathrm{res}^{G/N}_{H/N} \circ \Phi^N \simeq \Phi^N \circ \mathrm{res}^G_H \qquad \mathrm{res}^G_H \circ \mathrm{Inf}^N \simeq \mathrm{Inf}^N \circ \mathrm{res}^{G/N}_{H/N}$$
$$\mathrm{ind}^{G/N}_{H/N} \circ \Phi^N \simeq \Phi^N \circ \mathrm{ind}^G_H \qquad \mathrm{ind}^G_H \circ \mathrm{Inf}^N \simeq \mathrm{Inf}^N \circ \mathrm{ind}^{G/N}_{H/N}$$

□

Definition 1.7.4 *For a not necessarily normal subgroup $H \leqslant G$, we define geometric H-fixed points as*

$$\Phi^H := \Phi^H \circ \mathrm{res}^G_{N_G(H)} : \mathrm{Sp}_G \longrightarrow \mathrm{Sp}_{W_G(H)}.$$

Remark 1.7.5. One often finds the following functor referred to as the geometric H-fixed points.

$$\phi^H := \mathrm{res}^{W_G(H)}_e \circ \Phi^H \circ \mathrm{res}^G_{N_G(H)} \quad \simeq \quad \Phi^H \circ \mathrm{res}^G_H \quad : \quad \mathrm{Sp}_G \longrightarrow \mathrm{Sp}.$$

Example 1.7.6 By monoidality, $\mathrm{Inf}^N(\mathbb{S}_{G/N}) = \mathbb{S}_G$ and $\Phi^N(\mathbb{S}_G) \simeq \mathbb{S}_{G/N}$. Thus $F^N\mathrm{Inf}^N(\mathbb{S}_{G/N}) \neq \mathbb{S}_{G/N}$ in general, for instance when $N = G$, the right hand side is $\mathbb{S}$, while the left hand side is computed by Segal–tom Dieck splitting 1.3.4.

Again, the projection formula A.17 tells us that

$$F^N\mathrm{Inf}^N(-) \simeq - \otimes F^N(\mathbb{S}_G), \qquad \Xi^N\Phi^N(-) \simeq - \otimes \Xi^N(\mathbb{S}_{G/N}).$$

The first algebra is discussed in the previous example. The second example turns out to be something we've seen a few sections ago.

Theorem 1.7.7 (Geometric fixed points as isotropy separation) *Let $N \trianglelefteq G$ be a* closed *subgroup of G, then there is an equivalence of $\mathbb{E}_\infty$-algebras in Sp_G*

$$\Xi^N(\mathbb{S}_{G/N}) \simeq \mathbb{S}_G\langle\widetilde{\mathbf{E}\mathcal{P}_N}\rangle.$$

In particular, the algebra $\Xi^N(\mathbb{S}_{G/N})$ is idempotent, and the functor $\Phi^N : \mathrm{Sp}_G \to \mathrm{Sp}_{G/N}$ establishes $\mathrm{Sp}_{G/N}$ as a smashing localisation of Sp_G. The right adjoint Ξ^N is fully faithful with essential image equal to the category of $\mathcal{P}_N$-acyclic spectra.

Proof. By proposition 1.5.9, $\mathcal{P}_N$-free spectra are exactly generated by the orbits without N-fixed points, so that $\mathrm{Sp}_G^{\mathcal{P}_N\text{-free}} \subseteq \ker(\Phi^N)$. Hence Ξ^N lands in $\mathcal{P}_N$-acyclic spectra and the $\mathbb{E}_\infty$-algebra $\Xi^N(\mathbb{S}_{G/N})$ is an $\mathbb{S}_G\langle\widetilde{\mathbf{E}\mathcal{P}_N}\rangle$-algebra.

Using compatibility of restriction with Ξ^N, the computation of Proposition 1.5.4 implies that the map from $(\mathbb{S}_G)^H$ to either of their categorical H-fixed points is the projection onto the same tom Dieck summands. Therefore $\mathbb{S}_G\langle\widetilde{\mathbf{E}\mathcal{P}_N}\rangle \to \Xi^N(\mathbb{S}_{G/N})$ is an isomorphism.

We want to show that the counit $\mathrm{id} \xleftarrow{\varepsilon} \Phi^N \circ \Xi^N$ is an isomorphism. Because

$$\mathrm{coind}_G^{G/N} : G\text{-Set}^\omega \to G/N\text{-Set}^\omega$$

is essentially surjective, the functor Ξ^N is conservative. It thus suffices to show that

$$\Xi^N \xleftarrow{\Xi^N\varepsilon} \Xi^N \circ \Phi^N \circ \Xi^N$$

is an isomorphism. By the triangle equalities, this is equivalent to the unit

$$\Xi^N \xrightarrow{\eta\Xi^N} \Xi^N \circ \Phi^N \circ \Xi^N$$

being an isomorphism. By the projection formula, this is isomorphic to

$$\Xi^N \simeq \Xi^N(-) \otimes \mathbb{S}_G \Longrightarrow \Xi^N(-) \otimes \Xi^N(\mathbb{S}_{G/N}) \simeq \Xi^N(-) \otimes \mathbb{S}_G\langle\widetilde{\mathbf{E}\mathcal{F}}\rangle,$$

which is an isomorphism, as Ξ^N lands in $\mathcal{P}_N$-acyclic spectra. □

Corollary 1.7.8 *The two notions of (geometric/categorical) inflation and fixed points are related by:*

$$\Phi^N \simeq F^N(- \otimes \mathbb{S}_G\langle\widetilde{\mathbf{E}\mathcal{P}_N}\rangle), \qquad \Xi^N \simeq \mathrm{Inf}^N(-) \otimes \Xi^N(\mathbb{S}_{G/N}).$$

Proof. This follows from the projection formula (A.17) and the equivalences

$$\Phi^N \simeq F^N \circ \Xi^N \circ \Phi^N, \qquad \Xi^N \simeq \Xi^N \circ \Phi^N \circ \mathrm{Inf}^N.$$ □

In [AMR17], the authors showed that genuine equivariant spectra for *compact Lie groups* can be recovered from their geometric fixed points. In particular, geometric fixed points are jointly conservative. For finite groups, this can be deduced from the above identification.

Proposition 1.7.9 *If G is finite, then the functors $\phi^H : \mathrm{Sp}_G \to \mathrm{Sp}$, where $H \leqslant G$, are jointly conservative.*

Proof following [CMNN24, A.9]. Recall that $\phi^H = \Phi^H \circ \mathrm{res}^G_H$. Suppose $\phi^H(X) = 0$ for all $H \leqslant G$, we show that $X = 0$. By induction on the order of G, we can assume the proposition is true for all proper subgroups $H < G$, and we have that $\mathrm{res}^G_H(X) = 0$ for all proper subgroup H. In other words, $X \in \mathrm{Sp}_H$ is $\mathcal{P}_G$-acyclic. By theorem 1.7.7, we have $X = \Xi^G\Phi^G(X) = 0$. □

We will see in the next chapter that the above fails for infinite profinite groups.

2. Explorations of equivariant spectra for infinite groups

This chapter is devoted to phenomena which are particular to Sp_G for *infinite* (profinite) groups, as well as exploration of related constructions.

As always, finite always means finite discrete, and subgroups are always closed.

Before we begin, some comment is due on why one might be interested in this theory.

- Devinatz and Hopkins [DH04, Theorem 1] have constructed a "Bredon" G-ring spectrum from the Lubin–Tate theories E_n for $G = \mathbb{G}_n$ the Morava stabiliser group. Their resulting spectrum is Borel and exhibits $\mathbb{S}_{K(n)} \to E_n$ as a $K(n)$-local pro-Galois extension.

  The study of such pro-Galois extensions has motivated multiple related work, [DQ16; Qui13; BD10] to list a few.

- K-theory of automorphisms and of Laurent polynomial rings both carry Verschiebung and Frobenius morphisms, making them into $\widehat{\mathbb{Z}}$-Mackey functors. It's natural to put the "free orbit" $\mathbb{Z}$ into the picture, and try to develop a theory of $\mathbb{Z}$-Mackey functors. But one discovers that $\mathrm{KAut}(-)$ and $\mathrm{K}(-[t^{\pm}])$ live in different categories. The category $\mathrm{Sp}_{\widehat{\mathbb{Z}}}$ is a common ground on which both exist, although in Chapter 3 we will show see that they are both *integral topological Cartier modules* in the sense of McCandless [McC23].

- A recent result of Efimov [E3, Sec. 9], using the work of McCandless [McC23], showed that the TR-theory of a connective ring spectrum A acts on its nil-K-theory. Efimov established both spectra in terms of mapping spectra in the category $\mathrm{Mot}_{\mathrm{loc}}$ of localising motives, involving the reduced motive $\widetilde{\mathcal{U}}^{\mathrm{loc}}(\mathbb{S}[t])$ of the affine line. We also note that Blumberg–Gepner–Tabuada [BGT16] showed previously that the reduced affine line in $\mathrm{Mot}_{\mathrm{add}}$ corepresents endomorphism K-theory.

  Classically, the module structure on homotopy groups is given by the action of endomorphism K-theory $\mathrm{KEnd}(A)$ on nil-K-theory $\mathrm{KNil}(A)$. The former is a ring spectrum, whose π_0 is the rational Witt vectors $\mathbb{W}_{\mathrm{rat}}(\pi_0 A)$. This module structure is continuous with respect to the discrete topology on $\pi_\bullet\mathrm{KNil}$ and the usual topology on $\mathbb{W}_{\mathrm{rat}}$, and therefore extends to the completion of $\mathbb{W}_{\mathrm{rat}}$, that being the big Witt vectors. A classical textbook account is [KBook, §II.7.4, §IV.6.7].

  There should a notion of completeness and torsion of the related $\widehat{\mathbb{Z}}$- or $\mathbb{Z}$-Mackey functors.

We will not elaborate on the first point in this manuscript, and will come back to to ideas related to the latter two in chapter 3.

2.1. Genuine spectra for (infinite) profinite groups

Many have observed that Sp_G for profinite G is "assembled" from all of its finite quotients in a specific way.

Proposition 2.1.1 *For a profinite group G, let $\mathcal{B}_G$ be the cofiltered poset of open normal subgroups of G. Then Sp_G identifies with the filtered $\mathrm{Pr}^L_{\mathrm{st},\omega}$-colimit of*

$$\begin{array}{ccc} \mathcal{B}_G^{op} & \xrightarrow{I} & \mathrm{Pr}^L_{\mathrm{st},\omega} \\ U & \longmapsto & \mathrm{Sp}_{G/U} \\ (U \leqslant V) & \longmapsto & (\mathrm{Inf}^{V/U} : \mathrm{Sp}_{G/V} \to \mathrm{Sp}_{G/U}) \end{array}$$

or equivalently, the $\widehat{\mathrm{Cat}}$-limit along their right adjoints, namely categorical fixed point functors.

Proof. The category G-Set^ω is the filtered colimit of G/U-Set^ω along inflation. Hence $\mathrm{Span}(G)$ is the filtered colimit of $\mathrm{Span}(G/U)$ by Lemma 1.2.2. The claim follows as the functor $\mathrm{Sp} \otimes \mathcal{P}_\Sigma(-) : \mathrm{Cat}^{\amalg} \to \mathrm{Pr}^L$ preserves colimits, giving the equivalent description as a $\widehat{\mathrm{Cat}}$-limit.

Moreover the (non-full) subcategory $\mathrm{Pr}^L_{\mathrm{st},\omega} \hookrightarrow \mathrm{Pr}^L$ is closed under colimits, so that the colimit is also one in $\mathrm{Pr}^L_{\mathrm{st},\omega}$. Indeed, in a Pr^L-colimit of a diagram compactly generated categories, if the diagram consists of functors which preserve compact objects, then the generators of the constituents remain compact in the colimit, and form a system of generators. □

Remark 2.1.2. Inflation between span categories is symmetric monoidal, hence the above colimit is in fact a Pr^L-colimit of symmetric monoidal categories.

2.1.1. Genuine G-spectra with trivial geometric fixed points

We discuss the first structural divergence between Sp_G for finite and infinite profinite groups: The joint conservativity of geometric fixed points fails for infinite profinite groups.

Intuitively, for any infinite group G_0 with profinite completion G, we should be able to freely induce from from naive G_0-spectra into genuine G-spectra. We can expect these to have zero geometric fixed points for all nontrivial open subgroups.

Rather than identifying non-zero objects in Sp_G which are $\mathcal{P}_N$-free for every open $N \trianglelefteq G$, it turns out to be easier to identify those which are $\mathcal{F}$-free for *every* non-empty family of open subgroups.

Theorem 2.1.3 *For $X \in \mathrm{Sp}_G$, the following are equivalent:*

1. *X is $\mathcal{F}$-free for every non-empty family $\mathcal{F}$ of open subgroups.*
2. *X is $\mathcal{F}_N$-free for every open normal subgroup $N \trianglelefteq G$.*
3. *For every pair of subgroups $N \leqslant H \leqslant G$ with N normal in G, the transfer on X induce an isomorphism $(X^N)_{hH/N} \to X^H$.*

The category of such spectra, which we denote by $I_{\mathcal{F}}$, is a localising $\otimes$-ideal of Sp_G. As a category, it is equivalent to

$$I_{\mathcal{F}} \simeq \lim_{N \trianglelefteq G} \mathrm{Sp}^{\mathbf{B}G/N} \quad \text{along homotopy orbits.}$$

In particular, $I_{\mathcal{F}} \neq 0$.

A dual statement holds with "Borel" replacing "free", and homotopy fixed points replacing homotopy orbits.

Proof. A fortiori item 1 implies 2, and conversely every non-empty family of open subgroups contains an open normal subgroup, so 2 implies 1.

Adams isomorphism for the family $\mathcal{F}_N$ (Lemma 1.5.13) shows that X is $\mathcal{F}_N$-free iff for all $H \leqslant G$, the transfers induce an equivalence $X^{H\cap N}_{hH/(H\cap N)} \simeq X^H$. With H and N running over all appropriate subgroups, we see that 2 and 3 are equivalent.

The subcategory $I_{\mathcal{F}}$ is a localising $\otimes$-ideal, since it is an intersection of localising $\otimes$-ideals. The limit description follows from Proposition 2.1.1 and item 3. □

If G is finite, then $I_{\mathcal{F}}$ is exactly the subcategory of free G-spectra, which is equivalent to $\mathrm{Sp}^{\mathbf{B}G}$. In general, the homotopy N-orbit functors on naive G-spectra provides a functor

$$\mathrm{Sp}^{\mathbf{B}G} \to \lim_{N\trianglelefteq G} \mathrm{Sp}^{\mathbf{B}G/N} \simeq I_{\mathcal{F}}.$$

Proposition 2.1.4 *If G is infinite, then the functor $\mathrm{Sp}^{\mathbf{B}G} \to I_{\mathcal{F}}$ is not an equivalence.*

Proof. Consider $\mathbb{Q}[G] \in \mathrm{Sp}^{\mathbf{B}G}$ with free G-action. It suffices to show that the unit transformation in the adjunction

$$\mathbb{Q}[G] \to \lim_{N\trianglelefteq G} \mathbb{Q}[G/N]$$

is not an equivalence. On π_0, the right hand side contains the element given by

$$\left(\sum_{gN\in G/N} \frac{[gN]}{[G:N]}\right)_{N\trianglelefteq G},$$

which is not in the image from the left hand side. □

2.2. Geometric G-spectra

We consider another way of "assembling" the categories $\mathrm{Sp}_{G/N}$ of G. Recall that $\mathrm{Sp}_{G/N}$ is equivalent to the category of $\mathbb{S}_G\langle\widetilde{\mathbf{E}\mathcal{P}_N}\rangle$-modules in Sp_G. This produces a cofiltered inverse system of symmetric monoidal localisations under Sp_G:

$$\begin{array}{ccc}
\{-\infty\}\cup\mathcal{B}_G & \xrightarrow{\gamma_-} & \mathrm{Pr}^L \\
-\infty & \longmapsto & \mathrm{Sp}_G \\
U & \longmapsto & \mathrm{Sp}_{G/U} \\
(-\infty \leqslant U) & \longmapsto & (\Phi^U : \mathrm{Sp}_G \twoheadrightarrow \mathrm{Sp}_{G/U}) \\
(U \leqslant V) & \longmapsto & (\Phi^{V/U} : \mathrm{Sp}_{G/U} \twoheadrightarrow \mathrm{Sp}_{G/V})
\end{array}$$

It's easy to come up with the following constructions in analogy to Proposition 2.1.1.

Definition 2.2.1 *For a profinite group G with the lattice of open normal subgroups $\mathcal{B}_G$, we define the category $\mathrm{Sp}^{\mathrm{geo}}_G$ of **geometric G-spectra** to be the Pr^L-limit of the diagram $\gamma|_{\mathcal{B}_G}$. The induced symmetric monoidal left adjoint functor and its lax symmetric monoidal right adjoint are denoted*

$$\Phi : \mathrm{Sp}_G \rightleftharpoons \mathrm{Sp}^{\mathrm{geo}}_G : \Xi.$$

Roughly speaking, a geometric G-spectrum X consists of $X_N \in \mathrm{Sp}_{G/N}$ for each open normal $N \trianglelefteq G$, together with coherent isomorphisms $\Phi^{V/U} X_U \simeq X_V$ for all $U \subseteq V$. One sees that Φ takes a G-spectrum Y to its geometric fixed points $\left(\Phi^N(Y)\right)_{N\trianglelefteq G}$, and Ξ takes a geometric G-spectrum $(X_N)_{N\trianglelefteq G}$ to $\lim_N \Xi^N X_N \in \mathrm{Sp}_G$.

Definition 2.2.2 *We call the monoidal unit of $\mathrm{Sp}_G^{\mathrm{geo}}$ the **geometric** G-sphere $\mathbb{S}_G^{\mathrm{geo}}$.*

*For any $M \in \mathrm{Sp}_G$, we define the **geometric completion of** M as $\widehat{M} := \Xi\Phi(M)$. We write $\widehat{\mathbb{S}}_G[X]$ for the geometric completion of $\mathbb{S}_G[X]$. In particular, $\widehat{\mathbb{S}}_G$ is called the **geometrically complete G-sphere**.*

From the discussion above, combined with Segal-tom Dieck splitting 1.3.4, we immediately obtain the following.

Corollary 2.2.3 *We have for $M \in \mathrm{Sp}_G$,*

$$\widehat{M} \simeq \lim_{N\trianglelefteq G} M \otimes \mathbb{S}_G\langle\widetilde{\mathbf{E}\mathcal{P}_N}\rangle.$$

For $X \in \mathrm{An}_G$, the G-fixed points of $\widehat{\mathbb{S}}_G[X]$ is given by the limit

$$(\widehat{\mathbb{S}}_G[X])^G \simeq \lim_{U\trianglelefteq G \text{ open}} \simeq \prod_{(H)} \mathbb{S}[X^H]_{\mathrm{h}W_G(H)}. \qquad \square$$

Remark 2.2.4. Dress and Siebeneicher [DS88] showed that the kernels of the surjections

$$f^N : A(G) \to A(G/N), [X] \mapsto [X^N]$$

defines a filtration on the Burnside ring of a profinite group G, whose completion provides a notion of G-typical Witt vectors (with $\mathbb{Z}$ coefficients), in that it recovers the p-typical Witt vectors for $G = \mathbb{Z}_p$ and big Witt vectors for $G = \widehat{\mathbb{Z}}$.

The above computation thus shows that $\pi_0(\widehat{\mathbb{S}_G})$ is the Dress–Siebeneicher completed Burnside ring $\widehat{A}(G)$.

Remark 2.2.5. The monoidal unit of $\mathrm{Sp}_G^{\mathrm{geo}}$ is hardly ever compact. Indeed, unless G is already finite,

$$\mathrm{map}\left(\mathbb{S}_G^{\mathrm{geo}}, \bigoplus_{\mathbb{Z}} \mathbb{S}_G^{\mathrm{geo}}\right) \simeq \prod_{(U)\leqslant G} \bigoplus_{\mathbb{Z}} \mathrm{K}(G/U\text{-Set}) \neq \bigoplus_{\mathbb{Z}} \prod_{(U)\leqslant G} \mathrm{K}(G/U\text{-Set}) \simeq \bigoplus_{\mathbb{Z}} \mathrm{map}\left(\mathbb{S}_G^{\mathrm{geo}}, \mathbb{S}_G^{\mathrm{geo}}\right).$$

2.2.1. Geometric localisation of Sp_G

We could have taken all possible fancy limits of $\mathrm{Sp}_{G/N}$, but we will consider in this subsection a construction more "internal" to the category Sp_G itself.

Definition 2.2.6 *Let $I_{\mathcal{P}} \subseteq \mathrm{Sp}_G$ be the kernel of $\Phi : \mathrm{Sp}_G \to \mathrm{Sp}_G^{\mathrm{geo}}$, which is a localising $\otimes$-ideal:*

$$I_{\mathcal{P}} := \bigcap_{N\trianglelefteq G} \mathrm{Sp}_G^{\mathcal{P}_N\text{-free}}.$$

*The corresponding monoidal localisation L^{geo} is called the **geometric localisation**.*

Note that if G is finite, then $\mathcal{P}_e$ is the empty family and $I_{\mathcal{P}} = 0$. But if G infinite, then $I_{\mathcal{F}} \subseteq I_{\mathcal{P}}$, so neither is trivial by the result of subsection 2.1.1.

By construction, there is a factorisation

$$\mathrm{Sp}_G \rightleftarrows L^{\mathrm{geo}}\mathrm{Sp}_G \underset{\Xi}{\overset{\Phi}{\rightleftarrows}} \mathrm{Sp}_G^{\mathrm{geo}},$$

so that $\widehat{(-)}$ lands in L^{geo}-local G-spectra.

Remark 2.2.7. The localisation L^{geo} is a localisation without finite acyclics, in that there are no compact L^{geo}-acyclic G-spectra:

$$I_{\mathcal{P}} \cap \mathrm{Sp}_G^{\omega} = 0.$$

Indeed, by Proposition 2.1.1 and the remark following it, a nonzero compact G-spectrum is of the form $\mathrm{Inf}^N(X)$ for some $0 \neq X \in \mathrm{Sp}^{\omega}_{G/N}$, whence $\Phi^N\mathrm{Inf}^N(X) = X \neq 0$.

2.3. A completion theorem for Sp_G

We attempt a comparison between $L^{\mathrm{geo}}\mathrm{Sp}_G$ and $\mathrm{Sp}_G^{\mathrm{geo}}$, subject to a finiteness condition on G.

Definition 2.3.1 *A profinite group G is said to be $\mathcal{P}$-compact if for any open normal $N \trianglelefteq G$, the family $\mathcal{P}_N$ of open subgroups has only finitely many maximal elements under inclusion.*

We will see in Lemma 2.3.13 that topologically finitely generated pro-p-groups are $\mathcal{P}$-compact.

Definition 2.3.2 *A set of spectra is called* uniformly bounded below *if there exists $N \in \mathbb{Z}$ such every spectrum in the set is N-connective.*

Let Sp_G^- denote the full subcategory of Sp_G on those G-spectra X, whose $\left\{X^H\right\}_{H\leqslant G}$ is uniformly bounded below.

Similarly, let $\mathrm{Sp}_G^{\mathrm{geo},-}$) be the full subcategory of $\mathrm{Sp}_G^{\mathrm{geo}}$ on those geometric G-spectra X, whose $\left\{\phi^H(X)\right\}_{H\leqslant G}$ is bounded below.

By the following lemma, a geometric G-spectrum X is uniformly bounded below iff the set of its categorical fixed points $\left\{(X_N)^H\right\}_{N\trianglelefteq G, H\leqslant G/N}$ is uniformly bounded below. Since $I_{\mathcal{F}} \neq 0$ for infinite G, this characterisation fails for Sp_G.

Lemma 2.3.3 *Let G be a finite group and X a G-spectrum. Then X^H is connective for every subgroup H iff $\phi^H(X)$ is connective for every subgroup H.*

Proof. The "only if" part is clear, as $\phi^H(X)$ is a colimits of categorical fixed points. The converse follows from induction on the order of G. When G is trivial, so is the claim. Suppose the converse holds for all proper subgroups of G, then the induction hypothesis implies that X^H is connective for every proper subgroup H. Isotropy separation sequence

$$\operatorname*{colim}_{G/H\to \mathbf{E}\mathcal{P}_G} X^H \longrightarrow X^G \longrightarrow \phi^G(X)$$

then shows that X^G is also connective. □

Theorem 2.3.4 *If G is a $\mathcal{P}$-compact group, then $L^{\mathrm{geo}} : \mathrm{Sp}_G \to \mathrm{Sp}_G$ preserves the subcategory Sp_G^-. Moreover, the adjunction $\mathrm{Sp}_G \rightleftharpoons \mathrm{Sp}_G^{\mathrm{geo}}$ restricts to an equivalence $L^{\mathrm{geo}}\mathrm{Sp}_G^- \simeq \mathrm{Sp}_G^{\mathrm{geo},-}$.*

In particular, for $X \in \mathrm{Sp}_G^-$ we have $L^{\mathrm{geo}}(X) \simeq \widehat{X}$, and for $X \in \mathrm{Sp}_G^{\omega}$, we have $L^{\mathrm{geo}}X \simeq X \otimes \widehat{\mathbb{S}}_G$.

The proof will be given at the end of subsection 2.3.2.

Remark 2.3.5. From the perspective of tt-geometry, this result is analogous to the *Chromatic Convergence Theorem* of Hopkins–Ravenel, in that they identify, for specific objects, an inverse limit of (smashing) localisations in terms of another localisation.

2.3.1. Bounded below spectra and their (co)limits

The reason one needs the uniformly bounded below condition is that they allows us to commute *finite dimensional* limits with *finite-type* colimits of spectra. The author learned this trick from [AN21, Lemma 2.11], but see also [NP26, Lemma 2.35].

Definition 2.3.6 *A category I is **cofinally of finite type** if there exists a finite type simplicial set I_0 and a cofinal map $I_0 \to I$. A G-space X is said to be of* finite type *if its category of elements $(\mathrm{Orb}_G)_{/X}$ is cofinally of finite type.*

*Similarly we speak of **coinitially finite dimensional** categories.*

Lemma 2.3.7 *Let $F : \mathcal{C} \to \mathcal{D}$ be an exact functor between presentable stable categories equipped with t-structures. Suppose that F sends $\mathcal{C}_{\geqslant 0}$ to $\mathcal{D}_{\geqslant -N}$ for some $N \in \mathbb{Z}$. If $\mathcal{D}$ is left t-complete, then F preserves uniformly bounded below colimits which are cofinally of finite type.*

Proof. Consider $\alpha : J \to \mathcal{C}_{\geqslant 0}$, where J is a simplicial set of finite type. Let $J(n)$ be the n-skeleton of J, then each $J(n)$ is finite, and by exactness of F we have

$$\operatorname{colim}_J F\alpha = \operatorname*{colim}_n \operatorname{colim}_{J(n)} F\alpha = \operatorname*{colim}_n F \operatorname{colim}_{J(n)} \alpha,$$

Moreover, as $J(n)$ is obtained by attaching copies of Δ^n to $J(n-1)$ along $\partial\Delta^n$, the cofiber of

$$\operatorname*{colim}_{J(n-1)} \alpha \longrightarrow \operatorname*{colim}_{J(n)} \alpha$$

is a sum of $\Sigma^n \alpha(-)$[5] and thus n-connective. It follows that the cofiber of

$$F \operatorname{colim}_{J(n-1)} \alpha \longrightarrow F \operatorname{colim}_J \alpha$$

is $n-N$-connective. In the colimit, the canonical map

$$\operatorname*{colim}_n F \operatorname{colim}_{J(n)} \alpha \longrightarrow F \operatorname*{colim}_n \operatorname{colim}_{J(n)} \alpha = F \operatorname{colim}_J \alpha$$

is ∞-connective and thus an isomorphism by left t-completeness of $\mathcal{D}$. □

Lemma 2.3.8 *Let J be a small category of coinitially dimension N, then the limit functor $\lim_J : \mathrm{Sp}^J \to \mathrm{Sp}$ sends $\mathrm{Sp}^J_{\geqslant 0}$ to $\mathrm{Sp}_{\geqslant -N}$.*

Proof. Limits along J are again computed by pullbacks and simplex attachments. Since infinite products of connective objects in Sp are connective, the decrease in connectivity is bounded by the dimension of J, just as in the previous proof. □

[5] It's a fun exercise that for $\beta : \Delta^n \to \mathcal{C}$, there is a cofiber sequence $\operatorname{colim}_{\partial\Delta^n} \beta \to \operatorname{colim}_{\Delta^n} \alpha (\simeq \alpha(n)) \to \Sigma^n \alpha(0)$.

2.3.2. Finite type colimits and $\mathcal{P}$-compactness

Unsurprisingly, $\mathcal{P}$-compactness is tailored to make the previous subsection apply.

Lemma 2.3.9 *For a profinite group G, the following are equivalent:*

1. *G is $\mathcal{P}$-compact.*
2. *Every family $\mathcal{F}$ of open subgroups, such that there are only finitely many subgroups $H \leqslant G$ with $H \notin \mathcal{F}$, has only finitely many maximal elements.*
3. *For every family $\mathcal{F}$ as in (2), the G-anima $\mathbf{E}\mathcal{F}$ is of finite type.*

Proof. To see that 1 is equivalent to 2, we note that every $\mathcal{F}$ as in 2 contains $\mathcal{P}_N$, where N is the intersection of $H \notin \mathcal{F}$. Thus maximal elements of $\mathcal{F}$ are either not in $\mathcal{P}_N$ or maximal in $\mathcal{P}_N$, either of which there are only finitely many.

We prove that 2 implies 3. Let L be the (finite) intersection of the maximal elements of $\mathcal{F}$, which is an open normal subgroup of G, so G/L is finite. Since $H \in \mathcal{F}$ iff H is contained in some maximal element M of $\mathcal{F}$, we see that

$$\mathbf{E}\mathcal{F} \simeq \mathrm{Inf}^L \mathbf{E}\mathcal{K}, \quad \mathcal{K} = \{H \leqslant G/L : H \subseteq M/L \text{ for some maximal } M \in \mathcal{F}\}.$$

It follows that $\mathbf{E}\mathcal{K}$ is a finite type G/L-space, as any orbit category for a finite group is a finite $(1,1)$-category. Its inflation $\mathbf{E}\mathcal{F}$ is finite type G-space by Lemma 2.3.10 below.

To see that 3 implies 2, note that any isotropy of a G-space must be subconjugate to the isotropy of its 0-cells. If $\mathbf{E}\mathcal{F}$ is of finite type, then maximal elements are among the isotropy of its 0-cells. □

Lemma 2.3.10 *Let $f : C \to D$ be a functor between small categories, and $X \in \mathrm{PSh}(C)$ a presheaf. Consider the left Kan extension $f_\sharp X \in \mathrm{PSh}(D)$ and the following square between their corresponding right fibrations over C and over D.*

$$\begin{array}{ccc} C_{/X} & \xrightarrow{\alpha} & D_{/f_\sharp X} \\ {\scriptstyle \mathrm{RFib}}\downarrow & & \downarrow{\scriptstyle \mathrm{RFib}} \\ C & \xrightarrow{f} & D \end{array}$$

Then the map α is cofinal.

Proof. Let $A := C_{/X}$ and $B := D_{/f_\sharp X}$. Write $\mathrm{Map}_{/B}(-,-)$ for the mapping spaces in $\mathrm{Cat}_{/B}$ etc. Recall the following definition of cofinality.

$$\text{For every } (E \to B) \in \mathrm{RFib}(B), \text{ there holds } \mathrm{Map}_{/B}(B, E) \simeq \mathrm{Map}_{/B}(A, E).$$

For such $E \to B$, also $E \to D$ is a right fibration. Now for any $E \in \mathrm{RFib}(D)$, definition of $f_\sharp$ shows that α induces an isomorphism:

$$\mathrm{Map}_{/D}(A, E) \simeq \mathrm{Map}_{/C}(A, f^* E) \simeq \mathrm{Map}_{/D}(B, E).$$

Thus it also induces $\mathrm{Map}_{/B}(B, E) \simeq \mathrm{Map}_{/B}(A, E)$ by the following pullback for every $T \in \mathrm{Cat}_{/B}$.

$$\begin{array}{ccc} \mathrm{Map}_{/B}(T, E) & \longrightarrow & \mathrm{Map}_{/D}(T, E) \\ \downarrow & \ulcorner & \downarrow \\ * \simeq \mathrm{Map}_{/B}(T, B) & \longrightarrow & \mathrm{Map}_{/D}(T, B) \end{array}$$

□

Proposition 2.3.11 *Finite groups are $\mathcal{P}$-compact. Let G be a $\mathcal{P}$-compact profinite group.*

1. *G is topologically finitely generated.*
2. *Any open subgroup $H \leqslant G$ is $\mathcal{P}$-compact.*

Proof. Clearly finite groups are $\mathcal{P}$-compact, as each of them has only finitely many subgroups.

1. If G is $\mathcal{P}$-compact, then $\mathcal{P}_G$ has only finitely many maximal elements. For each maximal open subgroup $M < G$, choose an element $x_M \in G \smallsetminus M$. Then the elements (x_M) topologically generate G.
2. Let $I \trianglelefteq H$ be an open normal subgroup. By Lemma 2.3.9, item 2, we may assume furthermore that I is *normal in* G. Then maximally elements of $\mathcal{P}_{I\leqslant H}$ is of the form $M \cap H$ for some maximal element $M \in \mathcal{P}_{I\leqslant G}$. □

We can now prove Theorem 2.3.4.

Lemma 2.3.12 *Let G be a $\mathcal{P}$-compact group. For all open $N \trianglelefteq G$, the functor $(- \otimes \widetilde{\mathbf{E}\mathcal{P}_N}) : \mathrm{Sp}_G \to \mathrm{Sp}_G$ commutes with uniformly bounded below limits of (coinitially) finite dimension.*

Proof. By isotropy separation, it suffices to show that $(- \otimes \mathbf{E}\mathcal{P}_N)$ commutes with such limits. Adams isomorphism implies that for all $K \leqslant G$:

$$(- \otimes \mathbf{E}\mathcal{P}_N)^K \simeq \operatorname*{colim}_{G/H \to \mathbf{E}\mathcal{P}_N} (- \otimes G/H)^K.$$

The functor $(-\otimes G/H)^K$ commutes with arbitrary limits, and the colimit is cofinally of finite type by Lemma 2.3.9, so that Lemma 2.3.7 and 2.3.8 apply. □

Proof of theorem 2.3.4. A compact G-spectrum $X \in \mathrm{Sp}_G^{\omega}$ is uniformly bounded below, and the projection formula $\widehat{X} \simeq X \otimes \widehat{\mathbb{S}}_G$ (A.17) holds for dualisables unconditionally. Thus it suffices to prove the first paragraph.

Let $X \in \mathrm{Sp}_G^-$ and $(Y_N)_{N\trianglelefteq G} \in \mathrm{Sp}_G^{\mathrm{geo},-}$. We need to show that

1. $L^{\mathrm{geo}}(X)$ is bounded below and equivalent to $\widehat{X}$.
2. For all open $N \trianglelefteq G$, $(\lim_{I\trianglelefteq G} Y_I) \otimes \widetilde{\mathbf{E}\mathcal{P}_N} \simeq Y_N$.

We now prove part 1. It suffices to show that $X \longrightarrow \widehat{X}$ induces an equivalence after tensoring with $\widetilde{\mathbf{E}\mathcal{P}_N}$.

$$\widehat{X} \otimes \widetilde{\mathbf{E}\mathcal{P}_N} \overset{\text{def.}}{=\!=} \left(\lim_{I\trianglelefteq G} X \otimes \widetilde{\mathbf{E}\mathcal{P}_I}\right) \otimes \widetilde{\mathbf{E}\mathcal{P}_N} \overset{(!)}{=\!=} \left(\lim_{I\trianglelefteq G} X \otimes \widetilde{\mathbf{E}\mathcal{P}_I} \otimes \widetilde{\mathbf{E}\mathcal{P}_N}\right)$$
$$\overset{\text{as long as } I\subseteq N}{=\!=} \left(\lim_{I\trianglelefteq G} X \otimes \widetilde{\mathbf{E}\mathcal{P}_N}\right) = = X \otimes \widetilde{\mathbf{E}\mathcal{P}_N}.$$

Proposition 2.3.11 tells up that G is topologically finitely generated, so it admits a coinitial sequence of open normal subgroups by [Wil24, Theorem 2.3.7]. Therefore the limit is coinitially one dimensional, so that Lemma 2.3.12 justifies (!).

Thus $X \to \widehat{X}$ is an L^{geo}-equivalence into an L^{geo}-local object. Therefore $L^{\mathrm{geo}}(X) \simeq \widehat{X}$, and is in particular uniformly bounded below.

The proof of part 2 is essentially the same argument

$$\left(\lim_{I\trianglelefteq G} Y_I\right) \otimes \widetilde{\mathbf{E}\mathcal{P}_N} \overset{\text{Lemma 2.3.12}}{=\!=} \left(\lim_{I\trianglelefteq G} Y_I \otimes \widetilde{\mathbf{E}\mathcal{P}_N}\right) \overset{\text{as long as } I\subseteq N}{=\!=} \left(\lim_{I\trianglelefteq G} Y_N\right) = = Y_N. \qquad \square$$

2.3.3. Nilpotence and $\mathcal{P}$-compactness

A lot of, but not all interesting groups are $\mathcal{P}$-compact.

Lemma 2.3.13 *If G is $\mathcal{P}$-compact, it is topologically finitely generated. Conversely if G is topologically finitely generated, pro-nilpotent, with only finitely many prime numbers dividing its order, then G is $\mathcal{P}$-compact. In particular, finitely generated pro-p-groups are $\mathcal{P}$-compact.*

The proof of the above will span the subsection.

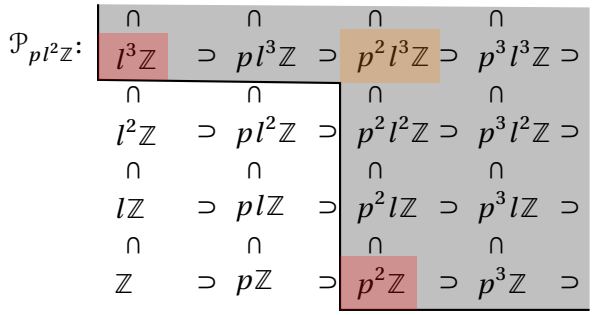


Figure 2.1.: The group $\mathbb{Z}_p \times \mathbb{Z}_l$ is $\mathcal{P}$-compact: $\mathbf{E}\mathcal{P}_{p^a l^b\mathbb{Z}}$ inflated from $\mathbb{Z}/p^{a+1}l^{b+1}\mathbb{Z}$

This result is hardly sharp: Any finite group is $\mathcal{P}$-compact, and it's not hard to see that the "n-adic dihedral group" $(\mathbb{Z}_n \rtimes C_2)$ for a *finite* integer n is also $\mathcal{P}$-compact.

Question 1 Is $\mathcal{P}$-compactness closed under finite extensions with finite kernel or cokernel? Does virtually pro-nilpotence (+ finite support on Spec $\mathbb{Z}$) imply $\mathcal{P}$-compactness?

We refer to the textbook [Wil24] for general results for profinite groups. Recall the following standard facts from finite group theory:

Proposition 2.3.14
1. *Let G be a finite group and $H \leqslant G$ a p-subgroup. If $p \mid [G : H]$, then there exists a subgroup $H \subseteq K \leqslant G$ with $[K : H] = p$.*
2. *If G is a finite nilpotent group, then every Sylow subgroup of G is normal and G is canonically the product of its Sylow subgroups G_p.*

It follows that a pro-nilpotent group is canonically the product of its pro-p-Sylow subgroups. □

The first property fails for solvable groups. For instance A_4 admits no subgroup of order 6.

Lemma 2.3.15 *Let G be a pro-nilpotent group with only finitely many primes dividing its order, then there exists an integer m, such that for every proper open subgroup $H < G$, there is a subgroup $K \supset H$ with $1 < [K : H] \leqslant m$.*

Proof. Take m to be the largest prime factor of $|G|$. The subgroup H is a product of p-subgroups of G, and for some prime $p \leqslant m$, the Sylow p-subgroup of $H_p \neq G_p$. The previous proposition provides a subgroup $K_p \leqslant G_p$ containing H_p with $[K_p : H_p] = p$, then we may take K to be $HK_p \leqslant G$. □

Proof of lemma 2.3.13. Suppose that G is topologically finitely generated pro-nilpotent, such that $|G|$ admits a largest prime factor m, Let $N \trianglelefteq G$ be an open normal subgroup of index n. We show that the maximal elements of $\mathcal{P}_N$ have index $\leqslant mn$, so that there can be only finitely many of such.

Suppose $H \in \mathcal{P}_N$ has index $> mn$, then there exists $K \supset H$ such that $1 < [K : H] \leqslant m$. Consequently $m[G : K] \geqslant [G : H] > mn$, so that $K \not\supseteq N$, and H is not maximal in $\mathcal{P}_N$. □

2.4. Quasifinitely genuine spectra, or Mackey profunctors

This section reviews and studies the variant $\mathrm{Sp}_G^{\mathrm{qfgen}}$[6] introduced in [KMN23, Section 4]. We prove a comparison of $\mathrm{Sp}_G^{\mathrm{qfgen},-}$ with $\mathrm{Sp}_G^{\mathrm{geo},-}$ for finitely generated groups in general.

Definition 2.4.1 *For a profinite group G, the category of* quasifinite *G-sets is the limit of categories:*

$$G\text{-Set}^{\mathrm{qf}} := \lim_{U \in \mathcal{B}_G} G/U\text{-Set}^{\omega},$$

where the limit is taken along coinduction along $G/V \twoheadrightarrow G/U$.

Equivalently, a quasifinite G-sets X is a continuous, discrete G-set such that for each open subgroup $U \leqslant G$, the set of U-fixed points $X^U \subseteq X$ is finite.

As coinduction preserves pullbacks, the category $G\text{-Set}^{\mathrm{qf}}$ admits pullbacks, so that we may speak of $\mathrm{Span}(G\text{-Set}^{\mathrm{qf}})$.

Definition 2.4.2 ([KMN23, 4.5]) *The category of* quasifinitely genuine *G-spectra, denoted by $\mathrm{Sp}_G^{\mathrm{qfgen}}$, is the full subcategory of $\mathrm{Fun}(\mathrm{Span}(G\text{-Set}^{\mathrm{qf}})^{op}, \mathrm{Sp})$ spanned by those functors M which are* very additive, *in the sense that for each $X = \coprod_i G/H_i \in G\text{-Set}^{\mathrm{qf}}$, restriction along $G/H_i \hookrightarrow X$ induce an equivalence*

$$M(X) \xrightarrow{\sim} \prod_i M(G/H_i).$$

We also let $\mathrm{Sp}_G^{\mathrm{qfM}} := \mathcal{P}_\Sigma(\mathrm{Span}(G\text{-Set}^{\mathrm{qf}}); \mathrm{Sp})$ denote the quasifinite spectral G-Mackey functors.

Proposition 2.4.3 ([KMN23, Lemma 4.6, Warning 4.10, Proposition 4.19]) *The category $\mathrm{Sp}_G^{\mathrm{qfgen}}$ is a presentably stable category, generated by the set of G-orbits and is an accessible, symmetric monoidal localisation of $\mathrm{Sp}_G^{\mathrm{qfM}}$ which we call* very additivisation, *denoted $(-)_{\mathrm{vadd}}$.* □

Remark 2.4.4. Similar to Lemma 1.2.5, quasifinite G-sets are self-dual in $\mathrm{Sp}_G^{\mathrm{qfgen}}$, which are however not compact (unless G is finite). Indeed, infinite coproducts of very additive functors are not computed underlying.

A quasifinitely genuine spectrum M admits the additional structure of infinite sums of transfers: The fibers of a map $X \to G/H$ between quasifinite G-sets is in general a quasifinite H-set, and M admits a transfer map

$$\prod_{G/U_\alpha \subseteq X} M^{U_\alpha} \longrightarrow M^H \quad \text{extending} \quad \bigoplus_{G/U_\alpha \subseteq X} M^{U_\alpha} \longrightarrow M^H.$$

2.4.1. Isotropy separation in $\mathrm{Sp}_G^{\mathrm{qfgen}}$

Following [KMN23, Section 4.3], we study isotropy separation in $\mathrm{Sp}_G^{\mathrm{qfgen}}$, and show that geometric fixed points commute with uniformly bounded below limits of finite cohomological dimension without finiteness assumptions on the group.

It'll be convenient to consider some auxiliary categories.

[6] We've changed notations to be consistent with the current manuscript.

Construction 2.4.5 *Consider the category* $\mathrm{An}_G^{\mathrm{qf}} := \mathcal{P}_\Sigma(G\text{-Set}^{\mathrm{qf}})$*, equipped with Day convolution. One has the following diagram of symmetric monoidal functors*

$$\begin{array}{ccccccc} \mathrm{An}_G & \xrightarrow{\mathbb{S}_G[-]} & \mathrm{Sp}_G & \longrightarrow & \mathrm{Sp}_G^{\mathrm{qfgen}} & \xrightarrow{\Phi} & \mathrm{Sp}_G^{\mathrm{geo}} \\ \downarrow & & \downarrow & \nearrow_{(-)_{\mathrm{vadd}}} & & & \\ \mathrm{An}_G^{\mathrm{qf}} & \xrightarrow[\mathbb{S}_G^{\mathrm{qf}}[-]]{} & \mathrm{Sp}_G^{\mathrm{qfM}} & & & & \end{array}$$

There is an (-1)*-truncated presheaf* $\mathbf{E}^{\mathrm{q}}\mathcal{F} \in \mathrm{An}_G^{\mathrm{qf}}$ *corresponding to the subcategory* $G\text{-Set}_{\mathcal{F}}^{\mathrm{qf}}$ *of* quasifinite $\mathcal{F}$*-sets, i.e. quasifinite* G*-sets with* $\mathcal{F}$*-isotropy. There is a complementary idempotent algebra* $\widetilde{\mathbf{E}^{\mathrm{qf}}\mathcal{F}}$ *in* $\mathrm{An}_{G,*}^{\mathrm{qf}}$.

Proposition 2.4.6 (cf. [KMN23, Lemma 4.28]) *Let* $\mathcal{F}$ *be a family of open subgroups of* G.

1. *The subcategory of* $\mathbf{E}^{\mathrm{q}}\mathcal{F}$*-comodules in either* $\mathrm{Sp}_G^{\mathrm{qfM}}$ *or* $\mathrm{Sp}_G^{\mathrm{qfgen}}$ *is the localising subcategory generated by the quasifinite* $\mathcal{F}$*-sets.*
2. *The map* $\mathbf{E}\mathcal{F} \to \mathbf{E}^{\mathrm{q}}\mathcal{F}$ *in* $\mathrm{An}_G^{\mathrm{qf}}$ *becomes an equivalence after suspension to* $\mathrm{Sp}_G^{\mathrm{qfgen}}$*. Thus in* $\mathrm{Sp}_G^{\mathrm{qfgen}}$*, the idempotent algebras* $\widetilde{\mathbf{E}\mathcal{F}}$ *and* $\widetilde{\mathbf{E}^{\mathrm{q}}\mathcal{F}}$ *are also equivalent.*
3. *If* $\mathcal{F}$ *contains all but finitely many open subgroups, then* $\widetilde{\mathbf{E}^{\mathrm{q}}\mathcal{F}}$*-modules in* $\mathrm{Sp}_G^{\mathrm{qfM}}$ *are automatically very additive.*

Bear in mind that the orbits no longer generate $\mathrm{An}_G^{\mathrm{qf}}$ or $\mathrm{Sp}_G^{\mathrm{qfM}}$.

Proof. Part 1. follows from the same proof as Proposition 1.5.9. For part 2., it suffices by very-additivity to take only the orbits as generators, By the same proof again, they generate exactly the $\mathbf{E}\mathcal{F}$-comodules.

For the last part, $\widetilde{\mathbf{E}^{\mathrm{q}}\mathcal{F}}$-modules in $\mathrm{Sp}_G^{\mathrm{qfM}}$ are exactly those Mackey functors which vanish on quasifinite $\mathcal{F}$-sets. If $\mathcal{F}$ contains all but finitely many subgroups, then any quasifinite G-set is a union of a quasifinite $\mathcal{F}$-set and finitely many orbits, and thus such Mackey functors are automatically very additive. □

Corollary 2.4.7 ([KMN23, cf. Corollary 4.30]) *For a family* $\mathcal{F}$ *containing all but finitely many open subgroups, we can compute* $X \otimes \widetilde{\mathbf{E}\mathcal{F}}$ *in* $\mathrm{Sp}_G^{\mathrm{qfgen}}$ *as* $X \otimes \widetilde{\mathbf{E}^{\mathrm{q}}\mathcal{F}}$ *in* $\mathrm{Sp}_G^{\mathrm{qfM}}$*. The sequence*

$$X \otimes \mathbf{E}\mathcal{F} \longrightarrow X \longrightarrow X \otimes \widetilde{\mathbf{E}\mathcal{F}} \quad \text{in } \mathrm{Sp}_G^{\mathrm{qfgen}}.$$

coincides with the sequence

$$X \otimes \mathbf{E}^{\mathrm{q}}\mathcal{F} \longrightarrow X \longrightarrow X \otimes \widetilde{\mathbf{E}^{\mathrm{q}}\mathcal{F}} \quad \text{in } \mathrm{Sp}_G^{\mathrm{qfM}}.$$

Proof. The tensor product in $\mathrm{Sp}_G^{\mathrm{qfgen}}$ is the very-additivisation of the tensor product in $\mathrm{Sp}_G^{\mathrm{qfM}}$. Proposition 2.4.6 item 2 implies that $\widetilde{\mathbf{E}^{\mathrm{q}}\mathcal{F}}$ and $\widetilde{\mathbf{E}\mathcal{F}}$ become equivalent in $\mathrm{Sp}_G^{\mathrm{qfgen}}$, and item 3 shows that very-additivisation is redundant. □

This is good news, since colimits in $\mathrm{Sp}_G^{\mathrm{qfM}}$ are computed underlying. In order to compute geometric fixed points, it remains to describe $\mathbf{E}^{\mathrm{q}}\mathcal{F}$.

Let $\mathcal{M}$ be a *subset* of $\mathcal{F}$ containing all maximal elements up to conjugacy and consider the quasifinite $\mathcal{F}$-set

$$S := \coprod_{M \in \mathcal{M}} G/M \in G\text{-Set}_{\mathcal{F}}^{\mathrm{qf}} \overset{y}{\hookrightarrow} \mathrm{An}_G^{\mathrm{qf}}.$$

We stress that S is the Yoneda image of one single quasifinite G-set, not the coproduct in $\mathrm{An}_G^{\mathrm{qf}}$.

Lemma 2.4.8 ([KMN23, Lemma 4.37]) *The geometric realisation in* $\mathrm{An}_G^{\mathrm{qf}}$ *of*

$$\Delta^{op} \longrightarrow \mathrm{An}_G^{\mathrm{qf}}, \quad [n] \longmapsto S^{\times(n+1)},$$

whose structure maps are given by projections and diagonals, is equivalent to $\mathbf{E}^{\mathrm{q}}\mathcal{F}$.

Proof. The category $\mathrm{An}_G^{\mathrm{qf}}$ is closed under geometric realisation in $\mathrm{PSh}(G\text{-}\mathrm{Set}^{\mathrm{qf}})$, so we may work in the latter category. The simplicial object displayed above is the Čech nerve of $S \to *$, whose realisation computes $\tau_{\leqslant -1}(S)$ by [HTT, 6.2.3.4]. A quasifinite G-set admits a map to S precisely when it has $\mathcal{F}$-isotropy, so that $\tau_{\leqslant -1}(S) \simeq \mathbf{E}^{\mathrm{q}}\mathcal{F}$. □

Corollary 2.4.9 ([KMN23, cf. proof of 4.36]) *Suppose* $\mathcal{F}$ *contains all but finitely many open subgroups of* G *and let* S *be as above. For* $X \in \mathrm{Sp}_G^{\mathrm{qfgen}}$, *the* $\mathcal{F}$*-isotropy separation sequence (in* $\mathrm{Sp}_G^{\mathrm{qfM}}$*) takes the form*

$$\left\| \cdots \rightrightarrows X(S \times S) \rightrightarrows X(S) \right\| \longrightarrow X^G \longrightarrow (X \otimes \widetilde{\mathbf{E}\mathcal{F}})^G.$$

For $\mathcal{F} = \mathcal{P}_G$, *the above expression expands into*

$$\left\| \cdots \rightrightarrows \prod_{\substack{M_0, M_1 \leqslant G \\ \text{maximal open} \\ [g] \in M_0 \backslash G / M_1}} X^{M_0 \cap g M_1 g^{-1}} \rightrightarrows \prod_{\substack{M \leqslant G \\ \text{maximal open}}} X^M \right\| \longrightarrow X^G \longrightarrow \Phi^G(X). \qquad \square$$

Remark 2.4.10. The above shows that tensoring with $\widetilde{\mathbf{E}\mathcal{P}_N}$ in $\mathrm{Sp}_G^{\mathrm{qfgen}}$ commutes with uniformly bounded below limits of finite dimension. Indeed, the infinite product commutes with arbitrary limits, and geometric realisation commutes with uniformly bounded below limits of finite dimension.

The same trick performed in Sp_G results in direct sums in place of the products, so that a finiteness condition is necessary for the commutation with limits.

Corollary 2.4.11 *If* G *is topologically finitely generated, then geometric fixed points* $\Phi : \mathrm{Sp}_G^{\mathrm{qfgen}} \to \mathrm{Sp}_G^{\mathrm{geo}}$ *induces an equivalence between uniformly bounded below objects.*

If G *is* $\mathcal{P}$*-compact, then* $L^{\mathrm{geo}}\mathrm{Sp}_G^- \simeq \mathrm{Sp}_G^{\mathrm{qfgen},-} \simeq \mathrm{Sp}_G^{\mathrm{geo},-}$.

Proof. The final assertion is Theorem 2.3.4. Let X be a uniformly bounded below quasifinitely genuine G-spectrum. Knowing that in $\mathrm{Sp}_G^{\mathrm{qfgen}}$, sequential limits commute with $(-) \otimes \mathbf{E}\mathcal{P}_N$ unconditionally, the proof of 2.3.2 shows that $X \to \lim_N (X \otimes \widetilde{\mathbf{E}\mathcal{P}_N})$ induces an equivalence on all geometric fixed points, where both the domain and codomain are uniformly bounded below. By [KMN23, Proposition 4.25], geometric fixed points jointly detect equivalences for uniformly bounded below objects. □

2.4.2. Polygonic spectra

This subsection surveys the titular construction of [KMN23], which is a Nikolaus–Scholze-style variant of geometric C_{p^n}- and $\mathbb{Z}_p$-spectra. Although the essence of loc. cit. is in the integral case, we will restrict our attention to the p-typical situation for simplicity.

Definition 2.4.12 *For a prime* p *and* $0 \leqslant n \leqslant \infty$, *let* $\mathrm{PgcSp}_{\langle p^n \rangle}$ *be the category of the following datum:*

1. *a naive C_{p^k}-spectrum $X_k \in \mathrm{Sp}^{\mathbf{B}C_{p^k}}$ for $k = 0, .., n$, and*
2. *a map of naive $C_{p^{k-1}}$-spectrum $\alpha_k : X_{k-1} \to X_k^{tC_p}$ for $k = 1, .., n$.*

The uniformly bounded below subcategories, defined in the obvious way, are denoted $\mathrm{PgcSp}^-_{\langle p^n\rangle}$.
When $n = \infty$, we write PgcSp_p.

The reason why Nikolaus–Scholze-style definitions work is their Tate orbit lemma.

Lemma 2.4.13 ([NS18, II.4.1, II.6.1]) *For a bounded below Borel C_{p^n}-spectrum $X \in \mathrm{Sp}^{\mathrm{Borel}}_{C_{p^n}}$, the genuine $C_{p^{n-1}}$-spectrum $\Phi^{C_p} X$ is again Borel.* □

In the case $G = \widehat{\mathbb{Z}}$, [KMN23, Theorem 5.4] shows that $\mathrm{Sp}^{\mathrm{qfgen},-}_{\widehat{\mathbb{Z}}}$ is equivalent to the category of *integral* PgcSp^-. We review the p-typical situation.

Proposition 2.4.14 *For each finite n, there is a left adjoint $L : \mathrm{Sp}_{C_{p^n}} \to \mathrm{PgcSp}_{\langle p^n\rangle}$, sending*

$$Y \in \mathrm{Sp}_{C_{p^n}} \qquad \text{to} \qquad X_k := \Phi^{C_{p^{n-k}}}(Y)_{\mathrm{Borel}} \in \mathrm{Sp}^{\mathbf{B}C_{p^k}}$$

and the gluing maps

$$\alpha_k : X_{k-1} = \Phi^{C_p}(\Phi^{C_{p^{n-k}}}(Y))_{\mathrm{Borel}} \longrightarrow \Phi^{C_p}(X_k)_{\mathrm{Borel}} \overset{\text{Prop. 1.5.8}}{=\!=} X_k^{tC_p}.$$

The following squares commutes and produces in the limit a functor $L : \mathrm{Sp}^{\mathrm{geo}}_{\mathbb{Z}_p} \to \mathrm{PgcSp}_p$.

$$\begin{array}{ccccccc}
\cdots & \longrightarrow & \mathrm{Sp}_{C_{p^n}} & \xrightarrow{\Phi^{C_p}} & \mathrm{Sp}_{C_{p^{n-1}}} & \xrightarrow{\Phi^{C_p}} \cdots \longrightarrow & \mathrm{Sp} \\
 & & \downarrow{\scriptstyle L} & & \downarrow{\scriptstyle L} & & \| \\
\cdots & \longrightarrow & \mathrm{PgcSp}_{\langle p^n\rangle} & \xrightarrow{\mathrm{forget}_n} & \mathrm{PgcSp}_{\langle p^{n-1}\rangle} & \xrightarrow{\mathrm{forget}_{n-1}} \cdots \longrightarrow & \mathrm{Sp}
\end{array}$$

Each functor denoted L induces an equivalence between the respective (uniformly or not) bounded below subcategories.

Proof. It suffices to treat the case $n < \infty$. We inductively construct the functors L.
The trivial subgroup of C_{p^n} is the only subgroup not containing $C_p \leqslant C_{p^n}$, so that $\Phi^{C_p}(-)_{\mathrm{Borel}} \simeq (-)^{tC_p}$ by Proposition 1.5.8. Isotropy separation thus expresses $\mathrm{Sp}_{C_{p^n}}$ as the following *lax pullback* (Proposition A.5), where b_G denote Borelisation for typographical reasons.

$$\begin{array}{ccc}
\mathrm{Sp}_{C_{p^n}} & \xrightarrow{\Phi^{C_p}} & \mathrm{Sp}_{C_{p^{n-1}}} \\
{\scriptstyle b_{C_p^n}}\downarrow & \overset{\alpha}{\swarrow} & \| \\
\mathrm{Sp}^{\mathrm{Borel}}_{C_{p^n}} & \xrightarrow{\Phi^{C_p}} & \mathrm{Sp}_{C_{p^{n-1}}}
\end{array}$$

Composing in the lower right corner with $b_{C_{p^{n-1}}}$ induces a functor in to the following lax pullback.

$$\begin{array}{ccc}
\bullet & \longrightarrow & \mathrm{Sp}_{C_{p^{n-1}}} \\
\downarrow & \overset{\alpha}{\swarrow} & \downarrow{\scriptstyle b_{C_{p^{n-1}}}} \\
\mathrm{Sp}^{\mathrm{Borel}}_{C_{p^n}} & \xrightarrow{(-)^{tC_p}} & \mathrm{Sp}^{\mathrm{Borel}}_{C_{p^{n-1}}}
\end{array}$$

Since $b_{C_{p^{n-1}}}$ and $\mathrm{id}_{\mathrm{Sp}_{C^{p^{n-1}}}}$ preserve colimits[7], the induced functor $\mathrm{Sp}_{C_{p^n}} \to \bullet$ also preserves colimits, and is thus a left adjoint. (cf. [NS18, II.1.5(iv.)])

Using the induction hypothesis, the functor $L : \mathrm{Sp}_{C_{p^{n-1}}} \to \mathrm{PgcSp}_{\langle p^{n-1} \rangle}$ induces a left adjoint functor from $\bullet$ into the following lax pullback

$$\begin{array}{ccc} * & \longrightarrow & \mathrm{PgcSp}_{\langle p^{n-1} \rangle} \\ \downarrow & \overset{\alpha}{\swarrow} & \downarrow{\scriptstyle \mathrm{proj}_{n-1}} \\ \mathrm{Sp}^{\mathrm{Borel}}_{C_{p^n}} & \xrightarrow{(-)^{tC_p}} & \mathrm{Sp}^{\mathrm{Borel}}_{C_{p^{n-1}}} \end{array}$$

which is easily seen to be equivalent to $\mathrm{PgcSp}_{\langle p^n \rangle}$. Thus we've constructed the left adjoint L for the induction step, which, by induction, also has the stated form.

On bounded below objects, the target $\Phi^{C_p}(X_{\mathrm{Borel}})$ involved in the lax pullback is already Borel by the Tate orbit lemma 2.4.13. Thus L as constructed acts as an equivalence on bounded below objects. □

[7] The bottom edge $\Phi^{C_p} : \mathrm{Sp}^{\mathrm{Borel}}_{C_{p^n}} \to \mathrm{Sp}_{C^{p^{n-1}}}$ is not colimit preserving due to the implicit inclusion into $\mathrm{Sp}_{C^{p^n}}$.

3. Equivariant $\mathbb{K}$-theory, automorphism and endomorphism $\mathbb{K}$-theory

This chapter is a survey of K-theory Mackey functors. Their modern construction is by first establishing the appropriate Mackey functors valued in small or large categories, to which one can then apply K-theory, topological Hochschild homology or the universal additive or localising invariants of Blumberg–Gepner–Tabuada [BGT13] and Efimov [E1].

The technical underpinning of the first step is Barwick's unfurling construction, which extends a functor $\mathcal{C}^{\mathrm{op}} \longrightarrow \widehat{\mathrm{Cat}}$ along $\iota^{\mathrm{op}} : \mathcal{C}^{\mathrm{op}} \to \mathrm{Span}(\mathcal{C})^{\mathrm{op}}$, subject to an *adjointability* condition. More conceptually, this construction underlies a universal property of the span 2-category, which we review in section 3.2, and apply in section 3.3 to construct Mackey functors for the profinite group $\widehat{\mathbb{Z}}$, consisting of modules of Laurent polynomial rings ($\mathbb{G}_m$) and the category of automorphisms. A variation of the construction recovers endomorphism and automorphism categories twisted by an endofunctor, a special case of constructions found in [CLMPZ25; Nik18].

The second step involves determining the conditions needed for the Mackey functors valued in large categories to induce Mackey functors in K-theory. We will investigate the instances of equivariant K-theory for discrete groups in subsection 3.3.1 and endomorphism K-theory in subsection 3.3.2. Equivariant K-theory of small categories in this form has been used [CMNN24, §2] in the study of $T(n)$-local K-theory.

When the endofunctor is the identity, automorphism K-theory carries the additional structure of a *integral topological Cartier module* in the sense of [McC23], and the endomorphism K-theory is naturally an *epicyclic Mackey functor* in the sense of Definition 3.1.16.

Warning In this chapter, groups are no longer required to be profinite. In particular we allow the *discrete* group $\mathbb{Z}$. Similarly Orb_G may contain infinite orbits, and one must take care of the correct span categories, such as $\mathrm{Span}_{\text{ét}}(G)$.

3.1. From Borelisation to the action category

Let G be a (discrete) group and $\mathcal{C}$ a (potentially large) category with G-action. Through unstraightening, there is a coCartesian fibration $G\backslash\backslash\mathcal{C} \in \mathrm{coCart}(\mathbf{B}G)$ whose fiber at the base point of $\mathbf{B}G$ is identified with $\mathcal{C}$.

We recall the definition of a Borel G-object (1.4.3).

Definition 3.1.1 *A G-**category** is a $\widehat{\mathrm{Cat}}$-valued presheaf on Orb_G, the category of which is denoted $\widehat{\mathrm{Cat}}_G$.*

To a category $\mathcal{C}$ with G-action, its associated Borel G-category is denoted by

$$\underline{\mathcal{C}} \coloneqq b_G(\mathcal{C}) \in \widehat{\mathrm{Cat}}_G.$$

As per usual, the value of $\underline{\mathcal{C}}_G$ at G/H is called the homotopy H-fixed points of $\mathcal{C}$, denoted $\mathcal{C}^{\mathrm{h}H}$, which we describe now more explicitly.

The category $\widehat{\mathrm{Cat}}_G$ with the objectwise Cartesian monoidal structure is naturally enriched over $\widehat{\mathrm{Cat}}$, so that $\mathcal{C}^{\mathrm{h}H}$ is the mapping category

$$\underline{\mathcal{C}}(G/H) \simeq \mathrm{Fun}_{\widehat{\mathrm{Cat}}_G}(y(G/H), b_G(\mathcal{C})),$$

where $y_{G/H}$ is the Yoneda image regarded as a discrete G-category. Since b_G is right Kan extension along $\mathrm{free}^{\mathrm{op}} : \mathbf{B}G \hookrightarrow \mathrm{Orb}_G^{\mathrm{op}}$, by adjunction, we obtain that

$$\mathcal{C}^{\mathrm{h}H} \simeq \mathrm{Fun}_{\widehat{\mathrm{Cat}}^{\mathbf{B}G}}(\mathrm{free}^*(y(G/H)), \mathcal{C}).$$

Yoneda embedding corresponds under unstraightening to the functor

$$\mathrm{Orb}_G \hookrightarrow \mathrm{RFib}(\mathrm{Orb}_G) \hookrightarrow \mathrm{Cart}(\mathrm{Orb}_G), \quad G/H \longmapsto \big((\mathrm{Orb}_G)_{/(G/H)} \simeq \mathrm{Orb}_H \to \mathrm{Orb}_G\big),$$

the pullback of which to $\mathbf{B}G^{\mathrm{op}}$ is exactly the functor sending G/H to the right fibration over $\mathbf{B}G^{\mathrm{op}}$, classifying the right G^{op}-action on $\mathrm{Map}_{\mathrm{Orb}_G}(G/e, G/H) \simeq G/H$.

Definition 3.1.2 *For a group G and a G-set X, the* ***action groupoid*** *of X is the corresponding Kan fibration over $\mathbf{B}G$, denoted by $G\backslash\backslash X$. For a monoid M, the* ***action category*** *of an M-set Y is the left fibration $M\backslash\backslash Y$ over $\mathbf{B}M$. This defines a functor $M\backslash\backslash(-) : M\text{-Set} \to \mathrm{LFib}(\mathbf{B}M)$.*

Corollary 3.1.3 *The value of the Borel G-category $\underline{\mathcal{C}}$ on a G-orbit T is given by*

$$\underline{\mathcal{C}}(T) \simeq \mathrm{Fun}_{/\mathbf{B}G}(G\backslash\backslash T, G\backslash\backslash\mathcal{C}) \simeq \left\{ \textit{the category of functors} \quad \begin{array}{ccc} & & G\backslash\backslash\mathcal{C} \\ & \nearrow & \downarrow \\ G\backslash\backslash T & \longrightarrow & \mathbf{B}G \end{array} \quad \textit{over } \mathbf{B}G \right\}.$$

Proof. In the unstraightening correspondence, functors in $\widehat{\mathrm{Cat}}^{\mathbf{B}G}$ correspond to maps between coCartesian fibrations over $\mathbf{B}G$ which preserve coCartesian morphisms. Since $\mathbf{B}G$ and $G\backslash\backslash T$ are groupoids, coCartesian morphisms are exactly the isomorphisms, which are automatically preserved. □

Remark 3.1.4. When $\mathcal{C}$ is equipped with the trivial G-action, we have $\underline{\mathcal{C}}(G/H) \simeq C^{\mathbf{B}H}$, and $\underline{\mathcal{C}}$ extends to a presheaf on the category of connected finite groupoids, also called a "global category".

When $G = \mathbb{Z}$, the action groupoid construction sends maps between finite $\mathbb{Z}$-sets to positive degree maps between S^1. When $G = D_\infty$, the infinite dihedral group, the action groupoids of orbit with infinite cyclic isotropy (i.e. $D_\infty/n\mathbb{Z}$) are mapped to arbitrary non-zero[8] degree maps between S^1, over $\mathbf{B}D_\infty$.

Restricting actions on the finite $\mathbb{Z}$-set $\mathbb{Z}/r\mathbb{Z}$ to $\mathbb{N}$ and taking action category produces the following category of "walking r-fold endomorphisms".

Note that action groupoids of G-orbits are connected groupoids, whereas action categories are not in general connected, even for transitive M-sets.

Notation 3.1.5 *For $n \geqslant 1$, we let $\mathbb{T}_n$ denote the action category $\mathbb{N}\backslash\backslash\mathbb{Z}/n\mathbb{Z}$.*

[8] A choice of orientation for $D_\infty\backslash\backslash D_\infty/\mathbb{Z} \simeq S^1$ determines orientations for all other groupoids.

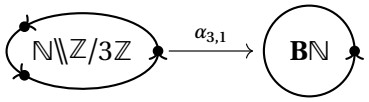


Figure 3.1.: $\mathbb{N}\backslash\backslash\mathbb{Z}/m\mathbb{Z}$ sitting over $\mathbf{B}\mathbb{N}$

Since $\mathbb{N}$ acts invertibly, the terminal map $\mathbb{Z}/n\mathbb{Z} \to *$ exhibits $\mathbb{T}_n$ as a Kan fibration over $\mathbf{B}\mathbb{N}$. Later we also have the opportunity to consider $\mathbb{T}_n$ as merely a category.

Construction 3.1.6 *For a category $\mathcal{C}$, equipped with an endofunctor F, let* $\mathrm{END}^{(r)}(\mathcal{C}, F) \to \mathbf{B}\mathbb{N}$ *denote the coCartesian unstraightening of*

$$\mathbf{B}\mathbb{N} \xrightarrow{(\mathbb{Z}/r\mathbb{Z}),(\mathcal{C},F)} \mathrm{An}^{op} \times \widehat{\mathrm{Cat}} \xrightarrow{\mathrm{Fun}(-,-)} \widehat{\mathrm{Cat}}$$

where the first factor classifies the right *fibration $\mathbb{T}_n \to \mathbf{B}\mathbb{N}$. This defines a functor*

$$\begin{aligned} \underline{\mathrm{END}} \quad : \quad & \widehat{\mathrm{Cat}}^{\mathbf{B}\mathbb{N}} \longrightarrow \mathrm{PSh}(\mathrm{Orb}_{\widehat{\mathbb{Z}}}; \mathrm{coCart}(\mathbf{B}\mathbb{N})) \\ & (\mathcal{C}, F) \longmapsto \left(\mathbb{Z}/r\mathbb{Z} \longmapsto \mathrm{END}^{(r)}(\mathcal{C}, F)\right). \end{aligned}$$

More explicitly, the coCartesian fibration $\mathrm{END}^{(r)}(\mathcal{C}, F) \to \mathbf{B}\mathbb{N}$ classifies endofunctor

$$\begin{aligned} V_r F \quad : \quad & \mathcal{C}^{\mathbb{Z}/r\mathbb{Z}} \longrightarrow \mathcal{C}^{\mathbb{Z}/r\mathbb{Z}} \\ & (c_0, c_1 \dots, c_{n-1}) \longmapsto (F c_{n-1}, F c_0, \dots, F c_{n-2}). \end{aligned}$$

Definition 3.1.7 *For a category $\mathcal{C}$ with endofunctor F, we define the* ***endomorphism $\widehat{\mathbb{Z}}$-category*** $\underline{\mathrm{End}}(\mathcal{C}, F)$ *as the sections of* $\underline{\mathrm{END}}(\mathcal{C}, \mathrm{F})$*:*

$$\underline{\mathrm{End}}(\mathcal{C}, F)(-) := \mathrm{Fun}_{\mathrm{coCart}(\mathbf{B}\mathbb{N})}(\mathbf{B}\mathbb{N}, \underline{\mathrm{END}}(\mathcal{C}, F)(-)).$$

The ***automorphism $\widehat{\mathbb{Z}}$-category*** $\underline{\mathrm{Aut}}(\mathcal{C}, F)$ *is defined as the category of coCartesian sections of* $\underline{\mathrm{END}}(\mathcal{C}, F)$.[9]
We write $\mathrm{End}^{(r)}(\mathcal{C}, F)$ *and* $\mathrm{Aut}^{(r)}(\mathcal{C}, F)$ *for their value respectively at the orbit $\mathbb{Z}/r\mathbb{Z}$.*
When $F = \mathrm{id}_{\mathcal{C}}$, we simply write $\underline{\mathrm{End}}(\mathcal{C})$ *and* $\underline{\mathrm{Aut}}(\mathcal{C})$.

To spell things out, the category $\mathrm{End}^{(r)}(\mathcal{C}, F)$ consists of the datum

$$\left(c_0, \dots, c_{r-1} \in \mathcal{C}; F(c_0) \xrightarrow{\alpha_0} c_1, \dots, F(c_{r-1}) \xrightarrow{\alpha_{r-1}} c_0\right),$$

with the appropriate mapping spaces between them. The category $\mathrm{Aut}^{(r)}(\mathcal{C}, F)$ is the full subcategory of $\mathrm{End}^{(r)}(\mathcal{C}, F)$ where each α_k is an isomorphism. This is equivalently the category consisting of pairs

$$(c, \beta : F^{\circ r}(c) \xrightarrow{\sim} c).$$

In the special case of $F = \mathrm{id}_{\mathcal{C}}$, we see that

$$\mathrm{End}^{(r)}(\mathcal{C}) \simeq \mathrm{Fun}(\mathbb{T}_n, \mathcal{C}), \qquad \mathrm{Aut}^{(r)}(\mathcal{C}) \simeq \mathcal{C}^{S^1},$$

and both $\widehat{\mathbb{Z}}$-categories carry a further *epicyclic* structure, which we now recall.

[9] This is well defined since the structure maps of $\underline{\mathrm{END}}(\mathcal{C}, F)$ are functors which preserve coCartesian morphisms.

3.1.1. Epicyclic objects

We recall Goodwillie's epicyclic objects after [McC23], albeit from a less combinatorial perspective. Epicyclic structures are designed to capture most of the interesting structure that's naturally present on $\underline{\mathrm{End}}(\mathcal{C})$.

Definition 3.1.8 (cf. [McC23, Example 2.1.11, Construction 2.1.1]) *The* ***epicyclic category*** $\widetilde{\Lambda}$ *is the subcategory of* Cat *consisting of* $\mathbb{T}_n$ *for* $n \geqslant 1$ *and maps which induce upon geometric realisations positive degree maps between* S^1.

Connes' ***cyclic category*** Λ *is the subcategory of* $\widetilde{\Lambda}$ *on the maps of degree 1.*

The classifying category of the ***Witt monoid*** $\mathbf{B}\mathbb{W}$ *is the subcategory of* An *consisting of self-maps of* S^1 *of positive degree.*

A functor from $\widetilde{\Lambda}^{op}$, Λ^{op} *or* $\mathbf{B}\mathbb{W}^{op}$ *to a category* $\mathcal{C}$ *is called an* ***epicyclic*** *object,* ***cyclic*** *object or object* ***with Frobenius lifts*** *respectively. But if* $\mathcal{C} = \widehat{\mathrm{Cat}}$, *we speak of* $\widetilde{\Lambda}$*-categories, rather than "epicyclic categories".*

Remark 3.1.9. The author believes that the description of $\widetilde{\Lambda}$ as the category of "essentially surjective" functors between $\mathbb{T}_n$ in [McC23, Definition 2.1.9] is a misnomer. Indeed, a [positive degree, non-decreasing self-map of S^1 preserving markings] need not be surjective on marked points, or vice versa. For instance, consider the constant map $\mathbb{T}_3 \to \mathbb{T}_1$ to the unique point which is essentially surjective, and any degree 1 map $\mathbb{T}_1 \to \mathbb{T}_3$.

Thus, action categories define a map $\mathrm{Orb}_{\widehat{\mathbb{Z}}} \to \widetilde{\Lambda}$, so that every $\widetilde{\Lambda}$-category has an underlying $\widehat{\mathbb{Z}}$-category.

Proposition 3.1.10 *The* $\widehat{\mathbb{Z}}$*-category* $\underline{\mathrm{End}}(\mathcal{C})$ *underlies a* $\widetilde{\Lambda}$*-category*

$$\underline{\mathrm{End}}(\mathcal{C}) \quad := \quad (\mathbb{T}_n \longmapsto \mathrm{Fun}(\mathbb{T}_n, \mathcal{C}))\,.$$

The $\widehat{\mathbb{Z}}$*-category* $\underline{\mathrm{Aut}}(\mathcal{C})$ *underlies a category with Frobenius lifts*

$$\underline{\mathrm{Aut}}(\mathcal{C}) \quad := \quad \left(S^1 \longmapsto \mathrm{Fun}(S^1, \mathcal{C})\right).$$

Proof. With the trivial $\mathbb{N}$-action on $\mathcal{C}$, the sections of $\underline{\mathrm{END}}(\mathcal{C})$ are just given by functors $\mathbb{T}_n \to \mathcal{C}$. A section is coCartesian precisely when the functor $\mathbb{T}_n \to \mathcal{C}$ factors through $\mathcal{C}^{\simeq}$, and therefore through the geometric realisation of $\mathbb{T}_n$, that being S^1. □

Remark 3.1.11. [McC23, Example 2.1.11] Let $\mathbb{Z}^\times$ and $\mathbb{N}^\times$ denote the multiplicative monoid of integers. Every self-map of S^1 is equivalent to $\left(z \mapsto \lambda z^k\right)$ for some $(\lambda, k) \in S^1 \times \mathbb{Z}^\times$ by a straight line homotopy. This recover the description in [McC23] of $\mathbb{W}$ as the topological monoid $S^1 \rtimes \mathbb{N}^\times$.

Similarly, every map $\mathbb{T}_{kn} \to \mathbb{T}_n$ in $\widetilde{\Lambda}$ factors as $\alpha_k \circ \delta$, where $\alpha_{k,n} : \mathbb{T}_{kn} \to \mathbb{T}_n$ is the Kan fibration $\mathbb{N}\backslash\backslash\mathbb{Z}/mn\mathbb{Z} \to \mathbb{N}\backslash\backslash\mathbb{Z}/n\mathbb{Z}$, and $\delta : \mathbb{T}_{kn} \to \mathbb{T}_{kn}$ is of degree 1. Any two such factorisations are uniquely isomorphic.

Proposition 3.1.12 ([McC23, Lemma 2.1.13]) *In the following commutative diagram of categories, every*

square is a pullback and every vertical functor is a Cartesian fibration.

$$\begin{array}{ccccc} \Lambda_\infty & \longrightarrow & \Lambda & \longrightarrow & \widetilde{\Lambda} \\ \downarrow & & \downarrow & & \downarrow \\ * & \longrightarrow & \mathbf{B}S^1 & \longrightarrow & \mathbf{B}\mathbb{W} \\ & & \downarrow & & \downarrow \\ & & * & \longrightarrow & \mathbf{B}\mathbb{N}^\times \end{array}$$

Here $\Lambda \to \widetilde{\Lambda}$ is the inclusion of degree 1 *maps, and Λ_∞ is the* paracyclic *category.* □

The cofinal map $\Delta^{\mathrm{op}} \to \Lambda_\infty^{\mathrm{op}}$ of [NS18, Theorem B.3] and [Kerodon, 02ZM] give the following.

Corollary 3.1.13 (cf. [McC23, Proposition 2.1.15]) *The left Kan extension along $\widetilde{\Lambda}^{op} \to \mathbf{B}\mathbb{W}^{op}$ and along $\Lambda^{op} \to (\mathbf{B}S^1)^{op}$ coincides respectively with what's known in the literature as the geometric realisation of epicyclic or cyclic objects.* □

McCandless showed [McC23, Lemma 3.2.1] that $\mathbf{B}\mathbb{W}$ is an orbital category, and that $\mathbf{B}\mathbb{W}$-Mackey functors give an integral version of Antieau–Nikolaus' topological Cartier modules. We note that $\widetilde{\Lambda}^{\amalg}$ admits pullbacks along the maps $\alpha_{k,n}$, so that one can speak of Mackey functors with transfers along such maps.

Definition 3.1.14 *Let $\widetilde{\Lambda}^{\amalg}$ denote the subcategory of* Cat *consisting of finite coproducts of $\mathbb{T}_n$, $n \geqslant 1$, and maps of component-wise positive degree.*

*A **regular covering** in $\widetilde{\Lambda}^{\amalg}$ is a morphism $\coprod_i \mathbb{T}_{m(i)} \to \coprod_j \mathbb{T}_{n(j)}$, where each factor $\mathbb{T}_{m(i)} \to \mathbb{T}_{n(j)}$ is a Cartesian morphism in the Cartesian fibration $\widetilde{\Lambda} \to \mathbf{B}\mathbb{N}^\times$.*

Regular coverings are exactly the maps of the form $\mathbb{N}\backslash\backslash X \to \mathbb{N}\backslash\backslash Y$ coming from maps $X \to Y$ of finite $\mathbb{Z}$-sets, or equivalently, functors in $\widetilde{\Lambda} \subseteq \mathrm{Cat}$ which are Kan fibrations.

Lemma 3.1.15 *The category $\widetilde{\Lambda}^{\amalg}$ admits pullbacks along regular coverings.*

Proof. Since coproducts in Cat are disjoint and universal, it suffices to show that the pullback along $\alpha_{k,b}$ of a morphism coming from $\widetilde{\Lambda}$ exists in $\widetilde{\Lambda}^{\amalg}$.

Every map $\mathbb{T}_a \to \mathbb{T}_b$ in $\widetilde{\Lambda}$ factors as a degree 1 map $\mathbb{T}_a \to \mathbb{T}_{lb}$ in Λ and some regular covering $\alpha_{l,b} : \mathbb{T}_{lb} \to \mathbb{T}_b$. The pullback of $\alpha_{k,b}$ along $\alpha_{l,b}$ is given by applying $\mathbb{N}\backslash\backslash(-)$ to the following pullback square of $\mathbb{Z}$-sets on the left.

$$\begin{array}{ccc} (\mathbb{Z}/\mathrm{lcm}(k,l)b\mathbb{Z})^{\amalg \gcd(k,l)} & \longrightarrow & \mathbb{Z}/lb\mathbb{Z} \\ \downarrow & \ulcorner & \downarrow \\ \mathbb{Z}/kb\mathbb{Z} & \longrightarrow & \mathbb{Z}/b\mathbb{Z} \end{array} \qquad \begin{array}{ccc} \mathbb{T}_{k'a} & \xrightarrow[\mathrm{Cart}]{\alpha_{k',a}} & \mathbb{T}_a \\ {\scriptstyle \exists! \delta_k}\downarrow & & \downarrow{\scriptstyle \delta} \\ \mathbb{T}_{k'b'} & \xrightarrow[\mathrm{Cart}]{\alpha_{k',b'}} & \mathbb{T}_{b'} \\ \downarrow & & \downarrow \\ * & \xrightarrow{k'} & * \end{array} \qquad \begin{array}{c} \widetilde{\Lambda} \\ \\ \mathbf{B}\mathbb{N}^\times \end{array}$$

It thus remains to see that pullback of a degree 1 map $\delta : \mathbb{T}_a \to \mathbb{T}_{b'}$ along $\alpha_{k',b'} : \mathbb{T}_{k'b} \to \mathbb{T}_{b'}$ exists in $\widetilde{\Lambda}$. Using that $p : \widetilde{\Lambda} \to \mathbf{B}\mathbb{N}^\times$ is a Cartesian fibration, it's easy to check that the square on the right gives the pullback in $\widetilde{\Lambda}$. □

Definition 3.1.16 *The category of $\mathcal{V}$-valued* ***epicyclic Mackey functors*** *is defined as*

$$\mathrm{Mack}(\widetilde{\Lambda};\mathcal{V}) \coloneqq \mathcal{P}_\Sigma(\mathrm{Span}_\alpha(\widetilde{\Lambda}^{\amalg});\mathcal{V}),$$

where α denotes the subcategory of regular coverings.

The category of $\mathcal{V}$-valued ***topological Cartier modules*** *(cf. [McC23, Definition 3.2.2])*

$$\mathrm{Mack}(\mathbf{B}\mathbb{W};\mathcal{V}) \coloneqq \mathcal{P}_\Sigma(\mathrm{Span}(\mathbf{B}\mathbb{W}^{\amalg});\mathcal{V}).$$

When $\mathcal{V} = \mathrm{Sp}$, we introduce the following shorthands.

$$\mathrm{Sp}_{\widetilde{\Lambda}} \coloneqq \mathrm{Mack}(\widetilde{\Lambda};\mathrm{Sp}) \qquad \mathrm{TCart} \coloneqq \mathrm{Mack}(\mathbf{B}\mathbb{W};\mathrm{Sp}).$$

Corollary 3.1.17 *Geometric realisation $\widetilde{\Lambda}^{\amalg} \to \mathbf{B}\mathbb{W}^{\amalg}$ preserves disjoint unions and pullbacks along regular coverings. It thus induces a strongly continuous functor $\mathrm{Sp}_{\widetilde{\Lambda}} \to \mathrm{TCart}$.*

Proof. Disjoint unions are obviously preserved. Pullbacks in $\mathbf{B}\mathbb{W}^{\amalg}$, the subcategory of An consisting of unions of positive-degree self-covering of S^1, is treated in parallel as in the previous lemma by replacing $\mathbb{N}\backslash\!\backslash(-)$ with $\mathbb{Z}\backslash\!\backslash(-)$. Using this description, such pullbacks are evidently preserved by geometric realisation. The rest follows from the functoriality of $\mathcal{P}_\Sigma(-)$. □

3.2. Adjointability and the span 2-category

The next step is to turn the G-categories $\underline{\mathcal{C}}_G$, $\underline{\mathrm{End}}(\mathcal{C},F)$ and $\underline{\mathrm{Aut}}(\mathcal{C},F)$ into $\widehat{\mathrm{Cat}}$-valued G-Mackey functors, which we denote by the same symbols.

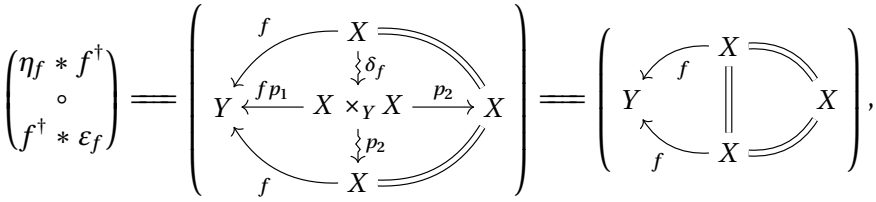
Slogan(Gaitsgory–Rozenblyum): The span 2-category $2\mathrm{Span}_E(\mathcal{C})$ is obtained from $\mathcal{C}$ by freely adding right adjoints to 1-morphisms in E, subject to the Beck–Chevalley condition.

As an instructive example, let $f : X \to Y$ be a map in $\mathcal{C}$ with pullbacks. The adjunction $f_\dagger \dashv f^\dagger$ in $2\mathrm{Span}(\mathcal{C})$ can be exhibited by the following unit-counit pair:

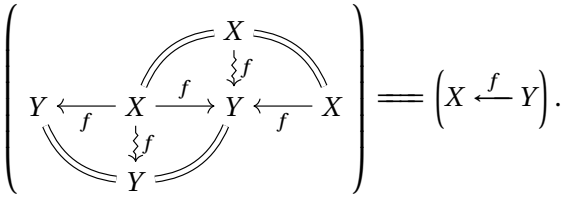
$$\eta_f \coloneqq \left(\begin{array}{ccccc} & & X & & \\ & \overset{p_1}{\nwarrow} & \downarrow{\scriptstyle \delta_f} & \overset{p_2}{\nearrow} & \\ X & & X\times_Y X & & X \end{array} \right) : \begin{pmatrix} \mathrm{id}_X \\ \Downarrow \\ p_{2\dagger}p_1^\dagger \end{pmatrix} \qquad \varepsilon_f \coloneqq \left(\begin{array}{ccccc} & \overset{f}{\swarrow} & X & \overset{f}{\searrow} & \\ Y & & \downarrow{\scriptstyle f} & & Y \\ & & Y & & \end{array} \right) : \begin{pmatrix} f_\dagger f^\dagger \\ \Downarrow \\ id_Y \end{pmatrix}.$$

$$\simeq \left(\begin{array}{ccccc} & & X & & \\ & \overset{f}{\swarrow} & \downarrow{\scriptstyle f} & \overset{f}{\searrow} & \\ X & & Y & & X \end{array} \right) : \begin{pmatrix} \mathrm{id}_X \\ \Downarrow \\ f^\dagger f_\dagger \end{pmatrix},$$

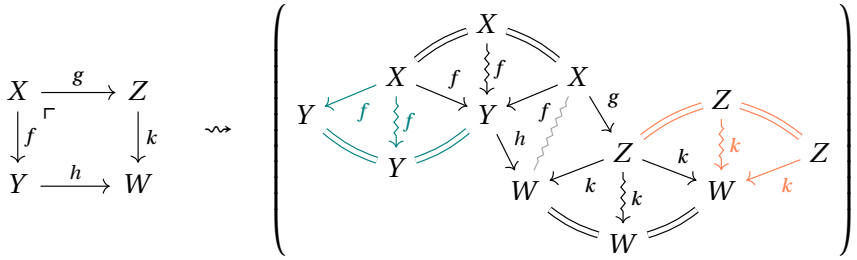
One of the triangle equalities $(f^\dagger \varepsilon_f) \circ (\eta_f f^\dagger) = \mathrm{id}_{f^\dagger}$ is witnessed by the following computation:



which we suggestively depict as



Given a pullback square in $\mathcal{C}$ as on the left below, the *vertical right mate* of the commutative square $k_\dagger g_\dagger \simeq h_\dagger f_\dagger$ in $2\mathrm{Span}(\mathcal{C})$ (black part on the right middle), given by the composite 2-cell on the right, is known as its (vertical right) Beck–Chevalley transformation.



In the form of a formula: $\mathrm{BC}(\alpha) : g_\dagger f^\dagger \overset{\eta_k \cdot\cdot}{\Longrightarrow} k^\dagger k_\dagger g_\dagger f^\dagger = k^\dagger h_\dagger f_\dagger f^\dagger \overset{\cdot\cdot \varepsilon_f}{\Longrightarrow} k^\dagger h_\dagger.$

Here the colored parts in the formula and diagram correspond. Thus, as opposed to a mere base change isomorphism $k^\dagger h_\dagger \simeq g_\dagger f^\dagger$ in the span 1-category, the span 2-category specifically forces $\mathrm{BC}(\alpha)$ to be an isomorphism. One says that the pullback square on the left becomes *vertically right adjointable* in $2\mathrm{Span}(\mathcal{C})$.

Thus, given a functor $\mathcal{C} \to \widehat{\mathrm{Cat}}$, satisfying the existence of right adjoints, and the right adjointability condition, one might expect a uniquely determined extension $2\mathrm{Span}(\mathcal{C}) \to 2\mathrm{Cat}$. This is indeed the case, and manifests in the form of the following theorem.

Theorem 3.2.1 (Hermida–Gaitsgory–Rozenblyum–MacPherson–Stefanich [CLR25, Theorem 2.5 and 3.12]) *For a pullback pair $(\mathcal{C}, E)$, the inclusion $\mathcal{C}^{op} \to 2\mathrm{Span}_E(\mathcal{C})^{1-op}$ is the initial functor from $\mathcal{C}$ into a 2-category which is left adjointable along E, in the sense that for any 2-category $\mathscr{X}$, there is an equivalence of functor 2-categories*

$$2\mathrm{Fun}^{E\text{-}ladj.}(\mathcal{C}^{op}, \mathscr{X}) \xleftarrow[\sim]{(-)^{op,*}_\dagger} 2\mathrm{Fun}(2\mathrm{Span}_E(\mathcal{C})^{op}, \mathscr{X}),$$

where the left hand side denotes the 2-category of functors which sends pullbacks along E to squares in $\mathscr{X}$ that are left adjointable in the E-direction.

Moreover, when $= 2\widehat{\mathrm{Cat}}$ *is the* 2*-category of* 1*-categories, the* 2*-categorical* ***unfurling*** *construction*

$$\mathrm{Fun}^{E\text{-}ladj.}(\mathcal{C}^{op}, \mathscr{X}) \xleftarrow{\Upsilon} \mathrm{Fun}(2\mathrm{Span}_E(\mathcal{C})^{op}, \mathscr{X})$$

provides an explicit inverse equivalence between the functor 1*-categories.* □

We briefly recall how unfurling is defined for 1-categories.

Unfurling Given a pullback pair $(\mathcal{C}, E)$ and a functor $p : \mathcal{D} \to \mathcal{C}$, such that $p^{-1}(E) \to E$ is a Cartesian fibration, let $\mathcal{D}_E^{p\text{-Cart}}$ of p-Cartesian morphisms over E. Then $(\mathcal{D}, \mathcal{D}_E^{p\text{-Cart}})$ is a pullback pair and p is a morphism of pullback pairs [MackI, Proposition 11.2]. Barwick's unfurling construction is defined as

$$\Upsilon(p) : \mathrm{Span}_{\mathcal{D}_E^{p\text{-Cart}}}(\mathcal{D}) \to \mathrm{Span}_E(\mathcal{C}).$$

Theorem 3.2.2 (Barwick, [MackI, Lemma 11.5]) *Let* $p : \mathcal{D} \to \mathcal{C}$ *be a coCartesian fibration, classifying a functor* $D : \mathcal{C} \to \widehat{\mathrm{Cat}}$ *which is right adjointable along* E*. Let* $D(f)^R$ *denote the right adjoint to* $D(f)$ *for* f *in* E*. Then* p *is Cartesian along* E *and* $\Upsilon(p)$ *is a coCartesian fibration, classifying*

$$\begin{array}{ccccc} \mathbf{D} & : & \mathrm{Span}_E(\mathcal{C}) & \longrightarrow & \widehat{\mathrm{Cat}} \\ & & \left(\begin{array}{ccc} & d & \\ {\scriptstyle f}\swarrow & & \searrow{\scriptstyle g} \\ c & & e \end{array} \right) & \longmapsto & \left(D(c) \xrightarrow{D(f)^R} D(d) \xrightarrow{D(g)} D(e) \right). \end{array}$$

The proof [MackI, §12] is highly combinatorial and we cannot shed more light onto it.

History of the universal property of 2Span The above universal property is an ancient folklore in the $(2,2)$-categorical world, first appearing in literature as [Her00, Appendix A]) without proof. In the context of higher category theory, it is often attributed to Gaitsgory–Rozenblyum, but only formally established by MacPherson [Mac22] and Stefanich [Ste20].

There is a symmetric monoidal version, see [Mac22] and [CLL25b] for a much more generalised version. In recent years, spans have been used to accommodate six-functors, higher semiadditivity, equivariant algebra among many areas. See the introduction to [CLR25] for an incomplete list.

3.3. The categorical Mackey functors

We can now describe the construction of various K-theory Mackey functors.

Proposition 3.3.1 *Let* $\mathcal{C}$ *by a (large) semiadditive category.*

1. *Let* F *be an additive endofunctor of* $\mathcal{C}$*. The* $\widehat{\mathbb{Z}}$*-categories*

$$\underline{\mathrm{End}}(\mathcal{C}, F), \quad \underline{\mathrm{Aut}}(\mathcal{C}, F) \quad : \quad \mathrm{Orb}_{\widehat{\mathbb{Z}}}^{op} \to 2\widehat{\mathrm{Cat}}$$

are left and right adjointable, and extend to $\widehat{\mathbb{Z}}$*-Mackey functors*

$$\underline{\mathrm{End}}(\mathcal{C}, F), \quad \underline{\mathrm{Aut}}(\mathcal{C}, F) \quad : \quad \mathrm{Span}(\widehat{\mathbb{Z}}\text{-Set})^{op} \to \widehat{\mathrm{Cat}}.$$

2. *When F is the identity on $\mathcal{C}$, the $\widehat{\mathbb{Z}}$-Mackey functor $\underline{\mathrm{End}}(\mathcal{C})$ extends to an epicyclic Mackey functor, and $\underline{\mathrm{Aut}}(\mathcal{C})$ extends to a topological Cartier module.*

$$\underline{\mathrm{End}}(\mathcal{C}) : \mathrm{Span}_{\alpha}(\widetilde{\Lambda}^{\amalg})^{op} \longrightarrow \widehat{\mathrm{Cat}} \qquad \underline{\mathrm{Aut}}(\mathcal{C}) : \mathrm{Span}(\mathbf{B}\mathbb{W}^{\amalg})^{op} \longrightarrow \widehat{\mathrm{Cat}}$$

3. *Let G be a discrete group and $\mathcal{C}$ be equipped with a G-action. The associated Borel G-category $\underline{\mathcal{C}}_G$ is left adjointable along maps of finite fiber, and thus extends to a Mackey functor*

$$\underline{\mathcal{C}}_G : \mathrm{Span}_{\acute{e}t}(G)^{op} \to \widehat{\mathrm{Cat}}.$$

4. *More generally, for a profinite group G, let $\underline{\mathcal{C}}_G$ be an Orb_G-category, such that for every discrete quotient G/N, $\underline{\mathcal{C}}|_{\mathrm{Orb}_{G/N}}$ is G/N-Borel. Then there is a Mackey functor*

$$\underline{\mathcal{C}}_G : \mathrm{Span}(G)^{op} \to \widehat{\mathrm{Cat}}.$$

5. *Suppose the functors $\mathrm{Orb}_G^{op} \to \widehat{\mathrm{Cat}}$ above lands in Pr^R, and that the maps of finite fibers are sent into $\mathrm{Pr}^R \cap \mathrm{Pr}^L$. Then the functor $\mathrm{Orb}_G \to 2\mathrm{Pr}^L$ obtained by passing to left adjoints is right adjointable along maps of finite fiber, so that there is a "coMackey" functor with finite transfers.*

$$\underline{\mathcal{C}}_G : \mathrm{Span}_{\acute{e}t}(G) \to \mathrm{Pr}^L$$

Proof. We argue for item 1. Item 2 follows by a same argument, since the required adjointability is along the same maps. The items 3,5 and 4 are similar to item 1 but easier, since groups act by equivalences.

Let $p : \mathbb{N}\backslash\backslash\mathcal{C} \to \mathbf{B}\mathbb{N}$ be the coCartesian unstraightening of $(\mathcal{C}, F)$. In order to apply Theorem 3.2.1, we need the existence of left or right adjoint to

$$\alpha_{k,n}^* : \mathrm{End}^{(n)}(\mathcal{C}, F) \longrightarrow \mathrm{End}^{(kn)}(\mathcal{C}, F),$$

and verify that they satisfy base-change.

Consider the precomposition functor $\alpha_{k,n}^*$ between the coCartesian fibrations

$$\begin{array}{ccc} \mathrm{END}^{(n)}(\mathcal{C}, F) & \xrightarrow{\alpha_{k,n}^*} & \mathrm{END}^{(kn)}(\mathcal{C}, F) \\ & \searrow \quad \swarrow & \\ & \mathbf{B}\mathbb{N} & \end{array}$$

which is given by the diagonal map $\mathcal{C}^{\times n} \to \mathcal{C}^{\times kn}$ on the fibers.

The claim for $\underline{\mathrm{End}}(\mathcal{C}, F)$ would follow from the existence of a relative left or right adjoint in the sense of [HA, Definition 7.3.2.2]. For $\underline{\mathrm{Aut}}(\mathcal{C}, F)$, we additionally need these adjoints to preserve coCartesian morphisms.

By [HA, opposite of 7.3.2.6, 7.3.2.11], the relative right adjoint exists if right adjoints exist fiberwise, whereas for the relative left adjoint to exist, fiberwise left adjoints must exist, and they must additionally commute with the coCartesian transport functors. The latter condition translates to F preserving direct sums. Under this assumption, the relative adjoints exist and preserve coCartesian morphisms. Taking global sections gives the required adjoints between $\mathrm{End}^{(-)}(\mathcal{C}, F)$ and $\mathrm{Aut}^{(-)}(\mathcal{C}, F)$.

Furthermore, the adjoints obtained above are given by fiber-wise Kan extension, whence they automatically satisfy base-change. Thus the required adjointability is satisfied, and Theorem 3.2.1 (unfurling) applies to give the required extensions to $\mathrm{Span}(\widehat{\mathbb{Z}})^{\mathrm{op}}$. Since $\mathrm{Fun}(-, \mathcal{D})$ takes disjoint unions to products, the resulting functors are Mackey functors. This completes the proof of item 1. □

Note that if F does not preserve direct sums, we still obtain right-adjointability of $\underline{\mathrm{End}}(\mathcal{C}, F)$.

The subsections will be dedicated to examples, and determining the functorialities of various K-theory Mackey functors in the case of large categories.

3.3.1. G-modules and G-representations

We finally come to discuss the associated K-theory Mackey functors. Let G be a group and A a commutative ring. We might be interested in the representation theory of G in A-modules, and in particular the K-theory of them.

Definition 3.3.2 *Consider the category* A-Mod *of* A*-module spectra (a.k.a. the derived category* $\mathrm{D}(A)$*) with trivial* G*-action. Proposition 3.3.1 item 5 provides a* Pr^L*-valued* G*-coMackey functor with finite transfers, which we denote by*

$$\underline{A\text{-Mod}}_G : \mathrm{Span}_{ét}(G) \to \mathrm{Pr}^L.$$

Proposition 3.3.3 *The functor* $\underline{A\text{-Mod}}_G$ *factors through* $\mathrm{Pr}^L_{\mathrm{st},\omega}$*, and its value at the orbit* G/H *is equivalent to* $A[H]$-Mod.

Proof. The identification $A\text{-Mod}^{\mathrm{h}H} \simeq A[H]\text{-Mod}$ follows from the Schwede–Shipley theorem [HA, 7.1.2.6]. Consider a map $G/H \to G/K$ in Orb_G. Chasing through the definitions, we see the forward functoriality of $\underline{A\text{-Mod}}_G$ is given by induction $A[H]\text{-Mod} \to A[K]\text{-Mod}$, which is always in $\mathrm{Pr}^L_{\mathrm{st},\omega}$. Its right adjoint, module restriction, preserves compactness iff $A[K]$ is a perfect $A[H]$-modules, which happens when the map above has finite fiber. □

Applying the universal localising invariant $\mathcal{U}_{\mathrm{loc}} : \mathrm{Pr}^L_{\mathrm{st},\omega} \simeq \mathrm{Cat}^{\mathrm{perf}} \to \mathrm{Mot}_{\mathrm{loc}}$ of Blumberg–Gepner–Tabuada [BGT13], we thus obtain coMackey functor with finite transfers

$$\underline{\mathcal{U}_{\mathrm{loc}}(A)}_G \in \mathrm{Fun}^{\times}(\mathrm{Span}_{ét}(G), \mathrm{Mot}_{\mathrm{loc}}),$$

whose value at G/H is $\mathcal{U}_{\mathrm{loc}}(A[H])$.

Similarly, we can consider the small category $\mathrm{Perf}(A)$ instead of A-Mod.

Definition 3.3.4 *Consider* $\mathrm{Perf}(A) \in \mathrm{Cat}^{\mathrm{perf}}$ *with trivial* G*-action, we obtain through item 3 of Proposition 3.3.1 Mackey functors*

$$\underline{R(A)}_G \in \mathrm{Mack}_{ét}(G; \mathrm{Cat}^{\mathrm{perf}}) \qquad \textit{alongside} \qquad \underline{\mathcal{U}_{\mathrm{loc}}R(A)}_G \in \mathrm{Mack}_{ét}(G; \mathrm{Mot}_{\mathrm{loc}}).$$

Thus $(\underline{R(A)}_G)^H$ is the category of perfect A-modules with H action, in contrast to $(\underline{A\text{-Mod}}_G)^H$ being the perfect $A[H]$-modules.

3.3.2. Endomorphisms and Automorphisms

Let F be an additive endofunctor of $\mathcal{C}$. Then the transfer on the $\widehat{\mathbb{Z}}$-Mackey category $\underline{\mathrm{End}}(\mathcal{C}, F)$ along $\alpha_{k,n} : \mathbb{T}_{kn} \to \mathbb{T}_n$ acts as

$$\begin{aligned} \alpha_{k,n,\sharp} \quad : \quad & \mathrm{End}^{(kn)}(\mathcal{C}, F) \longrightarrow \mathrm{End}^{(n)}(\mathcal{C}, F) \\ & (c_i; \alpha_i : F(c_i) \to c_{i+1})_{i \bmod kn} \longmapsto \left(\bigoplus_{i \equiv j \bmod n} c_i, \bigoplus_{i \equiv j \bmod n} \alpha_i \right)_{j \bmod n}. \end{aligned}$$

On $\underline{\mathrm{Aut}}(\mathcal{C}; \mathrm{id})$, after identifications with $\mathcal{C}^{S^1}$, transfer and restriction along $\alpha_{k,1}$ act as

$$\alpha_{k,1,\sharp}(M, f : M \to M) \simeq \left(M^{\oplus k}, \begin{pmatrix} 0 & \cdots & \cdots & 0 & f \\ 1 & 0 & & \ddots & 0 \\ 0 & 1 & \ddots & & \vdots \\ \vdots & \ddots & \ddots & 0 & \vdots \\ 0 & \cdots & 0 & 1 & 0 \end{pmatrix} \right), \qquad \alpha_{k,1}^*(M, f) \simeq (M, f^{\circ k}).$$

These are classically known as the Verschiebung and Frobenius of modules equipped with an endomorphism, see for instance [KBook, Exercise II.7.12].

To conclude, we show that one obtains $\mathrm{Mot}_{\mathrm{loc}}$-valued $\widehat{\mathbb{Z}}$-Mackey functors when $\mathcal{C}$ is compactly generated and F is continuous.

Proposition 3.3.5 *Let J be a small category and $\mathcal{C}$ a category with an endofunctor F.*

1. *If $\mathcal{C}$ admits J-shaped limits, then $\mathrm{End}^{(r)}(\mathcal{C},F)$ admits J-shaped limits which are computed component-wise. If F preserve these limits, then the same holds for $\mathrm{Aut}^{(r)}(\mathcal{C},F)$.*
2. *If $\mathcal{C}$ admits J-shaped colimits and F preserves them, then J-shaped colimits in both $\mathrm{End}^{(r)}(\mathcal{C},F)$ and $\mathrm{Aut}^{(r)}(\mathcal{C},F)$ exist and are computed component-wise.*

Proof. The diagonal map $\delta : (\mathcal{C},F) \to (\mathcal{C}^J, F^J)$ of categories with endofunctor induces

$$\delta^* : \mathrm{END}^{(r)}(\mathcal{C},F) \longrightarrow \mathrm{END}^{(r)}(\mathcal{C}^J,F^J) \qquad \text{over} \qquad \mathbf{B}\mathbb{N}.$$

The existence criteria for relative adjoints [HA, 7.3.2.6, 7.3.2.11] implies the existence of a right adjoint δ_* relative $\mathbf{B}\mathbb{N}$

$$\mathrm{END}^{(r)}(\mathcal{C},F) \underset{\delta_*}{\overset{\delta_\sharp}{\rightleftarrows}} \mathrm{END}^{(r)}(\mathcal{C}^J,F^J) \quad (\text{via } \delta^*)$$

and, if F is continuous, a relative left adjoint $\delta_\sharp$. This proves the result for $\mathrm{End}^{(r)}(\mathcal{C},F)$. If F commutes with the limits, then the adjoints preserve coCartesian sections, whence the result for $\mathrm{Aut}^{(r)}(\mathcal{C},F)$. □

Corollary 3.3.6 *Let $\mathcal{C}$ be a presentable semiadditive category and F a continuous functor, Then the $\widehat{\mathbb{Z}}$-Mackey functors $\underline{\mathrm{End}}(\mathcal{C},F)$ and $\underline{\mathrm{Aut}}(\mathcal{C},F)$ take value in Pr^L.*

If $\mathcal{C}$ is κ-accessible then so is each $\mathrm{End}^{(r)}(\mathcal{C},F)$. If F furthermore preserves small limits, then also each $\mathrm{Aut}^{(r)}(\mathcal{C},F)$ is κ-accessible.

Proof. If $\mathcal{C}$ is presentable and F is continuous, then each $\mathrm{END}^{(r)}(\mathcal{C},F) \to \mathbf{B}\mathbb{N}$ is a presentable fibration as in [HTT, 5.5.3.2], so the category of sections and of coCartesian sections are both presentable by [HTT, 5.5.3.17, 5.5.3.14]. By semiadditivity, the adjoints $\alpha_{k,n,\sharp}$ are given by direct sums of components, which commutes with all colimits thanks to semiadditivity. Therefore the $\widehat{\mathbb{Z}}$-Mackey functors take value in Pr^L.

Thus, evaluation at the vertices $\mathrm{End}^{(r)}(\mathcal{C},F) \to \mathcal{C}^{\times r}$ preserves limits and colimits and is conservative. Therefore it admits a left adjoint which preserves κ-compactness and send a set of generators of $\mathcal{C}^{\times r}$ to generators of $\mathrm{End}^{(r)}(\mathcal{C},F)$. The same argument applies for $\mathrm{Aut}^{(r)}(\mathcal{C},F)$ if F preserves small limits. □

Corollary 3.3.7 *If $\mathcal{C}$ is compactly generated and stable, and F is continuous, then the $\widehat{\mathrm{Cat}}$-valued $\widehat{\mathbb{Z}}$-Mackey functor $\underline{\mathrm{End}}(\mathcal{C},F)$ factors through $\mathrm{Pr}^L_{\mathrm{st},\omega}$, and thus induces $\mathrm{Mot}_{\mathrm{loc}}$- and $\mathrm{Mot}_{\mathrm{add}}$-valued $\mathbb{Z}$-Mackey functors. If F preserves limits then the same holds for $\underline{\mathrm{Aut}}(\mathcal{C},F)$.*

When F is the identity, the above is also satisfied by the synonymous epicyclic Mackey category and topological Cartier module of Proposition 3.3.1 item 2.

Proof. The values of $\underline{\mathrm{End}}(\mathcal{C},F)$ are compactly generated presentable categories by the previous corollary. The left and right adjoint to restriction (along regular coverings) are given by finite direct sums over certain components, which coincide by semiadditivity. Therefore all the desired functors are strongly continuous, and thus lie in $\mathrm{Pr}^L_{\mathrm{st},\omega}$, on which $\mathcal{U}_{\mathrm{add}}$ and $Uloc$ are defined.

When F preserves small limits, the same arguments works for $\underline{\mathrm{Aut}}(\mathcal{C},F)$. The final paragraph follows, since all conditions are satisfied by the identity functor. □

A. Recollements, idempotent algebras, monoidal adjunctions and the projection formula

In this section, we collect some facts about localisations of stable categories which permeates our treatment throughout the chapters, including the notion of recollements and the special case induced by idempotent (co)algebras.

This appendix draws upon well-established literature, especially from the following.

- Recollements: the unpublished notes [Rez] by Rezk, based on a short note Barwick and Glasman, enriched with examples.
- Idempotent (co)algebras: Aoki's short paper [Aok25].
- Rigid-compactly generated categories: the paper [BDS16] on Neeman duality.

Semiorthogonal decompositions and recollements

A *semiorthogonal decomposition* of a stable category $\mathcal{C}$ is the datum of two full stable subcategories $\mathcal{A} \hookrightarrow \mathcal{C} \hookleftarrow \mathcal{B}$, such that:

1. for $a \in \mathcal{A}$, $b \in \mathcal{B}$, the mapping spectrum $\mathrm{map}_C(a, b)$ vanishes, and
2. each $c \in \mathcal{C}$ sits in a cofiber sequence $a \xrightarrow{\varepsilon} c \longrightarrow \eta b \xrightarrow{\delta} \Sigma a$, where $a \in \mathcal{A}$, $b \in \mathcal{B}$.

It follows from 1. that for each c, the cofiber sequence of 2. is unique up to contractible choice, and that setting $j(c) := a$, $p(c) := b$ defines a *right* adjoint j to the inclusion $i : \mathcal{A} \hookrightarrow \mathcal{C}$ and a *left* adjoint p to $q : \mathcal{B} \hookrightarrow \mathcal{C}$.

As a map in a stable category is an isomorphism iff its cofiber is 0, the situation can be depicted as follows.

Proposition A.1 *In the situation above, one has $i(\mathcal{A}) = p^{-1}(0)$ and $q(\mathcal{B}) = j^{-1}(0)$, i.e.*

1. *$c \in i(\mathcal{A})$ iff $p(c) = 0$ iff $\mathrm{map}_{\mathcal{C}}(c, q(b)) = 0$ for all $b \in \mathcal{B}$;*
2. *$c \in q(\mathcal{B})$ iff $j(b) = 0$ iff $\mathrm{map}_{\mathcal{C}}(i(a), c) = 0$ for all $a \in \mathcal{A}$.*

So that both sequences below are fiber sequences of stable categories:

$$\mathcal{A} \underset{j}{\overset{i}{\rightleftarrows}} \mathcal{C} \underset{q}{\overset{p}{\rightleftarrows}} \mathcal{B}, \tag{SOD}$$

where we understand each functor on top is left adjoint to the one below it. □

Proposition A.2 *The functor j admits a further right adjoint k iff q admits a right adjoint r, in which case there is a semiorthogonal decomposition*

$$\mathcal{B} \underset{r}{\overset{q}{\rightleftarrows}} \mathcal{C} \underset{k}{\overset{j}{\rightleftarrows}} \mathcal{A}..$$

Lemma A.3 (cf. proof of [NKP24, Lemma 2.3.15]) *Suppose a functor $U : \mathcal{D} \to \mathcal{E}$ admits both a left adjoint L and a right adjoint R. Then L is fully faithful iff R is, i.e. LU is a Bousfield colocalisation iff RU is a Bousfield localisation.*

Proof of lemma. Recall that the right transpose of a natural transformation $\lambda : F_1 \Rightarrow F_2$ between functors $F_i : \mathscr{C} \to \mathscr{D}$ with right adjoint G_i, $i = 1, 2$ is the transformation

$$(\rho : G_2 \Rightarrow G_1) := \left(\begin{array}{ccc} \mathscr{C} & \xleftarrow{G_1} & \mathscr{D} \\ & \overset{\eta_1}{\Rightarrow} \quad \overset{F_1}{\Rightarrow} \quad \overset{\lambda}{\Leftarrow} \quad \overset{F_2}{\Rightarrow} \quad \overset{\varepsilon_2}{\Rightarrow} & \\ \mathscr{C} & \xrightarrow[G_2]{} & \mathscr{D} \end{array} \right).$$

Transposition induces an equivalence between the $\mathrm{Fun}^{\mathrm{L}}(\mathscr{C}, \mathscr{D})^{\mathrm{op}}$ and $\mathrm{Fun}^{\mathrm{R}}(\mathscr{D}, \mathscr{C})$. [10]

Using the triangle equalities, one checks that the counit transformation $UR \Rightarrow \mathrm{id}_{\mathcal{E}}$ is the right transpose of the unit transformation $\mathrm{id}_{\mathcal{E}} \Rightarrow UL$, so that one is an isomorphism iff the other is. □

Proof of proposition A.2. Suppose that j admits a right adjoint k, then k is fully faithful by the lemma and it's easy to check that

$$r(c) := \mathrm{fib}(c \longrightarrow kj(c))$$

exhibits a right adjoint $r \vdash q$.

Now suppose that q admits a right adjoint r, one has a semiorthogonal decomposition

$$\mathcal{B} \underset{r}{\overset{q}{\rightleftarrows}} \mathcal{C} \underset{\hat{k}}{\overset{\hat{\jmath}}{\rightleftarrows}} \mathcal{D} := \ker(r).$$

We claim that the functors $\hat{\jmath}i : \mathcal{A} \to \mathcal{D}$ and $j\hat{k} : \mathcal{D} \to \mathcal{A}$ are mutually inverse under j and $\hat{\jmath}$.

Consider the following cofiber sequences, functorial $c \in C$.

$$ij(c) \longrightarrow c \longrightarrow qp(c), \quad qr(c) \longrightarrow c \longrightarrow \hat{k}\hat{\jmath}(c)$$

Substituting c in the first sequence with the second, one obtains commuting squares

$$\begin{array}{ccccc} ijqr(c) = 0 & \longrightarrow & ij(c) & \longrightarrow & ij\hat{k}\hat{\jmath}(c) \\ \downarrow & & \downarrow & & \downarrow \\ qr(c) & \longrightarrow & c & \longrightarrow & \hat{k}\hat{\jmath}(c) \\ \downarrow & & \downarrow & & \downarrow \\ qpqr(c) & \longrightarrow & qp(c) & \longrightarrow & qp\hat{k}\hat{\jmath}(c) \end{array}$$

[10] A coherent formulation would identify both as mapping categories between biCartesian fibrations over Δ^1 with specified fibers over $\partial\Delta^1$. For our purposes of detecting isomorphisms, it suffices to work with the homotopy category.

where each row and column is a cofiber sequence, and the top left term vanishes as $jq = 0$. It follows that $ij \simeq ij\hat{k}\hat{j}$ and the lower right square is exact.

Since i is fully faithful, one obtains $j \simeq (j\hat{k})\hat{j}$ and $\mathrm{id}_{\mathcal{A}} \simeq ji \simeq (j\hat{k})(\hat{j}i)$. In other words, $j\hat{k} : \mathcal{D} \to \mathcal{A}$ is a map under j and $\hat{j}$, and is left inverse to $\hat{j}i$. Similarly one obtains $\hat{j} \simeq \hat{j}ij$, from which the remaining of the claim follows. □

In the situation above, one obtains *two* fully faithful embeddings of $\mathcal{A}$ into $\mathcal{C}$, and the resulting structure is known as a *recollement* (on $\mathcal{C}$), which we depict as follows:

$$\begin{array}{ccccc} \mathcal{A}_{\vee} & \overset{i}{\hookrightarrow} & & \overset{p}{\twoheadrightarrow} & \\ \wr & \overset{\perp}{\underset{\perp}{\twoheadleftarrow}} j \multimap & \mathcal{C} & \overset{\perp}{\underset{\perp}{\twoheadleftarrow}} q \prec & \mathcal{B} \\ \mathcal{A}^{\wedge} & \overset{k}{\hookrightarrow} & & \overset{r}{\twoheadrightarrow} & \end{array} \tag{Rec}$$

Definition A.4 *In an recollement as in Rec, we refer to the categories and functors (after [MNN17]) as:*[11]

$\mathcal{A}_{\vee}$	*torsion objects*	$(-)_{\vee} \coloneqq ij : \mathcal{C} \to \mathcal{A}_{\vee} \subseteq \mathcal{C}$	*torsion part*
$\mathcal{A}^{\wedge}$	*complete objects*	$(-)^{\wedge} \coloneqq kj : \mathcal{C} \to \mathcal{A}^{\wedge} \subseteq \mathcal{C}$	*completion*
$\mathcal{B}$	*acyclic objects*	$(-)_{\mathcal{B}} \coloneqq qp : \mathcal{C} \to \mathcal{B} \subseteq \mathcal{C}$	*coacyclification*
		$(-)^{\mathcal{B}} \coloneqq qr : \mathcal{C} \to \mathcal{B} \subseteq \mathcal{C}$	*acyclification*

There are thus cofiber sequences

$$(-)_{\vee} \Longrightarrow \mathrm{id}_{\mathcal{C}} \Longrightarrow (-)_{\mathcal{B}} \overset{\delta_{\vee}}{\Longrightarrow} \Sigma(-)_{\vee} \qquad (-)^{\mathcal{B}} \Longrightarrow \mathrm{id}_{\mathcal{C}} \Longrightarrow (-)^{\wedge} \overset{\delta^{\wedge}}{\Longrightarrow} \Sigma(-)^{\mathcal{B}}$$

as well as the exact squares, occurred during the proof of Proposition A.2, called the fracture squares *of the recollement.*

$$\begin{array}{ccc} \mathrm{id}_{\mathcal{C}} & \Longrightarrow & (-)^{\wedge} \\ \Downarrow & & \Downarrow \\ (-)_{\mathcal{B}} & \overset{\alpha}{\Longrightarrow} & ((-)^{\wedge})_{\mathcal{B}} \end{array} \qquad \begin{array}{ccc} ((-)_{\vee})^{\mathcal{B}} & \Longrightarrow & (-)_{\vee} \\ \Downarrow \beta & & \Downarrow \\ (-)^{\mathcal{B}} & \Longrightarrow & \mathrm{id}_{\mathcal{C}} \end{array}$$

There are obvious functors between $\mathcal{C}$ and the categories consisting of such sequences, which are all equivalences.

Proposition A.5 (Fracture squares) *In a semiorthogonal decomposition as in SOD, the square*

$$\begin{array}{ccc} \mathcal{C} & \xrightarrow{p} & \mathcal{B} \\ {\scriptstyle j}\downarrow & \ulcorner\ {\scriptstyle \delta} & \downarrow{\scriptstyle q} \\ \mathcal{A} & \xrightarrow{\Sigma i} & \mathcal{C} \end{array}$$

is a lax pullback square of categories. Concretely this means that the square

$$\begin{array}{ccc} \mathcal{C} & \xrightarrow{\delta} & \mathcal{C}^{\Delta^1} \\ {\scriptstyle (j,p)}\downarrow & \ulcorner & \downarrow{\scriptstyle (t,s)} \\ \mathcal{A} \times \mathcal{B} & \xrightarrow{\Sigma i \times q} & \mathcal{C} \times \mathcal{C} \end{array}$$

[11] The names are merely suggestive. In specific situations, we often adopt more appropriate names.

is a pullback square, where $\delta : \mathcal{C} \to \mathcal{C}^{\Delta^1}$ *sends* $c \in \mathcal{C}$ *to the connective morphism* $qp(c) \to \Sigma ij(c)$.

In a recollement as in Rec, there are lax pullback squares

$$\begin{array}{ccc} \mathcal{C} & \xrightarrow{(-)_{\mathcal{B}}} & \mathcal{B} \\ {\scriptstyle (-)^\wedge}\downarrow & \swarrow_{\alpha} & \| \\ \mathcal{A}^\wedge & \xrightarrow{(-)_{\mathcal{B}}} & \mathcal{B} \end{array}, \qquad \textit{and} \qquad \begin{array}{ccc} \mathcal{C} & \xrightarrow{(-)^{\mathcal{B}}} & \mathcal{B} \\ {\scriptstyle (-)^\wedge}\downarrow & \swarrow_{\beta} & \| \\ \mathcal{A}_\vee & \xrightarrow{(-)^{\mathcal{B}}} & \mathcal{B} \end{array},$$

where α *and* β *is the marked morphism in the fracture squares above.*

Proof. We argue for the first fracture square. The rest is similar. The lax pullback is equivalent to the category of pullback squares

$$\begin{array}{ccc} P & \longrightarrow & b \\ \downarrow & & \downarrow{\scriptstyle \alpha} \\ a & \xrightarrow{\eta} & a_{\mathcal{B}} \end{array}$$

where $a \in \mathcal{A}^\wedge \subseteq \mathcal{C}$ and $b \in \mathcal{B} \subseteq \mathcal{C}$. It's routine to check that this square coincides with the fracture square of P. □

As a source for intuition, we provide two prototypes. See [Rez] for more examples.

Example A.6 (Arithmetic recollements and fracture squares) The category of rational spectra $\mathrm{Sp}_{\mathbb{Q}} \subseteq \mathrm{Sp}$ is both a left and a right Bousfield localisation of Sp. Consequently, there is a recollement

$$\mathrm{Sp}^{\mathrm{tor}} \underset{\mathrm{Sp}^{\mathrm{prof}}}{\overset{(-)_\vee}{\underset{(-)^\wedge}{\rightleftarrows}}} \mathrm{Sp} \underset{\mathrm{map}(-,\mathbb{Q})}{\overset{-\otimes\mathbb{Q}}{\rightleftarrows}} \mathrm{Sp}_{\mathbb{Q}}$$

Here the torsion (resp. complete) objects are exactly the torsion (resp. profinite complete) spectra.

The following exact square is known as the *arithmetic fracture square* for $X \in \mathrm{Sp}$.

$$\begin{array}{ccccc} X & \longrightarrow & X^\wedge & = & \prod_{p \text{ prime}} X_{\hat{p}} \\ \downarrow & & & & \downarrow \\ X_{\mathbb{Q}} & \longrightarrow & (X^\wedge)_{\mathbb{Q}} & = & \left(\prod_{p \text{ prime}} X_{\hat{p}}\right)_{\mathbb{Q}} \end{array}$$

The following prototypical recollement is ubiquitous in the study of topology.

Example A.7 ("Unstable recollement" on an ∞-topos) Consider the decomposition of a topological space, or more generally ([HTT, 7.3.2.10]), an ∞-topos $\mathfrak{X}$ into an open subspace/subtopos and its closed complement:

$$i : \mathfrak{U} \hookrightarrow \mathfrak{X} \hookleftarrow \mathfrak{Z} : j$$

Recall that a open subtopos $i_* : \mathfrak{U} \rightleftharpoons \mathfrak{X} : i^*$ is the slice ∞-topos $\mathfrak{X}_{/U}$ for some (-1)-truncated $U \in \mathfrak{X}$. There is a further left adjoint $i_! : \mathfrak{U} \to \mathfrak{X}$ to i^* corresponding to the forgetful functor $\mathfrak{X}_{/U} \to \mathfrak{X}$, which is fully faithful.

Comparing with [HTT, 7.3.2.4], the closed complement $\mathfrak{Z} := \mathfrak{X}/U$ is seen to correspond to the full subcategory of $\mathfrak{X}$ which is sent to the final object $U \in \mathfrak{U}$ under i^*, such that j_* is the inclusion. Passing to left adjoints, we obtain the following pushout in Pr^L.

$$\begin{array}{ccc} \mathfrak{U} & \xrightarrow{i_!} & \mathfrak{X} \\ \downarrow & & \downarrow{\scriptstyle j^*} \\ * & \longrightarrow & \mathfrak{Z} \end{array}$$

We depict this as the following "unstable recollement", for reasons which soon will become clear. Note that j_* doesn't admit a right adjoint unstably (unless $U = \emptyset$).

$$\begin{array}{ccccc} \mathfrak{U}_\vee & \xrightarrow{i_!} & & \xrightarrow{j^*} & \\ \Vdash & \overset{\perp}{\underset{\perp}{\longleftarrow}} i^* - & \mathfrak{X} & \overset{\perp}{\longleftarrow} j_* \prec & \mathfrak{Z} \\ \mathfrak{U}^\wedge & \xrightarrow[i_*]{} & & & \end{array}$$

Sheaves on $\mathfrak{X}$ valued in $\mathcal{V} \in \mathrm{Pr}^L_{\mathrm{st}}$ are defined as $\mathrm{Sh}(\mathfrak{X}; \mathcal{V}) = \mathrm{RFun}(\mathfrak{X}^{\mathrm{op}}, \mathcal{V}) \simeq \mathfrak{X} \otimes \mathcal{V}$. Tensoring with $\mathcal{V}$ preserves the above pushout and fully faithfulness of i_*, due to the internal adjunction $i_! \dashv i^*$ in Pr^L. We thus obtain the following recollement for any presentable stable $\mathcal{V}$.

$$\begin{array}{ccccc} \mathrm{Sh}(\mathfrak{U}; \mathcal{V})_\vee & \xrightarrow{i_!} & & \xrightarrow{j^*} & \\ \Vdash & \overset{\perp}{\underset{\perp}{\longleftarrow}} i^* - & \mathrm{Sh}(\mathfrak{X}; \mathcal{V}) & \overset{\perp}{\underset{\perp}{\longleftarrow}} j_* \prec & \mathrm{Sh}(\mathfrak{Z}; \mathcal{V}) \\ \mathrm{Sh}(\mathfrak{U}; \mathcal{V})^\wedge & \xrightarrow[i_*]{} & & \xrightarrow[j^!]{} & \end{array}$$

Remark A.8. The above example, applied to to $\mathfrak{X} = \mathrm{An}_G$ and $U = \mathbf{E}\mathcal{F}$, recovers a recollement on $\mathrm{Sp}_G^{\mathrm{Bredon}}$, but not Sp_G.

An important class of recollements involves a symmetric monoidal structure on $\mathcal{C}$, and are associated to *idempotent (co)algebras* therein. These will be studied in the next section of this appendix.

Idempotent (co)algebras

Let $(\mathcal{C}, \otimes, \mathbb{1})$ be a (potentially large) symmetric monoidal category. An $\mathbb{E}_0$-algebra $\mathbb{1} \xrightarrow{\eta} E$ is said to be **idempotent** if the maps

$$E \simeq \mathbb{1} \otimes E \xrightarrow{\eta \otimes \mathrm{id}_E} E \otimes E, \quad E \simeq E \otimes \mathbb{1} \xrightarrow{\mathrm{id}_E \otimes \eta} E \otimes E$$

are equivalences. Similarly we define **idempotent coalgebras** as idempotent algebras in $\mathcal{C}^{\mathrm{op}}$.

Example A.9 In a Cartesian monoidal category, every object is canonically a coalgebra, which is idempotent precisely when it is (-1)-truncated. In particular, the idempotent coalgebras in an ∞-topos are exactly the subobjects of the terminal object. In the case $\mathrm{Sh}(X)$ of sheaves on a topological space, these are open subsets of X, and in the case of An_G, the families of subgroups.

In equivariant homotopy theory, we see that $\mathrm{Cog}^{\mathrm{idem}}(\mathrm{An}_G)$ consists of the universal $\mathcal{F}$-spaces $\mathbf{E}\mathcal{F} \in \mathrm{An}_G$. The correspondence of downward-closed full subcategories with idempotent coalgebras holds in any presheaf topos.

Proposition A.10 ([HA, 4.8.2]) *An idempotent algebra E admits a canonical $\mathbb{E}_\infty$-algebra structure. The full category $\mathrm{CAlg}^{\mathrm{idem}}(\mathcal{C})$ of $\mathrm{CAlg}(\mathcal{C})$, consisting of idempotent algebras, is a (potentially large) poset which*

coincides with the lattice of smashing localisations L of $\mathcal{C}$, i.e. symmetric monoidal localisations for which the projection formula holds, in the sense that the morphism

$$L(\mathbb{1}) \otimes c \longrightarrow L(\mathbb{1}) \otimes L(c) \longrightarrow L(\mathbb{1} \otimes c) \simeq L(c)$$

is an equivalence for every $c \in \mathcal{C}$. The category of E-modules is full subcategory of $\mathcal{C}$, and coincides with the essential image of the corresponding localisation.

The dual statement holds for the category $\mathrm{CCog}^{\mathrm{idem}}$ of coidempotent coalgebras and smashing colocalisations.

Proof. This is a summary of [HA, 4.8.2.4/7/9]. The compatibility of a localisation with the monoidal structure, as well as the projection formula, are properties of a localisation. It follows that the category of such localisations is a (large) poset, and is equivalent to the idempotent algebras. □

When the symmetric monoidal structure is compatible with (co)limits, we have the following.

Lemma A.11 ([Aok25, Theorem 3.8]) *For two idempotent coalgebras E, F in $\mathcal{C}$, their meet $E \wedge F$ is given by $E \otimes F$.*

If $\mathcal{C}$ has pushouts and $\otimes$ preserves pushouts separately in each variable, then the join $E \vee F$ is given by the pushout $E \cup_{E \otimes F} F$. In particular $\mathrm{CCog}^{\mathrm{idem}}(\mathcal{C})$ is a finitary distributive lattice.

If $\mathcal{C}$ has all filtered colimits and tensoring preserves these in each variable, then the lattice $\mathrm{CCog}^{\mathrm{idem}}(\mathcal{C})$ in $\mathrm{Cog}_{\mathbb{E}_0}(\mathcal{C})$ is closed under, and thus admits, all filtered colimits. □

This lets us compute e.g. the join of two $\mathbf{E}\mathcal{F}$'s as a pushout in An_G or Sp_G.

Remark A.12 ([Aok25, Theorem 3.13]). In the presentable case, $\mathrm{CCog}^{\mathrm{idem}}$ is in fact small, and thus a complete distributive lattice.

Let $\mathcal{C}$ now be stably symmetric monoidal, then the category of $\mathbb{E}_0$-coalgebras $x \longrightarrow \mathbb{1}$ and $\mathbb{E}_0$-algebras $\mathbb{1} \longrightarrow y$ are equivalent through taking the (co)fiber.

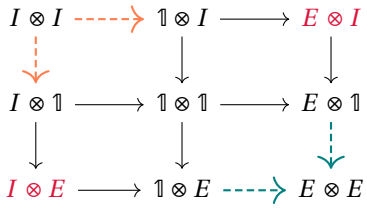
$$\begin{array}{rcl} \mathrm{Cog}_{\mathbb{E}_0}(\mathcal{C}) & \xleftrightarrow{\sim} & \mathrm{Alg}_{\mathbb{E}_0}(\mathcal{C}) \\ (x \to \mathbb{1}) & \longmapsto & (\mathbb{1} \to \mathrm{cofib}(x \to \mathbb{1})) \\ (\mathrm{fib}(\mathbb{1} \to y)) & \longmapsfrom & (\mathbb{1} \to y) \end{array}$$

Lemma A.13 *Let $(\mathcal{C}, \otimes, \mathbb{1})$ be a stably symmetric monoidal category, and $I \longrightarrow \mathbb{1} \longrightarrow E$ a cofiber sequence in $\mathcal{C}$. The following are equivalent:*

1. *$I \longrightarrow \mathbb{1}$ is an idempotent coalgebra.*
2. *$\mathbb{1} \longrightarrow E$ is an idempotent algebra.*
3. *$I \otimes E = 0$.*

Thus the above equivalence of $\mathbb{E}_0$-coalgebras and $\mathbb{E}_0$-algebras restrict to an equivalence between the posets $\mathrm{CCog}^{\mathrm{idem}}(\mathcal{C})$ and $\mathrm{CAlg}^{\mathrm{idem}}(\mathcal{C})$.

Proof. Stably monoidal means that $\otimes$ is biexact, so that each row and each column in the following commutative grid is a cofiber sequence.



Then the $\mathbb{E}_0$-algebra E being idempotent, i.e. the two arrows in the lower right being isomorphisms, is equivalent to the two copies of $I \otimes E$ being zero, which in turn is equivalent to the two arrows in the top left being isomorphisms, i.e. I being an idempotent coalgebra. □

Given an idempotent sequence $I \longrightarrow \mathbb{1} \longrightarrow E$ in a stably closed symmetric monoidal category $\mathcal{C}$ (e.g. presentable), one can associate the following recollement:

$$\mathrm{Comod}_I(\mathcal{C}) =: \mathcal{C}_I \underset{\underline{\mathrm{map}}(I,-)}{\overset{-\otimes I}{\rightleftarrows}} \mathcal{C} \underset{\underline{\mathrm{map}}(E,-)}{\overset{-\otimes E}{\rightleftarrows}} \mathcal{C}[I^{-1}] := \mathrm{Mod}_E(\mathcal{C})$$

where the markings with • indicate functors whose kernel is a (localising) $\otimes$-ideal. It follows that the localisations onto $\mathcal{C}^I$ and $\mathcal{C}[I^{-1}]$ are compatible with the monoidal structure in the sense of [HA, 2.2.1.9].

For instance, $\mathcal{F}$-isotropy separation on Sp_G (section 1.5) is of this form.

Monoidal adjunction, rigid compact generation and projection formula

We recall some statements on monoidal adjunctions and localisations.

Proposition A.14 ([HA, 1.4.4.7, 2.2.1.9]) *Let $\mathcal{C}$ be a stably symmetric monoidal category and*

$$\mathcal{C} \overset{L}{\twoheadrightarrow} L\mathcal{C} \overset{i}{\hookrightarrow} \mathcal{C}$$

an exact localisation. Then $L\mathcal{C} \subseteq \mathcal{C}$ is a stable subcategory, and there is a semiorthogonal decomposition

$$\ker(L) \underset{}{\overset{}{\rightleftarrows}} \mathcal{C} \underset{i}{\overset{L}{\rightleftarrows}} L\mathcal{C}.$$

If $\ker(L)$ is a $\otimes$-ideal of $\mathcal{C}$, then there is a symmetric monoidal structure on $L\mathcal{C}$ making L a symmetric monoidal functor. □

The symmetric monoidal structure structure on $L\mathcal{C}$ can be described as

$$c \otimes_{L\mathcal{C}} c' \simeq Lic \otimes_{L\mathcal{C}} Lic' \simeq L(ic \otimes_{\mathcal{C}} ic').$$

Proposition A.15 ([HA, 7.3.2.7/8]) *Let $F : \mathcal{C} \to \mathcal{D}$ be a symmetric monoidal functor with right adjoint $G : \mathcal{D} \to \mathcal{C}$. Then there is a relative adjunction $F^{\otimes} : \mathcal{C} \rightleftharpoons \mathcal{D}^{\otimes} : G^{\otimes}$ of maps of ∞-operads. In particular G underlies a lax symmetric monoidal functor.*

Moreover G is lax $\mathcal{C}$-linear, in that it underlies a lax $\mathcal{LM}$-monoidal functor $\mathrm{LMod}(\mathcal{D}) \to \mathrm{LMod}(\mathcal{C})$.

Remarks on loc. cit. In the original statement in [HA, 7.3.2.7] F should be a $\mathcal{O}^{\otimes}$-monoidal functor, rather than a mere map of ∞-operads (i.e. lax $\mathcal{O}^{\otimes}$-monoidal functor). One sees that the former condition is used during the proof. □

The relative adjunction means that the unit and counit morphisms are coherently compatible with the lax symmetric monoidal structures of FG and GF. For instance,

$$a \otimes b \xrightarrow{\eta_a \otimes \eta_b} GFa \otimes GFb \xrightarrow{\text{lax}} GF(a \otimes b) \quad \text{coincides with} \quad a \otimes b \xrightarrow{\eta_{a\otimes b}} GF(a \otimes b).$$

Lax linearity means concretely that there are coherent transformations such as

$$\text{laxLin.} : c \otimes Gd \longrightarrow GFc \otimes Gd \longrightarrow G(Fc \otimes d).$$

We say that the lax linearity is strict if laxLin. is an isomorphism for all $c \in \mathcal{C}$ and $d \in \mathcal{D}$.

If $c \in \mathcal{C}$ admits a dual $c^\vee$, the triangle equalities imply that

$$G(Fc \otimes d) \xrightarrow{\text{coev}_c} c \otimes c^\vee \otimes G(Fc \otimes d) \xrightarrow{\text{laxLin.}} c \otimes G(Fc^\vee \otimes Fc \otimes d) \xrightarrow{G\text{ev}_F} c \otimes Gd$$

is an inverse to laxLin.. We discuss the situation when the lax linearity is automatically strict, which covers every functor between genuine equivariant spectra of profinite groups which is a Mackey induction, which includes res^G_H, Φ^N and Inf^N.

We saw in Lemma 1.2.5 that a system of compact generators Sp_G are dualisable. By a thick subcategory argument, we see that the dualisable object of Sp_G are exactly the compacts.

A presentably stably symmetric monoidal category $\mathcal{C}$ is called *rigid-compactly generated* if it has a set of compact dualisable generators.

Proposition A.16 *Let $(\mathcal{C}, \otimes, \mathbb{1})$ be a presentably stably symmetric monoidal category.*

1. *If $\mathbb{1} \in \mathcal{C}^\omega$, then every dualisable object of $\mathcal{C}$ is compact.*
2. *If $\mathcal{C}$ is compactly generated and admits a set of compact and dualisable generators, then every compact object of $\mathcal{C}$ is dualisable.*

Proof.
1. If D is dualisable and $\mathbb{1}$ is compact, then $\text{Map}_{\mathcal{C}}(D, -) \simeq \text{Map}_{\mathcal{C}}(\mathbb{1}, D^\vee \otimes -)$ commutes with filtered colimits.
2. The full subcategory of dualisable objects is always a thick subcategory, and every compact object is a retract of a finite colimit of generators. □

Lemma A.17 (Projection formula,[BDS16, Theorem 1.3]) *Let $\mathcal{C}$ and $\mathcal{D}$ be rigid-compactly generated, $f^* : \mathcal{C} \to \mathcal{D}$ a colimit-preserving symmetric monoidal functor, with right adjoint f_*. Then f^* preserves colimits and the lax linearity of f_* is strict, i.e.*

$$f_*(d) \otimes c \xrightarrow{\text{laxLin.}} f_*(d \otimes f^*(c)) \qquad \textit{is an isomorphism for all } c \in \mathcal{C} \textit{ and } d \in \mathcal{D}.$$

Proof. The symmetric monoidal functor f^* preserves dualisables, and thus preserves compacts. Thus the right adjoint f_* preserves colimits. Thus both sides commute with colimits in the variable $c \in \mathcal{C}$, so the claim follows from the case where c is compact, which has already been checked above. □

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